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Master's Degree in Data Science for Economics

Enhancing Needs-Based Segmentation:
Using Text Data Mining Framework for Attribute Discovery and Validation. An application in the field of Destination Branding.

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In italiano:

A mia nonna Chiche, che rimane presente come sempre. Per il suo amore incondizionato, per accogliermi sempre con un sorriso e per sostenermi in ogni passo, anche da lontano. Oggi mi guida dal cielo e la sua forza mi accompagna ogni giorno.

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In English:

To my grandmother Chiche, who remains as present as ever. For her unconditional love, for always welcoming me with a smile, and for supporting me at every step, even from afar. Today she guides me from heaven, and her strength accompanies me every day.

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Franco Reinaldo Bonifacini

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Abstract

Traditional methodological approaches to tourism marketing research often produce fragmented or non-actionable results, limiting their practical usefulness for decision-making. This thesis addresses the question of *"How should a research framework be structured in order to produce reliable and actionable insights for tourism marketing, based on the identification and evaluation of relevant attributes of tourist behaviour?"*. It proposes an alternative framework aimed at improving how relevant tourism attributes are identified, analysed, and validated.

The proposed framework integrates text mining with qualitative exploration and quantitative validation into a single, coherent process. Qualitative insights from interviews and open-ended surveys are analysed using text-embedding and frequency-based methods to uncover new and less obvious attributes of tourist behaviour. These findings inform the design of a refined, closed-ended survey used to test how effectively these attributes capture diverse tourist preferences in a measurable and comparable way.

Results from clustering, random forest, and regression analyses show that this integrated approach produces meaningful and data-driven segmentation, confirming that combining text mining and computational techniques with qualitative and quantitative approaches increases both reliability and practical applicability.

Rather than focusing on a specific destination, the framework emphasises methodological development and replicability. Ascoli Piceno served as a contextual case, whose broader purpose is to support more reliable, data-driven insights for tourism marketing. Overall, the findings validate the proposed hypothesis: a mixed-method, text-supported research framework provides a more realistic and useful understanding of tourist decision making than traditional methodologies.

Keywords: Needs-based segmentation; Attribute discovery; Text mining; Natural language processing (NLP); Destination branding; Tourist behaviour; Mixed-methods approach; Clustering analysis; Random Forest; Multinomial Logit.

Chapter 1

Introduction

This chapter presents the context and motivation of this thesis, which aims to develop an efficient methodological framework for analysing tourist behaviour. It begins with an overview of the tourism sector, highlighting its economic importance and the structural changes brought by the Covid-19 pandemic. The discussion then introduces Ascoli Piceno as the reference case. The chapter continues by outlining the previous project that served as the starting point for this research, identifying its main limitations and the lessons learned.

1.1 Tourism Context

Tourism is recognized as one of the most important economic sectors in the world. In 2019, it represented a significant share of the global economy, accounting for 7% of global trade. However, despite its importance, tourism was also among the sectors most severely affected by the Covid-19 pandemic, which impacted not only global economies but also livelihoods, public services, and development opportunities in countries across all continents [1]. Four years after the onset of the pandemic, the sector shows strong signs of recovery. International tourist arrivals have returned to 99% of 2019 levels, indicating that tourism has bounced back from what is considered the worst crisis in its history [2]. As shown in Figure 1.1, alongside the rebound in arrivals, there has also been a notable increase in international tourism revenues, reaching USD 1.7 trillion in 2024 [3].

Moreover, Ascoli Piceno, a historic city located in the Marche region of central Italy and known for its rich cultural heritage, well-preserved medieval architecture, and local traditions [4], serves as a useful reference case for this thesis, which aims to develop an efficient analytical method for the tourism sector capable of producing reliable results through the identification of the most relevant attributes. Although it is a relatively small destination in terms of total arrivals, Ascoli Piceno has shown steady growth and clear potential for further development within the

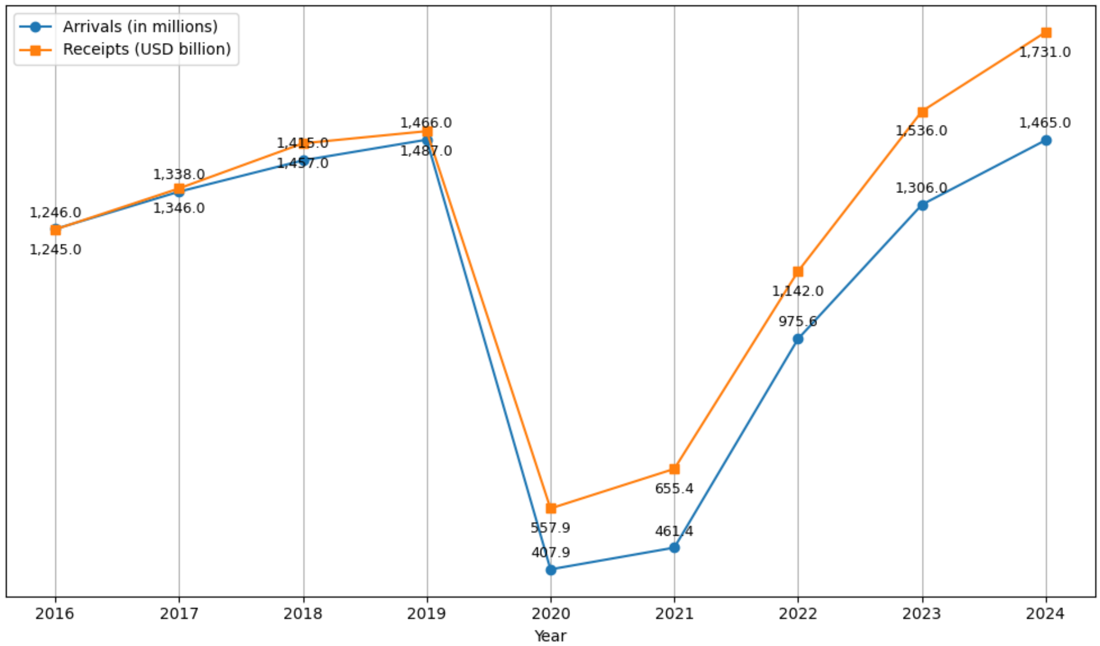


Figure 1.1: *International Tourist Arrivals and Receipts. Source: UNWTO.*

national tourism landscape. This highlights its marginal role in the Italian tourism market, despite its strong cultural and historical appeal, and reinforces the need to explore what types of tourists might be interested in discovering lesser-known destinations like Ascoli Piceno, as well as how they may differ from visitors to Italy's more popular cities [5].

1.2 Previous Work

This research originates from a project initially developed for the Marketing Analytics course within the Master's Degree in Data Science for Economics during the 2024/2025 academic year at the Università degli Studi di Milano. The project had a similar objective to the present work but was carried out over a shorter period and mainly aimed to apply the techniques taught during the course, rather than delivering an in-depth analysis of the city. The course lasted almost three months, while the project itself had a duration of around one month. Given these conditions, the earlier project presented several limitations, beyond the short timeframe, that are addressed differently and in greater depth in the current thesis.

First, there was a significant limitation in the qualitative interviews conducted during the initial stage of the project (see Appendix A for an example). The main goal of these interviews was to collect qualitative data about participants' tourist behaviour, which would then serve as the basis for defining the final attributes to be analysed in the quantitative survey. However, the interviews included a small number of predefined questions that were overly specific, leaving little room for respondents to describe their genuine behaviour in a tourist destination. This limitation may have also led to the omission of potential attributes that could have further enriched the study. These "open" questions were formulated so narrowly that interviewees often provided very short answers. In addition, the interviews lacked questions about real travel experiences, which resulted in a generalised understanding of tourist behaviour rather than experience-based insights.

Second, the following stage involved defining the main attributes. This phase also had limitations regarding the project's main goal, as it produced very general attributes (see Figure 1.2). Such generalisation led to a limited analysis of tourist behaviour, mainly because attributes such as price, food, culture, and accessibility did not clearly describe specific tourist behaviours or highlight the key aspects that a city could focus on. For instance, the attribute price could refer to food prices, accommodation costs, transportation, or event tickets, among others. Moreover, attributes like price might have been interpreted in opposite ways by different respondents (e.g., budget vs. premium travellers), leading to answers that seemed equally important but reflected contradictory meanings. This ambiguity prevented the attributes from effectively differentiating among tourist segments. Additionally, most interviews were not recorded, so the final transcriptions relied on the interviewer's notes, making it impossible for other team members to verify them or extract additional insights. Finally, no subsequent step was carried out to reinforce or validate the defined attributes, such as an open-ended survey that could have complemented the interviews and refined the attribute selection. Although the project's final outcome was sufficient for the course's goals, the definition of attributes could have been improved to achieve more precise and insightful results.

Previous Work Attributes
Accessibility/Public Transportation
Accessible Prices
Accomodation availability
Food
Historical background
Local events
Landscape
Availability of tourists' packs

Figure 1.2: *Final Attributes in Previous Work. Source: Author's own elaboration.*

Third, the next stage was the online survey, which aimed to obtain quantitative data for the final results and analysis. Although the outcomes were satisfactory and considered sufficient for the course's objectives, the total number of responses (80) was relatively low. A larger and more diverse sample would have further strengthened the validity and generalisability of the results. Some questions could also have been removed because they were already covered by others or did not contribute meaningfully. For example, respondents from Italy were asked an additional question: "From which Italian region are you?". This question not only failed to add value to the analysis but was ultimately not used. Another example is the question "How far would you be willing to travel for a weekend getaway?", which aimed to assess the distance respondents were willing to travel over a weekend. The main issue was that the question assumed a short trip, excluding the possibility of longer journeys, and again added little value due to its specificity.

Following the online survey, several questions related to Van Westendorp's Price Sensitivity Meter, explained in detail in Chapter 3, were included. These questions referred to very specific cases, such as the Quintana event, the Ascoliva Festival, a wine tasting, and a city tour. However, the model is designed to estimate willingness to pay for a representative product within the target market [6]. A more appropriate approach would have been to formulate a single, general question referring to a typical tourist package in order to obtain results that are more representative and comparable.

Fourth, the project was conducted as a group assignment, mainly during the winter break, which created challenges in terms of coordination. Each member faced different personal circumstances: some were engaged with other courses and exams, while others were travelling to their home countries. As a result, finding common times for meetings was often difficult, and overall progress was slower than expected.

Finally, the survey was written exclusively in English, which excluded some potential respondents who spoke Spanish, the third most widely spoken language in the world and commonly used across many regions. Although English is often regarded as a universal language, providing a Spanish version could have increased participation, particularly among older individuals less familiar with English. This would have enriched the dataset by including perspectives from a group that may have distinct travel behaviours and preferences.

In summary, while the original project successfully met the requirements of the course and provided an initial exploration of the research topic, these limitations constrained the depth, precision, and representativeness of its findings. The present thesis builds upon that foundation, addressing them through a longer working period, improved qualitative methods, the integration of text mining techniques, a refined survey design, and a larger and more diverse sample.

Chapter 2

Literature Review

This chapter presents a review of the main literature that informs this research. Its purpose is to provide the theoretical and methodological background necessary to understand the foundations of the study.

The chapter begins by discussing general research approaches, outlining qualitative and quantitative perspectives as well as their key strengths and limitations. It then introduces central concepts used to analyse tourist behaviour, including destination image formation, segmentation, and choice modelling. The review also examines common analytical techniques in tourism research such as discrete choice experiments and conjoint analysis, together with issues that may affect their results, including hypothetical bias.

Finally, the chapter considers more recent contributions that adopt data-driven methods. In particular, it highlights studies employing text and data mining to extract information from customer feedback, identify relevant attributes, and support segmentation.

2.1 Qualitative Approach

The qualitative approach relies on non-numerical evidence to generate insights and draw conclusions [7]. Such conclusions are based on rich, descriptive data, including words, narratives, and observations, often referred to as “soft” data [8]. This method is particularly suited to the in-depth exploration of a topic and is commonly employed to understand how individuals experience and interpret the world [9]. It allows the identification of previously unknown processes, provides explanations of why and how phenomena occur, and reveals the range of their effects [10]. In this approach, the researcher seeks to interpret situations without imposing pre-existing assumptions or expectations on the phenomenon or setting under investigation [11].

The literature also highlights several potential risks that should be considered and mitigated:

1. Compliance bias: refers to situations in which participants' responses may be influenced by their perception of what they believe is expected of them [12].

2. Offset bias: occurs when participants are encouraged to reflect on issues they had not previously considered or to perceive certain topics as more complex than they would under normal circumstances [12].

3. Speculation bias: arises when participants are asked to predict their own behaviour in hypothetical future scenarios. Research suggests that individuals are generally not accurate in forecasting their actions in situations they have never experienced [12].

2.1.1 Advantages of the Qualitative Approach

The qualitative approach seeks to preserve the voice and perspective of participants and can be adapted as new research questions emerge [9]. It is characterised by a flexible design, which allows the researcher to experiment with different techniques to determine what is most effective within the context of the project [13]. Furthermore, it is typically conducted in natural settings, where data collection takes place in real-world contexts or in ways that closely mirror participants' everyday environments. This facilitates the gathering of meaningful insights by listening to detailed descriptions of people's experiences, feelings, and opinions. Such information can help improve the design, testing, or development of systems, services, or products. In addition, using open-ended questions and flexible approaches makes it easier to come up with new ideas and to discover problems or opportunities that were not expected at the start of the study [9].

2.1.2 Disadvantages of the Qualitative Approach

On the other hand, qualitative research has several limitations. First, its real-world setting can make findings less reliable because many factors influencing the data cannot be controlled. Second, the strong role of the researcher in interpreting results introduces subjectivity, meaning that the same data could be analysed differently by different people, and the study cannot be exactly replicated [9]. This reinforces the long-standing criticism that qualitative research findings are not verifiable, and verifiability (the principle that empirical investigations should be capable of replication so that their findings can be confirmed or refuted) is a central tenet of scientific research. It is difficult to replicate qualitative studies in the same way as quantitative ones; however, this does not mean that qualitative research entirely lacks verifiability [14]. This is why it is important for the researcher to carefully document each step of the process, thereby creating an audit trail that allows other researchers to trace and review the original procedures if needed [15]. Third, the use of small samples limits the extent to which results can be generalised to a broader population, even when the analysis is rigorous [9]. This criticism may not be entirely justified, as the generalisability of results to an entire population is rarely the primary objective of the qualitative research approach [16]. Finally, qualitative research can be labour-intensive, as the review and interpretation of large amounts of textual data often require manual checking, despite the availability of software tools to assist with the process [9].

2.2 Quantitative Approach

The quantitative approach is defined as a research strategy that relies primarily on numerical evidence to draw conclusions [17]. Because of its focus on measurement and statistical analysis, it has often been described as a "hard" or "number-crunching" approach to research [18]. This approach involves the systematic collection and analysis of numerical data in order to identify patterns and averages, test relationships between variables, make predictions, and, when appropriate, generalise results to larger populations. It can be applied in various forms, including descriptive research, which summarises the characteristics of variables; correlational research, which explores relationships between variables; and experimental research, which examines cause-and-effect links under controlled conditions. Data collection methods range from surveys and structured observations to experiments and secondary data analysis, all of which require clear operational definitions to translate abstract concepts into measurable indicators. Once collected, the data are processed and subjected to statistical analysis to answer the research questions and test the hypotheses [19].

2.2.1 Advantages of the Quantitative Approach

Quantitative research offers several advantages, particularly in terms of standardisation and the potential to generalise findings to broader populations. One of its key strengths is replication, as the use of standardised data collection procedures and tangible definitions of abstract concepts allows studies to be repeated under similar conditions. This also facilitates direct comparisons of results, enabling research to be reproduced across different cultural contexts, time periods, or participant groups, with outcomes compared statistically. The accuracy of these measurement tools, such as questionnaires, can be strengthened by testing their reliability using objective measures prior to data collection [20]. Furthermore, the approach is also well suited to working with large samples, as quantitative data can be processed and analysed through consistent and reliable statistical procedures [19]. The knowledge gained from this group or subset of the population is representative of the entire population under study [21]. Finally, an additional advantage of the quantitative approach is that it is generally considered a relatively impersonal method of conducting research, which helps to minimise the influence of the researcher's personal biases on the outcomes of the study [22].

2.2.2 Disadvantages of the Quanitative Approach

Despite its strengths, quantitative research also has its own limitations. One of them is superficiality, as the use of precise and restrictive operational definitions may inadequately represent complex concepts. For example, an abstract notion such as "mood" may be represented by a single numerical score, whereas qualitative research could provide a richer explanation. A narrow focus can also be an issue, since predetermined variables and measurement procedures may lead to the omission of other relevant observations. Moreover, even with standardised procedures, structural bias can occur due to factors such as missing data, imprecise measurements, or inappropriate sampling methods, which may distort the findings [19]. Finally, the quantitative approach can be relatively inflexible, as there is little a researcher can do after discovering an error in the data collection instrument, such as a question that is ambiguous or misinterpreted [23].

2.3 Model of Destination Image Formation

Baloglu and McCleary's (1999) study is one of the most influential contributions to destination image research, offering both a conceptual framework and empirical evidence on how such images are formed. Their model shifts the focus from static descriptions of image, which dominated earlier studies, to the dynamic process through which potential tourists develop perceptions and impressions of destinations, particularly when they have not previously visited. They propose that destination image is a multidimensional construct composed of cognitive, affective, and global components. The cognitive, or perceptual, dimension reflects an individual's knowledge and beliefs about a destination's attributes, such as infrastructure, safety, cultural attractions, and value for money. The affective dimension refers to the emotional responses a destination evokes, such as whether it feels pleasant, exciting, or relaxing. Together, these components shape the overall image, which ultimately guides destination choice [24].

Destination image arises from two broad forces: stimulus factors and personal factors. Stimulus factors include external influences such as marketing communications, media portrayals, word of mouth, and prior experience. Personal factors, in contrast, are characteristics of the perceiver, covering sociodemographic attributes like age and education, as well as socio-psychological travel motivations, such as the desire for relaxation, excitement, knowledge, social interaction, or prestige. When individuals have not yet visited a destination, three determinants are particularly important in shaping image: tourism motivations, sociodemographic characteristics, and information sources. In their model, as shown in Figure 2.1, information sources act as stimulus variables, while motivations and sociodemographics represent personal characteristics [24].

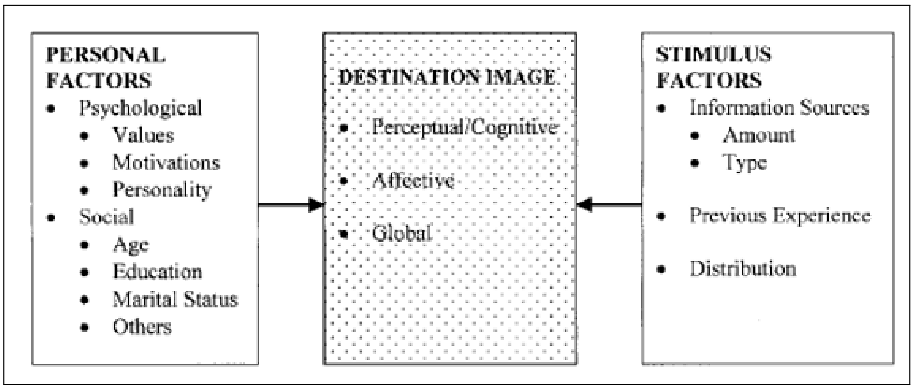


Figure 2.1: *General Framework of Destination Image Formation. Source: Baloglu and McCleary's (1999).*

To test these propositions, Baloglu and McCleary developed a path model that hypothesised causal relationships among these variables. They predicted that cognitive evaluations would significantly influence affective ones, and that both dimensions would in turn affect overall image, with affect expected to carry the stronger impact. They also proposed that the variety and type of information sources, along with age and education, would affect cognitive evaluations, while motivations would directly influence affective responses. In total, eight hypotheses were tested through path analysis, using survey data from 448 potential tourists evaluating four Mediterranean destinations (Turkey, Greece, Italy, and Egypt) [24].

The findings strongly supported the proposed model. Cognitive evaluations emerged as significant predictors of affective responses, with "quality of experience" standing out as the most powerful cognitive factor shaping emotions. Affective evaluations, in turn, had the strongest direct effect on overall image, underscoring their mediating role. The study also revealed that the variety of information sources used positively influenced cognitive evaluations, while word of mouth proved to be the single most powerful information source. Age and education, however, showed a negative influence on cognitive evaluations, with older and more educated respondents assessing destination attributes less positively. Motivations played a weaker role overall, with only relaxation and escape motives showing a significant, though negative, relationship with affect [24].

The study carries important implications for both theory and practice. Theoretically, it provides the understanding of destination image by integrating cognitive and affective components into a single causal framework, showing the process to be dynamic and multidimensional. Practically, it reaffirms the central role of information sources, particularly word of mouth, in shaping perceptions, suggesting that destinations should invest in strategies that encourage the sharing of positive experiences. Because motivations such as knowledge, prestige, and social interaction also influence image, marketers can adapt campaigns to tap into these psychological drivers. Finally, the negative relationship between age, education, and cognitive evaluations emphasises the need for targeted communication strategies that speak differently to diverse demographic groups. In essence, Baloglu and McCleary's work demonstrates that destination image is co-created by external stimuli and personal characteristics, with affective responses serving as the critical bridge between cognition and overall impression [24].

2.4 Segmentation and Response-Style Effects

Pesonen and Honkanen (2014) examine how tourists' response styles can influence segmentation outcomes when applying k-means cluster analysis. A response style refers to the tendency of individuals to answer questionnaires in a systematic way that is unrelated to the actual content. For example, consistently giving high or low ratings, or avoiding the extremes of a scale. These patterns, known as response-style effects, can significantly distort cluster solutions by grouping individuals based on their answering habits rather than their true motivations or attitudes. Since segmentation aims to form clusters that are internally homogeneous and externally distinct, ignoring response styles may lead to the production of segments that are artificial and unhelpful for marketing purposes [25].

The authors note that cluster analysis, particularly k-means, remains the most widely used technique in tourism segmentation research. However, this approach is highly sensitive to response styles. Past research has sometimes produced clusters resembling "want-it-all" tourists who rate everything highly, or "passive" tourists who rate everything poorly. In other words, these patterns that may reflect answering tendencies rather than meaningful market segments. Such results challenge the practical utility of segmentation for identifying actionable strategies [25].

To investigate this issue, the authors collected data via an online survey hosted on a Finnish cottage holiday website. Respondents rated 31 travel motivation items on a 7-point Likert scale. After data cleaning, 709 responses were retained, of which 25 motivation items were used for segmentation. The researchers applied k-means cluster analysis to four versions of the dataset [25]:

1. The original unstandardized data.

2. Data adjusted to remove acquiescence response style (ARS) through within-case standardization (row-centering), which highlights relative item importance per respondent.

3. Data adjusted to remove extreme response style (ERS) by dividing responses by each individual's standard deviation.

4. Data with both ARS and ERS removed.

The results revealed marked differences across the cluster solutions. Using the original data, the clusters largely mirrored response styles: some groups scored consistently high, others low, and some in the middle. This confirmed that k-means was capturing answering patterns rather than motivational differences. Once ARS was removed, however, the clusters displayed distinct motivational profiles, with respondents prioritising certain travel motives over others. This approach produced more interpretable and actionable segmentations. By contrast, removing only ERS offered little improvement, as the clustering still reflected broad high–low patterns. When both ARS and ERS were removed, the solution resembled the ARS-only result, suggesting that acquiescence was the dominant factor influencing clustering [25].

The study emphasises the methodological importance of addressing response-style bias in tourism segmentation research. Without accounting for these effects, researchers risk generating misleading clusters that do not represent real market segments. Pesonen and Honkanen conclude that correcting for ARS yields the most meaningful results in this case, and they stress the need for segmentation studies to explicitly report how response styles are handled. They further recommend that future research explore alternative segmentation methods and use simulated data to deepen understanding of how response styles shape clustering outcomes [25].

2.5 Representing Tourists' Heterogeneous Destination and Travel Party Choices

Wu, Zhang, and Fujiwara (2011) develop a modelling approach that jointly analyses tourists' choices of destination and travel party, recognising that these decisions are interdependent and vary across individuals. Travel decisions are rarely made in isolation; the choice of destination often depends on who tourists travel with, and the composition of the travel party can influence destination selection. The authors argue that treating these choices separately, as prior studies often did, overlooks the sequential and interactive nature of travel decisions. Their paper addresses this by integrating a latent class model with a nested logit model to capture both interdependency and heterogeneity in tourist decision-making [26].

The latent class model identifies unobserved tourist segments that follow different decision-making sequences. Some decide on a destination first, then their travel companions, while others choose their travel party before selecting a destination. Within each segment, the nested logit model reflects the sequential choice process based on random utility maximization. This combined approach allows the distinction of groups with distinct "decision hierarchies", offering a more realistic representation of travel behaviour [26].

The model uses survey data from over 2,000 tourists in Japan, covering 14 destination zones and three travel party categories: travelling alone, travelling with family, and travelling with friends or others. Variables influencing destination choice included travel time, destination attractiveness (measured by annual visitors), and number of tourist spots. Travel party choice was explained by gender, age, and marital status. Results showed all explanatory variables were statistically significant. Travel time had the strongest impact on destination choice, while marital status most strongly predicted travel party composition. Women and married individuals were more likely to travel with family or friends, whereas younger travellers leaned towards independent travel. Crucially, about half of tourists followed a decision sequence beginning with destination choice (latent class 1), while the other half started by selecting their travel party (latent class 2). This demonstrates the absence of a single dominant pattern and emphasises the importance of modelling heterogeneity rather than assuming uniform behaviour. The nested logit parameters confirmed significant interdependencies [26].

The authors conclude that this integrated latent class–nested logit model provides a more flexible and realistic framework for capturing tourist decision-making. For planners and marketers, the findings highlight the value of understanding destinations' sensitivity to travel party structure, enabling more targeted promotions.

The study also points to the importance of considering decision-making structures alongside sociodemographic segmentation. The authors suggest that future research should expand the model to include additional decision dimensions, such as accommodation and travel mode choices, and test its applicability across cultural contexts [26].

2.6 Discrete Choice Experiments in Tourism

Kemperman (2021) provides a comprehensive review of the use of discrete choice experiments (DCEs) in tourism, analysing 49 studies published between 2010 and 2020 in the five leading tourism journals. DCEs are widely employed to measure and predict preferences by presenting individuals with alternatives that vary in their attributes. This approach allows researchers to estimate the relative importance of different destination, product, or service features and to calculate willingness to pay for specific attributes. In tourism, DCEs have been applied to questions such as destination choice, travel mode, accommodation type, and activity selection [27].

The method is rooted in random utility theory, which assumes that individuals choose the option that maximises their utility. Utility consists of a systematic component, based on observable attributes, and a random component that captures unobserved factors. In tourism, this means that travellers are assumed to select the destination, activity, or service that provides the highest perceived benefit given their motivations and constraints. By presenting destinations or services as bundles of attributes such as price, quality, amenities, and location, and asking respondents to choose between hypothetical alternatives, researchers can estimate the marginal utility of each attribute [27].

DCEs can use both revealed and stated choice data. Revealed data come from actual behaviour such as bookings, while stated data are gathered in controlled experiments. The former reflects real-world decisions, whereas the latter allows researchers to anticipate preferences for products or services that do not yet exist. In practice, when market data are unavailable or lack variation, stated choice experiments often provide the only viable means of capturing likely behaviour [27].

A key strength of DCEs is their ability to simulate decision-making in a structured way. Tourists are presented with multiple choice scenarios in which attributes vary systematically, allowing researchers to isolate the impact of each factor. This design provides high internal validity, though external validity can be more limited since the choices are hypothetical. Kemperman's review shows that most studies still examine single, independent decisions, but relatively few investigate linked or sequential decisions, such as destination, transport, and accommodation together. In reality, these choices are interdependent [27].

Kemperman (2021) concludes that discrete choice experiments provide a powerful lens for understanding tourist preferences and predicting responses to innovations in tourism products and services. She argues that future research should design more realistic experiments, incorporate dynamic needs and social influences, and make use of emerging tools such as virtual reality and eye-tracking to enhance engagement and data quality. Such advancements would extend DCEs beyond predicting what choices tourists make towards offering richer insights into how and why those choices are made [27].

2.7 Conjoint Analysis in Tourism

Altin (2023) offers a comprehensive overview of how conjoint analysis has been applied in tourism research as a method for measuring tourist preferences. Conjoint analysis is a choice-based technique that allows researchers to understand how travellers value different attributes of destinations, services, or experiences. By decomposing a tourism product into key elements such as price, accommodation type, attractions, or amenities, it estimates the value respondents assign to each attribute. This makes it possible to capture the trade-offs travellers are willing to make when choosing between alternatives, providing useful insights for destination management organisations (DMOs), tour operators, and other industry stakeholders seeking to design competitive offerings [28].

Methodologically, conjoint analysis involves presenting respondents with sets of hypothetical profiles or “bundles” that combine different attribute levels. Respondents are asked to select their preferred option from each set. Statistical analysis of these choices produces utilities and relative importance estimates for each attribute, showing which features drive satisfaction and willingness to pay (WTP). In tourism, this approach has been applied to diverse products such as destination packages, activity mixes, accommodation types, and other experience-based offerings. Given that tourism products are typically multi-attribute in nature, conjoint analysis provides a structured way to reflect their combined influence on decision-making [28].

Altin (2023) [28] also highlights the practical advantages of conjoint analysis for industry practitioners. Well-designed studies can guide product development and market positioning by identifying which attributes matter most to particular traveller segments. By simulating realistic decision-making, conjoint analysis also supports market segmentation, differentiation, and competitive benchmarking, which are critical in a crowded and highly competitive industry [28].

The author also describes several considerations and challenges in applying conjoint analysis. First, the careful selection of attributes is essential: including too many risks overwhelming respondents, whereas omitting key features may reduce validity. Second, samples must be sufficiently representative to ensure generalisable results. Altin further notes the growing relevance of digital and adaptive conjoint approaches, which ease data collection and allow more flexible, personalised designs. As tourism evolves towards increasing complexity, through sustainability concerns, digital innovations, and shifting traveller expectations, conjoint analysis continues to adapt in order to capture these new dimensions of tourist choice [28].

Altin's work serves as a methodological anchor in tourism research by demonstrating the value of conjoint analysis for understanding preferences, informing product design, and supporting strategic decisions at both managerial and policy levels. By linking conjoint studies with broader frameworks of tourism behaviour, this contribution provides a roadmap for integrating preference modelling into the study of tourist decision-making [28].

2.8 Hypothetical Bias

Hensher (2010) reviews how hypothetical bias affects willingness-to-pay (WTP) estimates derived from stated preference (SP) methods and discrete choice experiments (DCEs). Hypothetical bias arises from discrepancies between what people report they would choose or pay in surveys and what they actually do in real-world situations. Since SP and DCE studies typically lack financial consequences, respondents may overstate their preferences or WTP, often producing inflated values. This poses a serious challenge in transport, environmental, and tourism research, where WTP estimates are commonly used to justify policies and investments [29].

The paper synthesises evidence comparing WTP from hypothetical surveys with values obtained from revealed preference (RP) studies or actual market behaviour. Some studies detect no difference, while others find that WTP from SP and DCEs tends to be much lower than real-market values. Hensher (2010) [29] argues that no single factor explains this variation and that outcomes depend on the type of good, survey design, and respondents' familiarity with the decision context [29].

To mitigate hypothetical bias, several strategies are discussed. One is anchoring surveys in real experiences, for example by "pivoting" choice sets around respondents' existing travel habits. This grounds choices in familiar conditions and narrows the gap between hypothetical and real decisions. Other techniques include adding opt-out alternatives, using "cheap-talk" scripts that warn about overstatement, or providing consequentiality scripts and honesty oaths to increase the perceived importance of responses. While these tools have shown promise, their effectiveness is context-dependent [29].

The author also highlights the role of calibration methods. Ex-ante approaches (instrument-based) seek to prevent bias during survey design, while ex-post techniques (statistical adjustments) correct results after data collection using confidence ratings or benchmark comparisons. For instance, calibration functions derived from observed market data have been applied to adjust WTP values, producing results more consistent with revealed behaviour [29].

A key contribution of the paper is its emphasis on reference alternatives in DCEs. Anchoring choice tasks to respondents' actual experiences, rather than purely hypothetical scenarios, was shown to reduce bias. Case studies in Australia and New Zealand demonstrated that mean values of travel time savings (VTTS) were significantly higher and more realistic when based on a reference commute compared with forced-choice scenarios. This suggests that reference-dependent designs yield WTP estimates that are more reliable for policy analysis [29].

Hensher (2010) [29] concludes that no universal remedy exists for hypothetical bias. Instead, researchers should use a combination of careful survey design, calibration techniques, and validation against real behaviour. The paper highlights the need for further research into hybrid data approaches and pivot-based experiments, which show strong potential for narrowing the gap between hypothetical and real-world valuations [29].

2.9 A Text Mining Based Approach for Mining Customer Attribute Data on Undefined Quality Problems

Zhu, Li, and Wu (2018) address the challenge of understanding customer perceptions of quality issues that are not easily defined through traditional evaluation methods. As market structures become increasingly complex, conventional tools such as SERVQUAL often fail to capture the full scope of customer concerns or provide actionable insights. To overcome this limitation the authors propose an unsupervised machine-learning text mining approach combined with a Recursive Neural Tensor Network (RNTN) to extract customer attribute data from unstructured textual sources [30].

Their framework focuses on the consumer perspective, demonstrating that customers exhibit a coherent "line of sight" when identifying quality problems, even when their feedback involves diverse and scattered attributes. The text-mining process enabled the identification of attribute clusters and revealed that approximately 61% of consumers converge on common strategies to address ambiguous quality issues through collective reasoning [30].

The study illustrates how advanced text-mining techniques can uncover hidden patterns in consumer feedback that traditional survey-based methods may overlook. By leveraging large volumes of unstructured customer data, this approach provides managers with data-driven insights into quality perception, thereby enhancing quality management and the detection of subtle or emerging concerns [30].

2.10 Text Mining for Product Attribute Extraction

Ghani et al. (2006) propose a method for extracting attribute–value pairs from textual product descriptions to enrich product databases. By conceptualising products as collections of attributes and values rather than as atomic entities, their approach improves the performance of applications such as demand forecasting, assortment optimisation, and product recommendation systems. The study addresses both explicit physical attributes (e.g., size, colour) and implicit semantic attributes (e.g., trendiness, brand appeal), focusing primarily on products in the apparel and sporting goods industries [31].

The authors formulate attribute extraction as a classification problem and employ a semi-supervised learning approach that requires minimal labelled data. Their model leverages large quantities of unlabelled text through co-training and dependency parsing to accurately identify relationships between attributes and their corresponding values. Furthermore, expert discussions informed the identification and categorisation of attributes according to explicit and semantic characteristics [31].

The proposed system's modular architecture (comprising data collection, seed generation, attribute extraction, value-relationship extraction, and user interaction) enhances both scalability and adaptability. Experimental results demonstrate strong accuracy in extracting attribute–value pairs, thereby facilitating the expansion, enrichment, and analytical use of large-scale product attribute databases [31].

2.11 Data Mining on Customer Segmentation: A Review

Kaur and Kaur provide a comprehensive review of data mining techniques applied to customer segmentation, a fundamental process for dividing customers into meaningful groups, based on shared characteristics. The paper emphasises the strategic importance of segmentation in modern marketing and customer relationship management, as it enables organisations to tailor their offerings and communication to the specific needs of each segment [32].

The authors examine a range of clustering algorithms, including K-means, DBSCAN, and fuzzy C-means, outlining their respective advantages and limitations in identifying distinct customer groups. They stress that effective segmentation depends on forming clusters that are internally cohesive yet externally distinct, while maintaining relevance to real-world customer profiles [32].

Beyond traditional clustering, the review explores advanced machine-learning approaches, such as neural network–based predictive models that leverage product reviews, purchasing behaviour, and temporal data to improve segmentation precision. The authors also highlight the role of feature extraction techniques, including unigram, bigram, and trigram analysis, in capturing customer sentiment and preferences from textual data [32].

Overall, the study demonstrates that the careful application of data mining techniques can yield valuable insights into customer behaviour, supporting the development of more targeted marketing strategies and enhancing customer satisfaction [32].

2.12 Unlocking insights: integrated text mining and interpretive structural modelling for enhanced user review analysis

Li, Liu, and Chen (2024) propose a novel approach that integrates text mining with Interpretive Structural Modelling (ISM) to analyse large volumes of user-generated reviews, enabling manufacturers and e-commerce platforms to better understand customer preferences and behaviours. The study applies two text-mining techniques (TF–IDF and Word2Vec) to extract meaningful keywords and explore semantic relationships within review texts. The extracted insights are visualised through knowledge graphs constructed using Neo4j, allowing the identification and exploration of thematic connections within the data [33].

To further extend the analysis, the authors employ ISM to reveal hierarchical and interdependent relationships among key product attributes derived from the reviews. Using a dataset of milk product reviews from Suning.com, six major elements influencing user feedback were identified: freshness of taste, discounted prices, logistics, customer repurchase, product packaging, and nutritional composition. These attributes were organised into three hierarchical layers representing their relative influence: discounted prices, repurchase, and logistics at the top; packaging and nutritional composition in the middle; and freshness of taste as the foundational factor [33].

This integrated text-mining-ISM framework improves the depth and interpretability of user review analysis by capturing both semantic meaning and structural relationships among attributes. The study demonstrates the potential of combining quantitative text analytics with structural modelling to generate actionable insights for product development, marketing optimisation, and consumer experience enhancement [33].

Chapter 3

Methodology and Results

This chapter describes the methodology used in this research and presents the main results obtained at each stage. The study follows a mixed-method approach, combining qualitative and quantitative techniques to better understand tourist behaviour and preferences.

The first part introduces a series of qualitative interviews, explaining how they were conducted and summarising the insights gathered from participants. These interviews were also analysed using text-mining techniques, including TF–IDF vectorisation and topic modelling, to identify recurring themes and behavioural patterns in participants' responses. The second part focuses on text mining applied to online accommodation reviews, outlining the analytical process and key findings. The chapter then presents an open-ended survey, including the methods used for data collection and analysis. Different analytical tools are applied, such as exploratory data analysis, price-sensitivity assessment, and text-based techniques. This stage also incorporates the discovery of complementary attributes through language-model assistance, which contributes to the refinement of the final set of attributes. The final part of the chapter describes a quantitative survey designed to validate and expand upon previous findings.

Overall, this chapter provides a detailed account of the methodological workflow and the results that guide the subsequent development of the thesis.

3.1 Qualitative Interviews

3.1.1 Methodology

The Interviews

The first step consisted of qualitative interviews, which were used to identify potential attributes related to tourist destinations. This was carried out through flexible, open-ended questions, mainly focusing on real-world experiences. Although flexible, a guide of predefined questions was created (see Figure 3.1) to provide a general structure for the interviews, while still allowing room to deviate and follow the flow of each individual conversation. Each new interview generated additional ideas and insights that informed subsequent ones, thus enriching the insights obtained. Furthermore, all interviews were conducted online and recorded, and each transcriptions were produced (see Appendix B for an example), allowing the subsequent analysis to be verified against the original data.

Main Questions	**In-depth Questions**
First, I'd like you to think about the last few times you went on vacation to a tourist destination. Can you walk me through the typical process you follow—from the moment you decide you want to travel until everything is planned?	• What do you consider when choosing the destination? • Do you look anywhere for help or inspiration? • How much time do you usually take for vacation?
Now, thinking about your last trip — was there anything you did differently from what you just described?	• If so, was the change for better or worse? • Why did you change or stop doing that?
Thinking about your recent trips, once you were at the destination, what did you do?	• Did you engage in any activities? • Was it a good experience? What did you like or dislike?
When it came time to eat, how did you decide where to go and what to eat?	• Was it the same for lunch and dinner? • Where did you search for places?
Before we finish, could you name a few of your most recent travel destinations?	• Which ones did you enjoy the most and the least? Why?
Finally, is there anything else you'd like to add or think is important to mention?	

If the interviewee feels comfortable: Please share your screen. Would you mind showing me how you would search for your next trip on your phone or computer, as if you were planning it right now?

Figure 3.1: *Guide for Qualitative Interview Questions. Source: Author's own elaboration.*

For these interviews, the aim was to gather perspectives from individuals of different generations, nationalities, and professional backgrounds. The sample included:

- Sex: 8 males and 2 females
- Age: median of 26 years, with a maximum equal to 41 years old, and a minimum equal to 21 years old.
- Nationalities: 5 Argentinians, 2 Spanish, 1 Italian, 1 Colombian, and 1 Iranian.
- Occupations: 5 employed people and 5 master's students (one of whom had worked for approximately 20 years before starting the degree).

The interviews were conducted via the Zoom platform and in each interviewee's original language, allowing them to clearly and fully describe their real experiences. In cases where the interviewer and interviewee did not share the same mother tongue, English was used as the common language. Among all the interviews, the only one conducted entirely in English was with the Iranian participant. Furthermore, all interviewees were informed about the recording process, with emphasis placed on the fact that no personal or sensitive information would be disclosed. The main objective at this stage was to understand the genuine process by which interviewees defined and prepared a trip. For this reason, participants were asked to share their screen (or, in cases where this was not possible, the interviewer shared the screen and followed the steps as described by the interviewee) and to demonstrate step by step, how they organised either a past trip or an upcoming one. While describing their planning process, they showed the resources they used and expressed both positive and negative impressions. When needed, the interviewer intervened to ask additional questions in order to explore certain topics in greater depth. Nevertheless, the main goal was to allow the interviewee to speak as freely as possible, and as mentioned earlier, without introducing any potential bias.

After this initial part, in which participants described their real-life experiences of trip planning, they were asked specifically about Ascoli Piceno. In this section, they were invited to demonstrate how they would prepare for a hypothetical trip to the city. They were asked to apply their usual, real-life trip planning process to the case of Ascoli Piceno, following the same steps, tools, and resources they normally use. The aim was to capture their genuine opinions about Ascoli Piceno and to identify potential attributes associated with it, both positive and negative. One of the most relevant aspects of this stage concerned their impressions when analysing the map and the geographical position of the city. In some cases, participants were also asked to compare Ascoli Piceno with Pistoia, a city of similar characteristics located near Florence. At this point, further attributes emerged, particularly when interviewees were asked to decide whether they would prefer to visit one city over the other, and to explain their reasons.

Finally, they were introduced to specific topics related to Ascoli Piceno, such as the olive ascolane and the Quintana festival, in order to gather insights about attributes associated with local events and cultural traditions. At the end of the interview, participants were given the opportunity to add any further comments, which often provided additional perspectives or reinforced the main points that had emerged throughout the session.

Manual Identification of Attributes

After all interviews were completed, the transcription-cleaning process was carried out. For the first two interviews, since the automatic transcription feature had not been enabled, the audio files were transcribed in Google Colab using

`WhisperX`, an enhanced transcription tool built on OpenAI's Whisper that incorporates speaker diarisation from Hugging Face to provide more accurate transcripts and speaker separation [34]. Once the raw transcriptions were obtained (either from Google Colab or Zoom), a manual verification process was conducted. During this stage, the transcripts were checked against the recordings and corrected where necessary, particularly in cases where words or phrases had been inaccurately transcribed. Finally, ChatGPT was used carefully to efficiently translate each transcription into English.

Following the transcription cleaning process, each transcription was carefully re-read, with notes taken on the most relevant insights and linked to potential attributes. This process was repeated for every interview, highlighting key phrases and extracting attributes. Once this analysis was completed, all unique attributes were compiled in an Excel sheet together with their corresponding frequencies. Frequency was defined as the number of interviews in which a given attribute appeared, with a maximum possible frequency of 10 (equal to the total number of interviews).

Text Mining and Topic Modelling

Complementing the previous manual analysis, a text mining process was conducted in Google Colab to reinforce the previously identified attributes and to determine whether any additional attributes could be extracted. Before proceeding with the analysis, a preprocessing stage was required. First, the Colab notebook was connected to the Google Drive account to access the raw interview transcripts. Each transcript was then cleaned in Python, by removing all interviewer segments, deleted metadata sections such as "Interviewee's information," and retained only the text spoken by the interviewee. Additional formatting operations were applied to remove redundant blank lines and extra spaces, ensuring a consistent and readable structure across all files. Once all transcripts were cleaned, the script automatically saved the processed versions in the same folder with a new filename prefix to distinguish them from the originals. Subsequently, another function imported all the cleaned files and concatenated them into a single DataFrame named `df_interviews`. This consolidated dataset served as the input for the text mining stage.

For the text preprocessing stage, the English language model `en_core_web_sm` from the `spaCy` library was employed for text processing. The Named Entity Recognition (NER) and syntactic parsing components were disabled to improve processing efficiency, as they were not required for this analysis. With this configuration, Algorithm 3 (see Appendix D) was implemented. In this algorithm, each text was first converted to lowercase, and all characters that were not letters, numbers, or whitespace were removed using regular expressions. The cleaned text was then passed to the `spaCy` NLP model, which performed tokenisation. Finally, if the parameter `stopwords` was set to "Yes," only alphabetic tokens that were not stopwords were retained, and each token was replaced by its lemma (i.e., the base or dictionary form of the word). Although the resulting dataset (`df_cleaned_interviews`) was ready for analysis, a final manual adjustment was performed to remove filler words that did not provide analytical value and were not captured as stopwords, such as "yes", "yeah", "yea", "uh", "um", "okay", "ok", "like", "dont", "didnt", "know", "thing", "things", "go", "get", "got", "just", "maybe", "kinda", "kind", "sorta", "mean", "ve", and "think".

Following text preprocessing, the cleaned tokens were transformed into numerical features using the Term Frequency–Inverse Document Frequency (TF–IDF) method, implemented through the `TfidfVectorizer` from the `scikit-learn` library. This approach quantifies the relative importance of each term across the collection of interviews, balancing its frequency within a document against its occurrence across all documents. Unigrams (single words), bigrams (two-word sequences), and trigrams (three-word sequences) were extracted using Algorithm 4 (see Appendix D). The TF–IDF scores were aggregated across all documents to obtain a global measure of each term's relative importance. To visualise the most relevant words in the corpus, a word cloud was generated for each case (single words, 2-grams, and 3-grams) using the `WordCloud` library, where word size corresponds to cumulative TF–IDF scores.

For the topic modelling stage, the `BERTopic` framework was employed, combining transformer-based embeddings with clustering and class-based TF–IDF representations. Prior to model training, each interview transcript was divided into smaller textual segments to improve topic coherence. The segmentation was performed using the `chunk_text` function (see Algorithm 5 in Appendix D), which tokenised each text into sentences using the `sent_tokenize` function from the `NLTK` library and then groups them into chunks of six sentences each. Only chunks longer than 30 characters were retained, resulting in a list of textual segments (`documents`) and their corresponding indices (`origin`). Sentence embeddings were then generated using the `all-MiniLM-L6-v2` model from the `Sentence Transformer` library, which encoded each chunk into a dense semantic vector. These embeddings were subsequently used as input for the `BERTopic` model, which clusters semantically similar texts and derives interpretable topic representations. This step enabled the identification of latent themes across participants' narratives, complementing the manual attribute extraction phase.

To refine the topic extraction process, a custom stopword list was defined by combining the default English stopwords from the `scikit-learn` library with domain-specific filler words and non-informative tokens such as "like", "yeah", "really", "actually", "basically", "um", "uh", "ok", "okay", "sort", "kind", "thing", "things", "so", "just", "well", "yes", "no", "don", "dont", "doesnt", "didnt", "im", "ive", "youre", "its", "id", "think", "know", "mean", "look", "get", "got", "going", "go", "place", "places", "city", "cities", "tourist", "tourists", "trip", and "travel". The `TfidfVectorizer` was configured to ignore tokens shorter than four letters and to discard words appearing in fewer than two documents. Additionally, the `KeyBERTInspired` representation model was applied to enhance topic interpretability through keyword relevance weighting.

The final model (see Algorithm 6 in Appendix D) was implemented with the parameters `low_memory=True` and `min_topic_size=8` to enhance computational efficiency and ensure that each topic contained a sufficient number of documents. The model was subsequently trained using the segmented documents and their corresponding embeddings, producing a set of topics (`topics`) and associated probabilities (`probs`) that represent the distribution of topics across all text segments.

Complementary Attribute Identification Using ChatGPT

Before proceeding to the next step, ChatGPT was also employed as a complementary tool to identify potential attributes by providing each interview transcription separately. The prompts (see Appendix E) were designed and implemented in two stages: first, to obtain an initial list of attributes, and subsequently, to refine and consolidate the results.

3.1.2 Results

Manual Identification of Attributes

After the interview transcriptions were properly cleaned, the main insights were extracted from each case (see Appendix C). Some phrases contained a single attribute, while others referred to multiple ones. For instance, in the first interview with the Italian respondent, he said: "...in that famous Tuscany trip, we also included the Palio di Siena...". This was linked to a general attribute: local events. On the other hand, in the second interview with an Argentinian respondent, he said: "...we looked a lot into what the transportation is like there, food prices, which beaches are recommended.". In this case, the phrase was related to three attributes: public transportation, food prices, and beaches. These initial attributes were not definitive but rather a starting point (see Figure 4.1 in Appendix F).

Text Mining and Topic Modelling

Regarding the text-mining analysis, the TF–IDF results provided valuable insights for refining and validating the list of attributes. Among the different n-gram configurations tested, the 3-grams (sequences of three consecutive words) produced the most meaningful expressions (see Figure 3.2), allowing for a clearer identification of recurring concepts and patterns relevant to tourist preferences. These multi-word combinations helped to capture contextual nuances that single words or bigrams could not, thereby contributing to a more accurate definition of the attributes. As illustrated in the following figure, the main terms and phrases extracted through TF–IDF analysis were closely aligned with the attributes identified in the previous manual stage, confirming the consistency and robustness of the attribute selection process.

Figure 3.2: *3-gram Word Cloud from Interviews. Source: Author's own elaboration.*

The topic modelling analysis revealed distinct thematic clusters that illustrated how travellers plan and experience their trips. Overall, the model identified eight coherent clusters representing key aspects of travel behaviour and preferences, as shown in Figure 3.3. Several clusters underscored the strong influence of digital discovery tools, as participants frequently mentioned using platforms such as Google and TripAdvisor as their primary sources of inspiration and information (Cluster 0: Digital Search & Food Inspiration). Culinary experiences, scenic beauty, and iconic landmarks emerged as central motivations, reflecting travellers' emphasis on authentic local food and visually appealing destinations (Cluster 1: Iconic Attractions & Scenic Destinations).

Financial considerations also played a critical role, with travellers demonstrating a clear preference for budget-conscious accommodation options such as hostels and Airbnb (Cluster 2: Budget-Friendly Lodging), as well as the systematic comparison of hotel offers and travel costs (Cluster 4: Hotel Comparison & Budget

Planning). The analysis further revealed a tendency towards multi-city itineraries and structured travel planning, where visitors combine well-known destinations with smaller towns or day trips to optimise time and variety (Cluster 3: Structured Tours & Multicity Itineraries; Cluster 5: Short Excursions from Urban Hubs).

In addition, certain clusters captured more specific travel patterns, such as the popularity of classic European city tourism focused on emblematic destinations (Cluster 6: Classic European City Tourism), and a broader category encompassing general travel exploration and destination diversity (Cluster 7: General Travel Exploration & Destination Variety).

Taken together, these clusters highlight the main drivers of contemporary travel behaviour: digital research, cost optimisation, local immersion, and multi-destination planning.

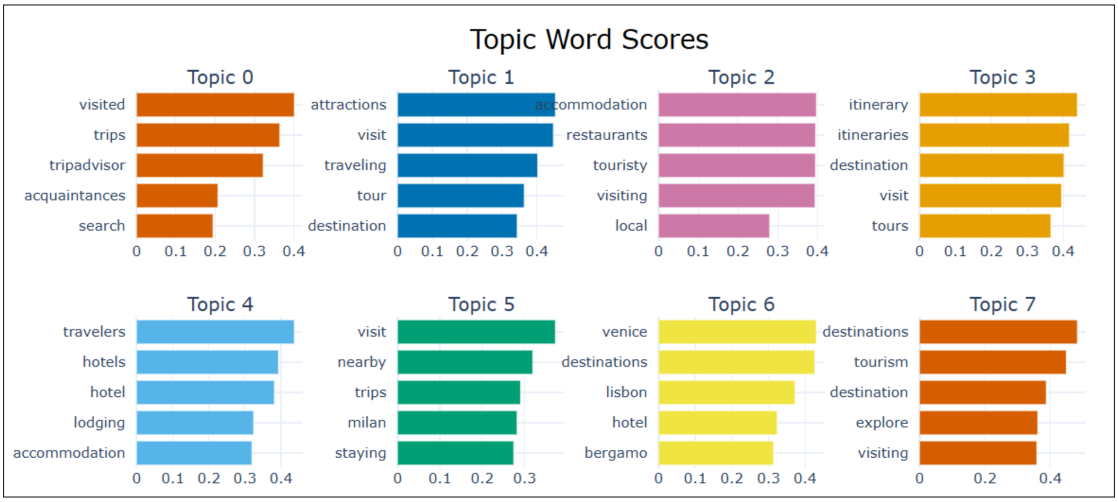


Figure 3.3: *BERTopic Clusters. Source: Author's own elaboration.*

Complementary Attribute Identification Using ChatGPT

The first attributes suggested by ChatGPT, as shown in Appendix G, were relatively broad and generic, often referring to general aspects of tourism such as "food," "culture," or "accommodation." While these terms reflected relevant dimensions of the travel experience, their lack of specificity risked creating conceptual overlap and ambiguity among categories. After applying the refinement prompt, the output became more detailed and contextually accurate. ChatGPT provided clearer, more actionable attribute definitions.

Final Attributes

Finally, the list of attributes was refined to enhance accuracy and clarity. Several attribute names were adjusted to prevent ambiguity or misinterpretation.

Supported by the results of the text mining analysis and the complementary suggestions provided by ChatGPT, the final ten attributes were carefully selected to ensure they were representative, clearly defined, and free from ambiguity or overlap.

Attribute
Distinctive local cuisine
Heritage ambience of the historic center
Availability of museums and art venues
Frequency of cultural events
Proximity to natural areas
Ease of reaching the city by public transport
Digital information availability
Accommodation type
Quality of accommodation options
Accessibility to touristic points

Figure 3.4: *Preliminary Set of Ten Attributes. Source: Author's own elaboration.*

3.2 Text Mining of Online Accommodation Reviews

3.2.1 Methodology

One of the datasets used for this analysis consisted of Airbnb reviews from various Italian cities. These data were obtained from the Inside Airbnb platform, which provides quarterly datasets for multiple regions and cities, available for free download [35]. The Italian cities included in this analysis were Bergamo, Bologna, Florence, Milan, Naples, Puglia, Rome, Sicily, Trentino, and Venice. All individual datasets were concatenated into a single dataset, resulting in a total of 7,141,844 records. The dataset contained the following variables: `listing_id`, `id`, `date`, `reviewer_id`, `reviewer_name`, `comments`, and `location`.

To optimise computational performance, a word count was calculated for each review, and only reviews up to the upper quartile (75%) of this distribution were retained. This filtering step reduced the dataset to 5,383,460 records. Since reviews were written in multiple languages, a language detection process was implemented using the Compact Language Detector 2 (`cld2`), which probabilistically detects over 80 languages in UTF-8 encoded text [36]. To ensure consistency in the subsequent text-mining process, only English-language reviews were retained, resulting in 2,800,779 entries (see Figure 3.5). To further optimise computational efficiency, a random 15% sample of the English reviews was used in the final analysis.

The remaining reviews were tokenised using the `word_tokenize` function from

```
lang
english    2800779
italian     866559
french      554692
spanish     328370
german      246455
             ...
tagalog          1
nyanja           1
oromo            1
quechua          1
khmer            1
Name: count, Length: 113, dtype: int64
```

Figure 3.5: *Compact Language Detector 2 Results. Source: Author's own elaboration based on Inside Airbnb datasets.*

the `NLTK` library. All tokens were converted to lowercase to maintain consistency and reduce redundancy caused by case differences. Stopwords and punctuation marks were then removed using the NLTK English stopword list and the `string` library, respectively. A custom cleaning function was applied to remove both types of noise, leaving only meaningful tokens. Subsequently, all words were lemmatised using the `spaCy` library to reduce them to their base or dictionary form (e.g., "running" → "run"). To optimise performance, as shown in Algorithm 1, texts were processed in batches of 2,000 using parallelisation across four cores. The cleaned and lemmatised tokens were then stored as a new variable in the dataset, producing a standardised textual representation suitable for further computational analyses, such as topic modelling and sentiment classification.

Algorithm 1 Lemmatizing Reviews

Input: Data frame *airbnb_reviews_sample* with column `clean_tokens`

Initialize empty list *lemma_token_list* ← [ ]
Create list *texts* ← join tokens in *airbnb_reviews_sample*[`clean_tokens`] with spaces
for each *doc* in NLP pipeline applied to *texts* with batch size 2000 and $n\_process = 4$ **do**
    Extract *lemmas* ← list of *token.lemma_* for each *token* in *doc*
    Exclude tokens that are punctuation or spaces
    Append *lemmas* to *lemma_token_list*
end for

Output: List *lemma_token_list* containing lemmatized tokens for each review

Following this text preprocessing, the cleaned tokens were converted into numerical features using the TF–IDF method, as described earlier. Both unigrams (single words) and trigrams (three-word sequences) were extracted using the following algorithm:

Algorithm 2 Computing TF-IDF Vectors

Input: List of documents *doc*

▷ — Compute TF-IDF for unigrams —

Initialize `vectorizer_unigrams` ← `TfidfVectorizer` with parameters:

- `lowercase = False`
- `token_pattern = r"(?u)\b\w+\b"`
- `ngram_range = (1,1)`
- `min_df = 5`, `max_df = 0.8`
- `max_features = 100,000`
- `dtype = float32`

$X_{unigrams}$ ← `vectorizer_unigrams` fit–transform on `doc`

$terms_{unigrams}$ ← `vectorizer_unigrams.get_feature_names_out()` ▷ — Compute TF-IDF for trigrams —

Initialize `vectorizer_trigrams` ← same parameters as above, but with `ngram_range = (3,3)`

$X_{trigrams}$ ← `vectorizer_trigrams.fit_transform(doc)`

$terms_{trigrams}$ ← `vectorizer_trigrams.get_feature_names_out()`

Output: TF-IDF matrices *X_unigrams* and *X_trigrams*, and their corresponding term lists

The TF–IDF scores were aggregated across all documents to obtain a global measure of each term's relative importance. To visualise the most relevant words in the corpus, a word cloud was generated using the `WordCloud` library, where word size corresponds to cumulative TF–IDF scores. Additionally, the top 20 terms were identified and presented in tabular form, ranked by their total TF–IDF weight.

The other dataset used in this work was the Booking.com Reviews Dataset, which contains a comprehensive collection of hotel reviews and ratings gathered from the popular travel booking platform, Booking.com. This dataset provides valuable insights into customers' experiences and perceptions of hotels across various destinations [37]. After importing the dataset, reviews containing the text "There are no comments available for this review" were removed, leaving a total of 18,988 valid records for analysis.

Each review was tokenised using the `word_tokenize` function from the `NLTK` library, and all tokens were converted to lowercase to ensure consistency. As in the Airbnb pipeline, stopwords (from NLTK's English stopword list) and punctuation marks (from Python's `string` library) were removed to retain only semantically meaningful words. The resulting tokens were then lemmatised as done with the previous dataset, using Algorithm 1. A final cleaning step was applied to remove generic or low-information terms such as "review" and "comment."

After preprocessing, the dataset was divided into two subsets based on the rating score:

- Positive reviews: ratings ≥ 7
- Negative reviews: ratings < 7

For each subset (positive and negative), a TF–IDF vectoriser was applied to transform the lemmatised text into a numerical representation, following the same steps as with Aribnb dataset (Algorithm 2). The resulting TF–IDF matrices were aggregated to identify the most relevant terms within positive and negative reviews. To visualise these patterns, word clouds were generated, with word size proportional to cumulative TF–IDF scores. Additionally, the top 20 terms were extracted and presented in tabular form for each subset, offering a clear comparison of the dominant lexical features that distinguish positive and negative customer experiences.

3.2.2 Results

After applying the TF–IDF method to the Airbnb dataset, the unigram analysis revealed that the most frequent terms were primarily related to the location of the property and the services offered by the accommodation. This suggests that, when writing reviews, guests tend to emphasise not only the geographical setting (e.g., location, walk, close) but also the specific amenities provided (e.g., clean, comfortable, or amenity). As shown in Figure 4.4 (see Appendix H), these two dimensions (location and services) emerged as dominant topics shaping guests' perceptions and experiences.

To gain a more detailed understanding, the same analysis was conducted using trigrams. As shown in Figure 3.6, the results further reinforced the centrality of location, with frequent expressions such as "place great location", "great location close", "within walking distance", and "close train station". In contrast, there were relatively few trigrams directly associated with services, and several of the most frequent ones still referred to location, such as "clean great location". This indicates that, even when considering multi-word expressions, guests consistently prioritised aspects related to the property's location over other features of the accommodation.

Turning to the Booking.com dataset, the unigram analysis of positive reviews revealed that the most frequent terms were again related to both the location of the property and the services provided by the accommodation. As shown in Figure 4.5 (see Appendix I), compared with the Airbnb reviews, a greater number of terms referred to specific services such as "breakfast", "reception", "garden",

Figure 3.6: *Top 3-grams from Airbnb Reviews. Source: Author's own elaboration based on Inside Airbnb datasets.*

and "bed". In addition, new location-related terms appeared, including "metro" and "restaurant".

In contrast, the unigram analysis of negative reviews (see Figure 4.6 in Appendix I) showed that, although "location" remained a highly frequent term, a larger share of the most common words referred to accommodation services, including "shower", "bathroom", "bed", and "staff".

The trigram analysis of positive reviews (see Figure 3.7) highlighted the growing importance of accommodation services, with common expressions such as "friendly helpful staff" and "room comfortable bed". Location, however, remained a crucial factor, with trigrams such as "within walking distance" and "close city center".

Figure 3.7: *Top 3-grams from Booking.com Positive Reviews. Source: Author's own elaboration based on Kaggle dataset.*

As shown in Figure 3.8, the trigram analysis of negative reviews revealed a stronger emphasis on location, with expressions such as "close train station", "5 minute walk", and "close city center". At the same time, several trigrams provided clearer insights into service-related concerns, including "4 star hotel", "friendly helpful staff", and "coffee make facility".

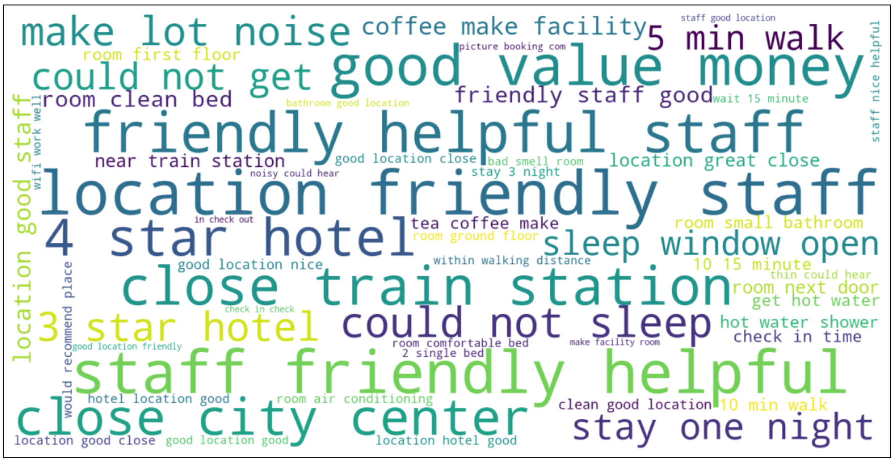


Figure 3.8: *Top 3-grams from Booking.com Negative Reviews. Source: Author's own elaboration based on Kaggle dataset.*

Based on the insights obtained from the textual analyses of both datasets, the list of attributes was refined and improved as follows:

Attribute	Observation
Accommodation services	Reviews frequently mentioned elements such as breakfast, room quality, staff, cleanliness, and shower facilities.
Availability of centrally located accommodation	Added as a new attribute, given that many reviews highlighted location; the word "centrally" provides a more precise segmentation.
Distinctive local cuisine	Retained due to interviews underscoring the importance of Italian food; the term "distinctive" emphasises the uniqueness of local dishes.
Density of attractions within walking distance	Refined to include "walking distance," thereby avoiding ambiguity and providing practical meaning.
Ease of reaching the city by public transport	Adjusted to specify "public transport," distinguishing it from private transport preferences.
Proximity to natural areas	Added based on interview references to beaches and mountains, capturing the importance for visitors who value nature.
Heritage ambience of the historic centre	Refined to emphasise the historic heritage dimension, while avoiding overlap with "central location" of accommodation.
Low tourist density	Introduced as a new attribute, mentioned in both interviews and the text analysis, reflecting a preference for less crowded areas.

Figure 3.9: *Refined Set of Preliminary Attributes. Source: Author's own elaboration.*

3.3 Open-ended Survey

3.3.1 Methodology

To further advance the analysis, an open-ended survey (see Appendix J) was conducted using Microsoft Forms to capture travellers' perceptions, decision processes, and experiences related to tourism. The survey was available in both English and Spanish and distributed to 25 respondents without disclosing the specific research purpose. Data collection concluded once all 25 responses had been received. This stage functioned as a bridge between the qualitative interviews and the final quantitative survey, serving both to test question clarity and to generate additional insights for refining the set of destination attributes.

The survey combined open-ended and structured questions. The open-ended questions invited respondents to narrate personal travel experiences (e.g., unsatisfactory trips, representative vacations, or memorable journeys) and to offer advice to their "past self" regarding travel decision-making. Scenario-based questions (scene choices) were also included to explore how travellers evaluate different destination settings. The structured questions focused on decision-making steps, missing factors in information searches, levels of information saturation, and perceptions of pricing. For the latter, a question based on Van Westendorp's Price Sensitivity Meter was included to estimate acceptable price ranges for a tourism package. This model offers a comprehensive multi-question framework that indirectly measures willingness to pay, identifying a range of acceptable prices rather than a single point estimate [6].

Once collected, responses in Spanish were translated into English using ChatGPT, followed by careful manual verification to ensure accuracy. The dataset was then imported into Google Colab for processing, where column names were standardised. An exploratory data analysis (EDA) was conducted to summarise key statistics and visualise general respondent characteristics. For the Van Westendorp model, inconsistent responses, such as cases in which the "too cheap" value exceeded the "bargain" value, were removed. Outliers were excluded using the interquartile range thresholds (`Q1 - 1.5 × IQR` and `Q3 + 1.5 × IQR`). This procedure enabled the estimation of price thresholds for "too cheap," "bargain," "expensive," and "too expensive," ultimately identifying the Optimal Price Point (OPP), by implementing the Algorithm 7 (see Appendix K).

The textual responses were processed using a standardised natural language preprocessing pipeline, which included:

- Conversion of text to lowercase.
- Removal of punctuation, special characters, and extra spaces.
- Normalisation of accented characters.
- Tokenisation and lemmatisation using `spaCy`.
- Stopword removal to retain only meaningful terms.

The analysis followed a text-mining approach combining both a semantic and frequency-based techniques. `SentenceTransformers` was used for sentence embeddings, to capture semantic similarity and group related responses, with cosine similarity calculated through the `networkx` library. TF–IDF vectorisation was subsequently applied within each semantic cluster to identify the most frequent words and expressions. Sentence embeddings were first used to group responses, followed by TF–IDF analysis within each group, for the following open-ended questions:

- "Tell us about a trip of yours that you were not satisfied with. What would you not do again, especially in the planning phase?" (similarity fixed greater than 0.7).
- "Which trip or vacation best represents your way of choosing how to travel? Tell us about the destination: how you chose it, why it reflects you, etc" (similarity fixed greater than 0.67).
- "If you had to teach someone your method of choosing, which 3 fundamental steps would you tell them to follow?" (similarity fixed greater than 0.7).
- "If instead you had to give 3 pieces of advice to your past self about organising a vacation, what would you say?" (similarity fixed greater than 0.7).
- "When you organised your last vacation, did you have a list of destinations in mind? How did you make your final choice? And among them, was there a destination you said, 'No, this is not for me'?" (similarity fixed greater than 0.7).

For other questions, only TF–IDF vectorisation was applied, using different n-gram configurations depending on the length and structure of responses:

- "Imagine organising a vacation. Sketch a short story of your choice: what would be the 3–4 main scenes?"
- "Which was the decisive piece of information you looked at last before choosing?"
- "When did you feel you had 'enough' information to decide? What made you stop searching? And if you had had more time, what would you have added to your research?"
- "What type of trip do you usually choose: a short trip, a medium-length trip, or a long one? Please explain the reason for your choice."

In some cases, additional steps were taken prior to text analysis. For the question "Which was the decisive piece of information you looked at last before choosing?", responses were divided into two groups (A and B) according to the answer to the preceding question, "If you had to choose right now, which of the two screens would you pick?". For the question "What type of trip do you usually choose: a short trip, a medium-length trip, or a long one? Please explain the reason for your choice", responses were manually labelled based on the preferred trip length (short, medium, long, or not defined). Finally, for the questions "If you had to teach someone your method of choosing, which 3 fundamental steps would you tell them to follow?" and "If instead you had to give 3 pieces of advice to your past self about organising a vacation, what would you say?", all steps were concatenated for each case, preserving their order, so that each respondent's full set of steps or pieces of advice could be analysed as a single continuous text.

Finally, using the same prompt described previously (see Appendix E), the raw dataset containing all responses was uploaded to ChatGPT. The model was asked to examine each column carefully and extract any tourist-related attributes identified within the answers. In addition, a series of personal observations and comparative notes were made to evaluate how ChatGPT's results aligned with the attributes previously obtained from the text mining process, and to determine whether any new attributes emerged from its output. This step allowed for a cross-validation between human interpretation and AI-assisted analysis.

3.3.2 Results

Exploratory Data Analysis (EDA)

The exploratory analysis of the open-ended survey dataset provided a general profile of the average respondent (see Figures 4.7 and 4.8 in Appendix L for additional visualisations). The results suggest that most participants were young adults between 25 and 35 years old, although a secondary group of respondents aged over 60 was also identified. The majority of respondents were female and employed, with South America representing the most common region of origin. In terms of travel behaviour, participants typically organised trips collaboratively and tended to travel with companions such as partners, family, or friends. As shown in Figure 4.9 (see Appendix L), the primary motivation for travel was exploring new cities, although relaxation was also cited as a complementary purpose.

Notable differences emerged across age groups. Younger respondents reported relying primarily on Google or similar platforms, and in some cases ChatGPT, for information searches, whereas older respondents showed a stronger reliance on travel blogs or travel agencies. Regarding decision-making patterns, respondents generally reported considering three to four destination options, and taking several days to evaluate alternatives before confirming their booking, as shown in Figure 4.10 (see Appendix L).

Van Westendorp Price-Sensitivity Analysis

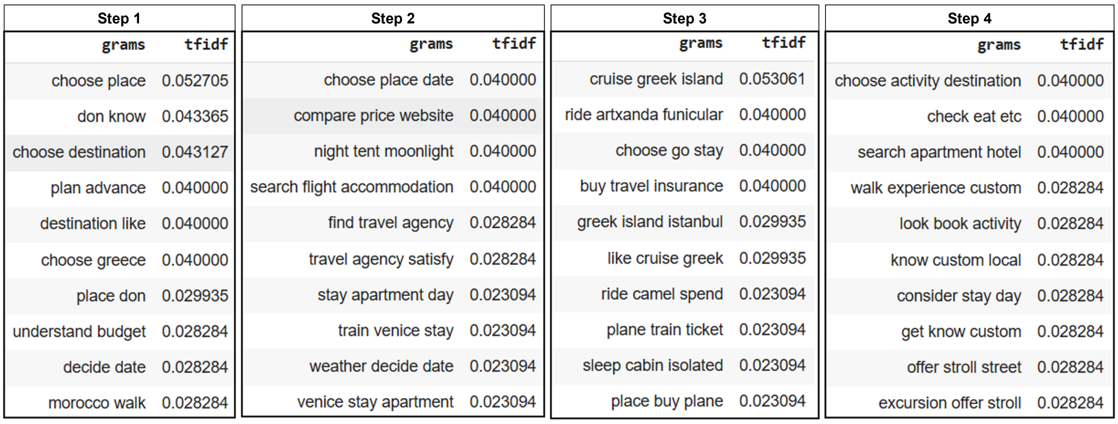
The analysis of price perceptions using the Van Westendorp method revealed notable differences across age groups. For the full sample (see Figure 4.11 in Appendix M), after removing outliers and inconsistent responses, the Optimal Price Point (OPP) was estimated at €250.06. When the analysis was segmented, clear differences emerged. Respondents aged 35 or younger reported a substantially lower OPP of €150.20, whereas those older than 35 displayed a much higher OPP of €279.46. For the latter group, outliers were retained because their removal resulted in no definable OPP. These findings indicate that younger respondents demonstrate greater price sensitivity and a lower willingness to pay for tourism products, whereas older respondents are generally more willing to spend, suggesting distinct consumer profiles relevant for market segmentation and pricing strategies.

Text Mining, Embeddings & TF-IDF

Among the responses to the question "Tell us about a trip of yours that you were not satisfied with. What would you not do again, especially in the planning phase?", two groups were identified. The first group, composed mainly of young adults (25–35), emphasised poor organisation as the main reason for dissatisfaction. The second group comprised respondents aged over 60, who mainly attributed their unsatisfactory experience to not booking tours conducted in their native language.

Furthermore, for the question "Which trip or vacation best represents your way of choosing how to travel? Tell us about the destination: how you chose it, why it reflects you, etc.", two groups were also identified. The first group included both young adults (25–35) and older participants (> 60). These respondents typically described trips in which they visited multiple destinations, most often in Europe, a continent that enables visiting several cities within a single trip. The second group comprised two respondents, both young adults, who expressed a preference for visiting multiple destinations, including cruises, while highlighting the importance of relaxation as a core element of their ideal trip.

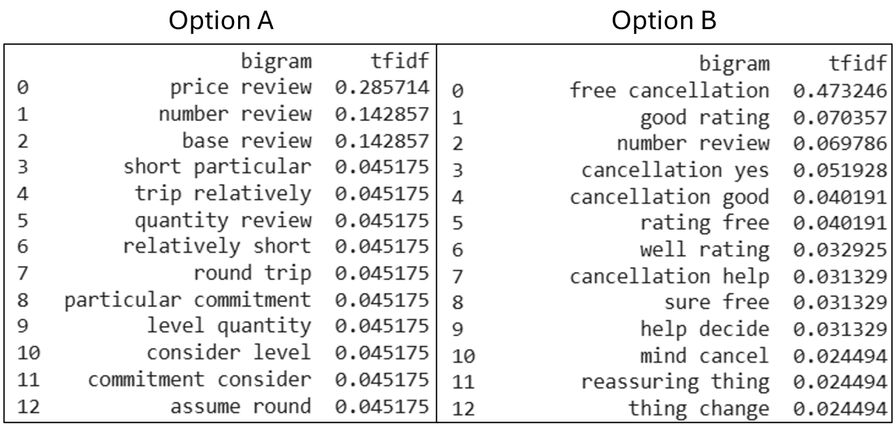
For the question "Imagine organising a vacation. Sketch a short story of your choice: what would be the 3–4 main scenes?", each step was analysed separately using TF–IDF vectorisation (see Figure 3.10). In the first stage, respondents primarily focused on defining the destination, often new or unexplored places, and aligning it with their available budget. The second stage was mainly described as planning logistics, such as defining travel dates, purchasing plane tickets, booking accommodation, and comparing different options. The third stage focused mainly on defining the itinerary and selecting activities at the destination. The optional fourth stage often repeated previous elements, such as defining accommodation or activities, confirming their importance in the final stages of trip planning.



Step 1		Step 2		Step 3		Step 4	
grams	tfidf	grams	tfidf	grams	tfidf	grams	tfidf
choose place	0.052705	choose place date	0.040000	cruise greek island	0.053061	choose activity destination	0.040000
don know	0.043365	compare price website	0.040000	ride artxanda funicular	0.040000	check eat etc	0.040000
choose destination	0.043127	night tent moonlight	0.040000	choose go stay	0.040000	search apartment hotel	0.040000
plan advance	0.040000	search flight accommodation	0.040000	buy travel insurance	0.040000	walk experience custom	0.028284
destination like	0.040000	find travel agency	0.028284	greek island istanbul	0.029935	look book activity	0.028284
choose greece	0.040000	travel agency satisfy	0.028284	like cruise greek	0.029935	know custom local	0.028284
place don	0.029935	stay apartment day	0.023094	ride camel spend	0.023094	consider stay day	0.028284
understand budget	0.028284	train venice stay	0.023094	plane train ticket	0.023094	get know custom	0.028284
decide date	0.028284	weather decide date	0.023094	sleep cabin isolated	0.023094	offer stroll street	0.028284
morocco walk	0.028284	venice stay apartment	0.023094	place buy plane	0.023094	excursion offer stroll	0.028284

Figure 3.10: *Steps for Organizing a Vacation. Source: Author's own elaboration.*

When analysing the question related to choosing between two options, most participants (72%) preferred option B, primarily due to the possibility of free cancellation. Some also mentioned reviews as a secondary factor. Conversely, those who chose option A (28%) cited price as the main decisive factor. Figure 3.11 displays the most frequent bigrams for each group.



Option A

	bigram	tfidf
0	price review	0.285714
1	number review	0.142857
2	base review	0.142857
3	short particular	0.045175
4	trip relatively	0.045175
5	quantity review	0.045175
6	relatively short	0.045175
7	round trip	0.045175
8	particular commitment	0.045175
9	level quantity	0.045175
10	consider level	0.045175
11	commitment consider	0.045175
12	assume round	0.045175

Option B

	bigram	tfidf
0	free cancellation	0.473246
1	good rating	0.070357
2	number review	0.069786
3	cancellation yes	0.051928
4	cancellation good	0.040191
5	rating free	0.040191
6	well rating	0.032925
7	cancellation help	0.031329
8	sure free	0.031329
9	help decide	0.031329
10	mind cancel	0.024494
11	reassuring thing	0.024494
12	thing change	0.024494

Figure 3.11: *Top Bigrams by Option (A and B). Source: Author's own elaboration.*

Moreover, the embedding similarity analysis for the question "If you had to teach someone your method of choosing, which 3 fundamental steps would you tell them to follow?" yielded a single cluster containing 11 responses. This group comprised mainly young adults (25–35) and a smaller proportion of older respondents (> 60). Both embedding similarity and TF–IDF analyses highlighted a consistent decision-making pattern: beyond defining the destination, respondents placed strong emphasis on checking recommendations, verifying accommodation availability, and planning the trip in advance. Based on these insights, the decision-making process for this group can be summarised in three steps:

1. Define & Learn: Decide on the type of trip and preferred destination, and gather initial information about the location.
2. Check & Compare: Review recommendations, compare accommodation and transport options, and verify availability.
3. Plan & Book: Create a detailed plan, book accommodations and key activities in advance, and prepare financially.

For the question "If instead you had to give 3 pieces of advice to your past self about organising a vacation, what would you say?", two groups were identified. The first group included a majority of older participants (> 60) but also a considerable number of young adults (25–35). The main themes extracted were as follows:

- Good preparation:
  - Read offers carefully and understand what is included.
  - Plan and book early, especially for flights and hotels.
  - Get informed from reliable sources, such as experienced travellers, review sites, and travel forums.
- Choose carefully:
  - Quality and reliability matter.
  - Choose hotels with at least 3 stars or good reviews.
  - Select transfers carefully.
- Spend Smart:
  - Be willing to pay a bit more for better quality and comfort.
  - Balance between booking in advance and leaving room for spontaneous activities.

The second group consisted primarily of young adults (25–35), along with a few older respondents (> 60). The main pieces of advice identified from this group were:

- Plan with time but leave some room for spontaneous experiences.
- Take your time to compare options.
- Book tours to get to know the city better.

- Check transportation.

The question "When you organised your last vacation, did you have a list of destinations in mind? How did you make your final choice?" produced two groups. The first comprised mainly young adults (25–35) and a few older participants (> 60). Most respondents began with a list of potential destinations, narrowing it down based on preferences such as activities, local events, and desired experiences. Some excluded destinations considered overly touristic, while a few made spontaneous choices without a predefined list, balancing planning with flexibility. The second group included only young adults. Their responses focused strongly on practical factors such as cost and time availability, indicating decisions guided more by feasibility and affordability than by emotional preference.

As shown in Figure 3.12, with regard to information saturation, most participants reported stopping their search once they had gathered essential information, such as dates, accommodation, transport, and prices, that made them feel confident in their decision. Avoiding information overload was a key trigger to stop searching. For example, one respondent noted that social media can sometimes create unnecessary information overload, promoting distrust rather than clarity. If given more time, some respondents indicated they would have explored nearby destinations, searched for better deals, or extended their stay, while a few felt no need to add anything further.

grams	tfidf
option price reference	0.040000
know exactly destination	0.028284
exactly destination want	0.028284
work tired search	0.023094
place didn city	0.023094
find nice campsite	0.023094
look carefully area	0.023094
find deal work	0.023094
campsite stay add	0.023094
agree place look	0.023094

Figure 3.12: *Top Trigrams for Information Saturation. Source: Author's own elaboration.*

Additionally, combined with the results presented above, responses to the question "Is there something important we haven't asked you but that influenced your decision? Tell us about it." provided additional insights for attribute refinement. Although few participants provided responses, some noteworthy comments included:

- "...the opportunity to learn something new. Whether it's history, local traditions, or even trying a completely new activity."

- "Ticket prices and promotions always influence my decision."

- "I listen to other travellers' stories and take what might interest me."

Finally, regarding the suggestions from ChatGPT, as shown in Figure 3.13, although most of the attributes were related to existing ones, there was one attribute that consistently appeared across the analyses and suggestions: Availability of guided tours in different languages.

Attribute	Observation
Centrally located accommodations.	Already in previous attributes
Accommodation within walking distance to key attractions.	Already in previous attributes
Accommodation's high quality (4/5 stars).	Already in previous attributes
Flexible booking options (free cancellation, easy date change).	Already in previous attributes
Accommodation near daily essentials (supermarket, pharmacy, ATM).	Already in previous attributes
Accommodation services (24h check-in or digital self check-in).	Already in previous attributes
Availability of authentic local restaurants with regional dishes.	Already in previous attributes
Restaurants with online or advance booking.	This is interesting, but should check if is relevant to include, taking into account that I already have something about restaurants.
Availability of local food markets/craft fairs.	Interesting, also saw this, but not mentioned by many tourists previously.
Availability of traditional, family-run restaurants.	This is also interesting, but again, related to local cuisine.
Possibility of cultural immersion (local rituals, traditions, or workshops).	Already in previous attributes
Availability of guided tours in different languages.	Clearly there is something about this matter, although was not frequent, with this approaches I reinforced the need of providing pre-fix activities so tourists avoid over-planning or FOMO (fearing not seeing something important)
Ready-made travel packages (include key attractions).	Can be combined with previous case
Local festivals.	Already in previous attributes and as mentioned before, this is stational
Walkable historic districts.	Can be related to walking distance of accommodation and attractions.
Diversity of small local shops/artisans.	Related to other attributes already mentioned.
Availability of hiking/trekking routes.	Already in previous attributes
Proximity to other cities/villages.	Already in previous attributes
Proximity to natural areas.	Already in previous attributes
Ease of reaching the city by public transport.	Already in previous attributes
Availability of car rentals.	Already in previous attributes
Touristic sites reachable on foot.	Already in previous attributes
Availabilkity of luggage storage near stations or airports.	This is really interesting attribute, yet it was not mentioned too much, and other attributes are more representative.

Figure 3.13: *Attributes Suggested by ChatGPT. Source: ChatGPT.*

3.4 Refined Final Attributes

After completing all stages of analysis, from the initial qualitative interviews to the open-ended survey, the process led to the consolidation of a refined set of attributes. These attributes represent the final version derived from multiple rounds of identification, validation, and refinement. They constitute the core variables to be included in the final survey for testing. For this research, the attribute "Ease of reaching the city by public transport" was manually replaced with "City located outside major tourist circuits", since, as mentioned earlier, Ascoli Piceno serves as the reference case and it was considered more relevant to test this new attribute.

To illustrate the progression, Figure 3.14 presents the evolution of the attributes from the previous project (the starting point) to the final set obtained through the methodology described in this work.

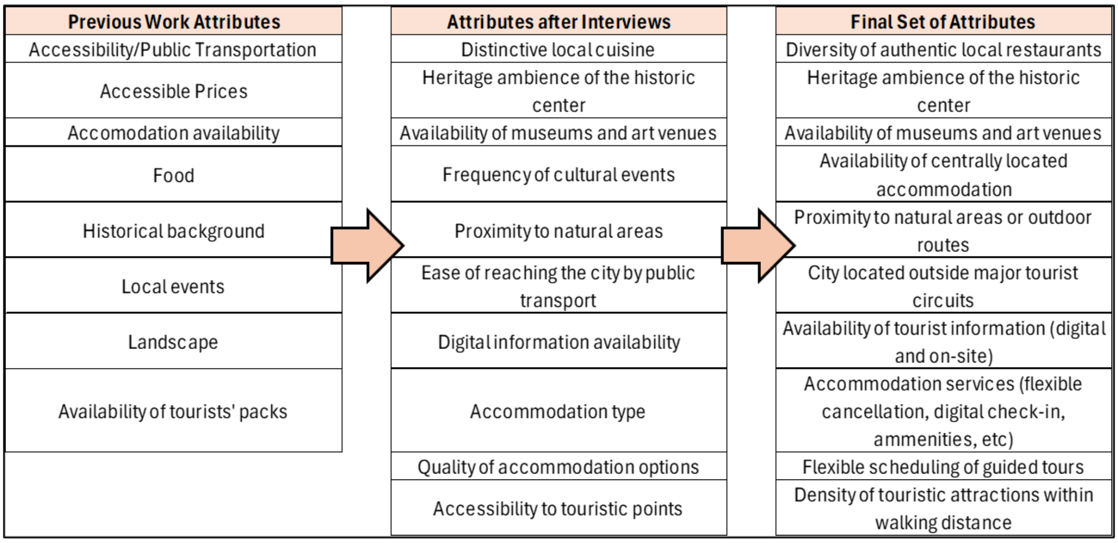


Figure 3.14: *Evolution to Final Attribute Set. Source: Author's own elaboration.*

3.5 Quantitative Survey

3.5.1 Methodology

Quantitative Survey

After completing the process of defining the final set of attributes, the quantitative survey (see Appendix N) was conducted to test this set and identify potential tourist clusters. In addition to asking respondents to rank the attributes according to their preferences, it was important to include Likert-scale questions to serve as a control for attribute preferences, as well as additional questions related to the attributes to further explain touristic behaviour. Some descriptive questions were also incorporated to characterise each cluster more precisely. The survey also included, as an illustrative example, a dedicated section with questions specifically referring to the city under analysis. This section was designed as a template that could be adapted to any destination applying the same methodology. In this study, it was implemented using the reference case of Ascoli Piceno.

The survey was completed by a total of 125 respondents, with the aim of capturing a diverse sample across different age groups, primarily individuals with the ability to plan and undertake trips, such as professionals or even students living in Europe (travel distances are generally shorter compared to larger or more distant regions).

Comparison of Ranking and Likert-Scale Responses

Firstly, after obtaining the dataset with all the answers, the ranking responses were reorganised so that each attribute corresponded to a separate column, containing the numerical rank assigned by each respondent. To enable comparison with the Likert-scale questions, the ranking values were then inverted (the attribute ranked first was assigned a value of 10, the second a value of 9, and so on) and subsequently rescaled to a 1–5 range. This transformation ensured comparability with the Likert scale, where values range from 1 (strongly disagree) to 5 (strongly agree).

Once both datasets were aligned on the same scale, a correlation analysis was conducted to evaluate the relationship between the ranking-based preferences and the Likert-scale responses for each attribute (see Appendix P). The non-parametric Spearman's rank correlation coefficient (ρ) was employed, as it measures the strength and direction of the monotonic relationship between two ranked variables [38]. Statistical significance was assessed through p-values, and significance levels were labelled using standard notation ($*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$).

Determining the Optimal Number of Clusters (K)

The ten final attributes, using their original ranking values, were employed as input variables for the clustering analysis. The inverted and rescaled versions were used exclusively for validation purposes, such as the consistency check with the Likert-scale questions.

In the previous work, the clustering procedure was conducted using the K-Means algorithm; however, given the ordinal nature of the ranking data, a more suitable combination involves the Gower distance and the K-Medoids algorithm. Unlike K-Means, which relies on Euclidean distance and is therefore appropriate only for continuous and numerical data, K-Medoids is more robust to noise and can handle mixed or ordinal data types, as it minimizes a sum of pairwise dissimilarities rather than squared distances [39]. The use of Gower distance enables the measurement of similarity across variables of different scales and types, including ordinal and categorical attributes [40]. This makes the Gower + K-Medoids approach particularly suitable for clustering survey data, as it preserves the relative ordering of attributes while maintaining interpretability and robustness to outliers [41, 42].

To determine the optimal number of clusters (k), three complementary methods were applied:

- Elbow method (Algorithm 6 in Appendix D): evaluates the reduction in within-cluster dissimilarity as k increases [43].
- Silhouette score (Algorithm 6 in Appendix D): assesses how well each observation fits within its assigned cluster compared to others [44].

- Davies–Bouldin Index (DBI) (Algorithm 6 in Appendix D): measures intra-cluster cohesion and inter-cluster separation [45].

The combination of these criteria ensured a robust and data-driven determination of the most appropriate clustering structure.

Principal Component Analysis (PCA)

After completing the clustering analysis, the cluster labels were adjusted for clarity by reindexing them from 0–3 to 1–4. This change was made solely for presentation purposes, as it provides a more intuitive numbering convention when discussing and visualising the resulting groups.

To further explore the internal structure of the data and assess the separability of the identified clusters, a Principal Component Analysis (PCA) was conducted. PCA is a widely used linear dimensionality reduction technique that transforms the original variables into a new set of uncorrelated components, ordered according to the amount of variance they explain in the dataset [46]. This transformation allows high-dimensional data to be represented in two or three dimensions while retaining as much of the underlying information as possible.

In this study, PCA was applied to the scaled ranking data to evaluate whether the resulting components could visually distinguish the clusters obtained through the Gower + K-Medoids procedure. The main objective was not dimensionality reduction for analysis itself, but rather to obtain a visual representation that could help interpret and validate the clustering results.

Random Forest for Attribute-Importance Estimation

To complement the clustering results and gain a deeper understanding of the role played by each attribute in differentiating the identified segments, a Random Forest Classifier was applied. This supervised learning algorithm is particularly suitable for estimating the relative importance of predictor variables in classification problems, as it evaluates how much each variable contributes to reducing impurity across multiple decision trees [47]. In this context, the Random Forest was used to determine which attributes most strongly influence cluster definition, providing interpretability to the segmentation model.

The analysis was implemented using the `RandomForestClassifier` from the `scikit-learn` library. The input dataset (`X`) consisted of the rankings for each of the ten attributes, while the target variable (`y`) represented the cluster assignment obtained from the K-Medoids model. To identify the most appropriate number of trees, a sensitivity analysis was performed by testing several values for the parameter `n_estimators` (ranging from 10 to 500). For each configuration, a five-fold cross-validation procedure was used to compute the mean classification accuracy, ensuring that the model's performance was stable and not dependent on a single random partition.

Once the optimal number of estimators was selected, the final model was trained using all observations. The resulting feature importance scores were then extracted and ranked to identify which attributes contributed most to distinguishing between clusters. These importance values reflect the average decrease in Gini impurity caused by splits on each variable across all trees in the ensemble. Finally, the distribution of attribute importance was visualised through a stem plot, providing an interpretable representation of the relative weight of each variable in the clustering structure.

Predictive Modelling

To further understand how individual and behavioural characteristics influence cluster membership, a Multinomial Logit (MNL) model was applied. This model allows for the estimation of the probability that a respondent belongs to each of the four clusters identified previously, based on a set of explanatory variables [48]. The dependent variable (`cluster`) represents the cluster assignment, while the independent variables correspond to respondents' socio-demographic features and behavioural preferences related to most attributes analysed earlier.

Before estimating the model, a data preparation and transformation process was conducted to ensure consistency and interpretability. The selected dataset included both descriptive variables (such as `age_group`, `gender`, `residence`, and `occupation`) and behavioural variables describing decision-making patterns, such as `define_accommodation`, `accommodation_search`, `less_crowded_willingness`, `extra_travel_time`, and `architectural_interest`.

Categorical variables were recoded to numerical formats to facilitate model estimation. For example, the variable `define_accommodation` was coded on a scale from 1 to 3, representing different booking behaviours (advance booking, last-minute booking, or booking directly at the destination). Similarly, ordinal variables such as `extra_travel_time` were transformed into numerical values representing the midpoint of each time range (e.g., 1–2 hours $\rightarrow$ 1.5). Demographic variables were also encoded numerically: `gender` (male = 1, female = 0), and `residence` (categorical values from 1 to 8).

Several survey questions allowed multiple selections. To handle this structure, each response option was expanded into a binary column (1 if selected, 0 otherwise). To reduce dimensionality and enhance interpretability, these variables were then grouped into broader conceptual categories, such as:

- Accommodation search: grouped into digital platforms such as Booking, Airbnb, social media, and traditional channels such as travel agencies, personal recommendations.

- Architectural interest: grouped into historical architecture (e.g., medieval, renaissance, religious buildings) and contemporary cultural interest (e.g., museums, modern architecture).

- Restaurant choice: grouped into personal choice (e.g., walking by, location-based) and social influence (e.g., reviews, recommendations).

- Natural environment type: simplified into four binary variables (Beach, Natural parks, Mountains, Lakes/Rivers).

After all transformations, a correlation matrix was generated to identify potential multicollinearity among predictors. Additionally, the Variance Inflation Factor (VIF) was computed for each variable to quantify redundancy and ensure model stability [49]. Variables with excessively high VIF values were candidates for removal or aggregation, although the majority remained within acceptable limits (VIF < 5).

The final model was estimated using the `MNLogit` function from the `statsmodels` package. This model was chosen due to its suitability for modelling categorical outcomes with more than two unordered alternatives [48]. The estimated coefficients provide insights into how each explanatory variable affects the likelihood of belonging to one cluster compared to the baseline category. This approach helps to interpret behavioural patterns across tourist segments and identify key differentiating factors in destination-related decisions.

3.5.2 Results

Quantitative Survey

The final quantitative survey collected a total of 125 responses. This sample provides a diverse representation of individuals from different age groups, genders, occupations, and geographical regions, allowing for a broad perspective on tourism preferences and behaviour patterns.

In terms of age distribution (see Figure 3.15), most respondents were between 26 and 35 years old (51 participants), followed by those aged 18–25 (26 participants) and those over 55 (19 participants). This indicates a clear predominance of young adults within the sample. Regarding gender, the distribution was relatively balanced across male and female respondents, with no substantial disproportion in representation.

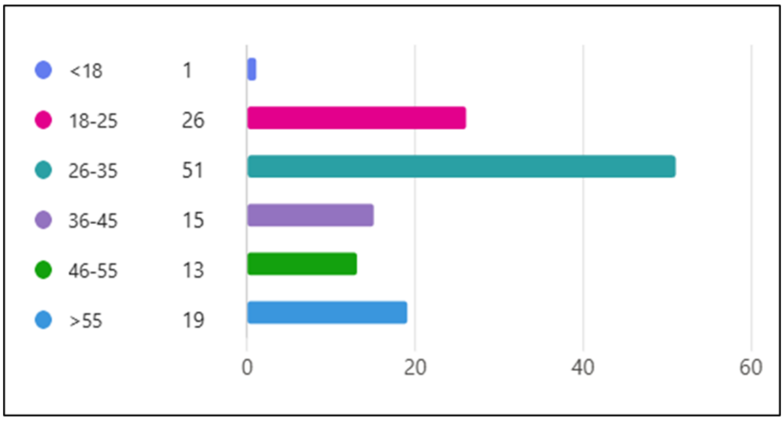


Figure 3.15: *Number of Respondents by Age Range. Source: Author's own elaboration.*

With respect to the region of residence, the survey was mainly aimed to get respondents from South America and Europe. This latter group was separated into Italian and non-Italian respondents, as mentioned earlier, the reference case is Ascoli Piceno. This distinction is important, as it allows the comparison between national and international tourism. As shown in Figure 3.16, most respondents were located in South America (54) and Italy (44), followed by participants from Europe excluding Italy (23). This distribution aligns with the recruitment strategy, which focused on reaching individuals familiar with the European travel context, while maintaining an international perspective.

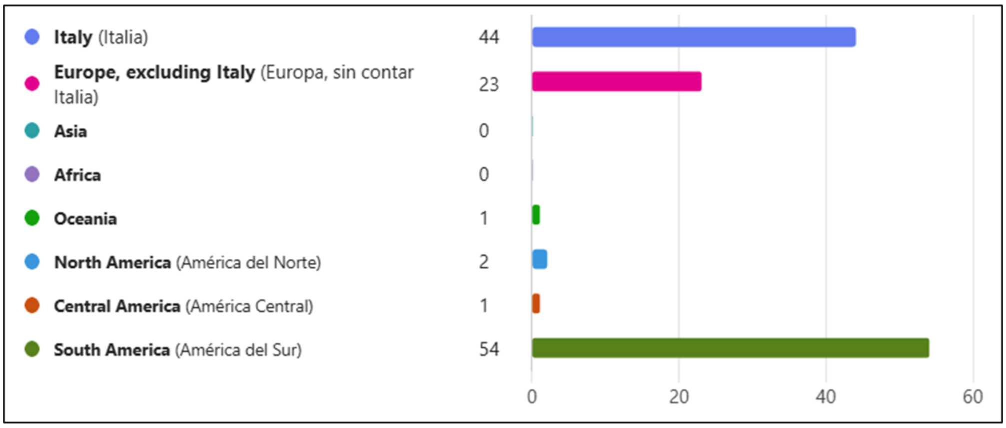


Figure 3.16: *Number of Respondents by Residence. Source: Author's own elaboration.*

Finally, regarding occupational status (Figure 3.17), employed individuals accounted for the largest share of the sample (56 participants), followed by students (31) and self-employed professionals (27). This variety of employment situations reflects different travel capacities and motivations among respondents.

Overall, the demographic composition of the sample ensures a balanced mix of respondents across key sociodemographic variables, providing a solid basis for the subsequent analyses of tourist segmentation and behavioural patterns.

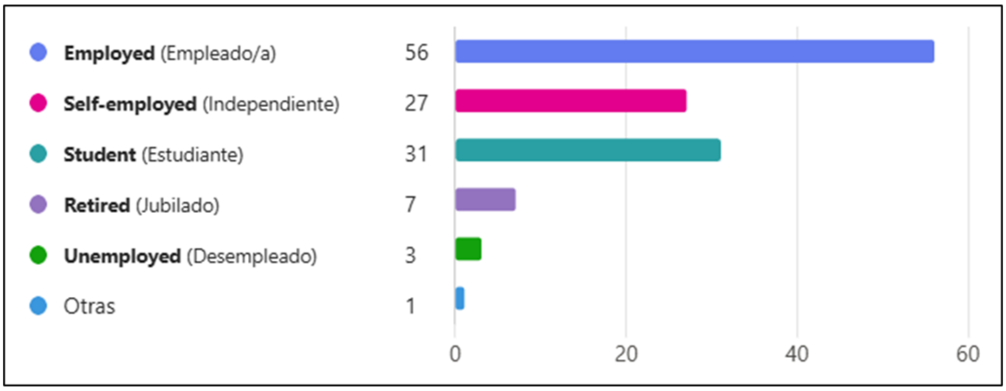


Figure 3.17: *Number of Respondents by Profession. Source: Author's own elaboration.*

Comparison of Ranking and Likert-Scale Responses

The inclusion of Likert-scale questions proved valuable as a control mechanism to assess whether respondents answered the ranking section consciously rather than arbitrarily. As shown in Figure 3.18, the correlation between the ranking values and their corresponding Likert-scale evaluations was both positive and statistically significant for all attributes, with Spearman's ρ values ranging between 0.40 and 0.60. This indicates a consistent alignment between the two measures, confirming that participants generally evaluated the attributes coherently. The only exception was the manually modified attribute, "City located outside major tourist circuits", which displayed a weaker correlation ($\rho = 0.208$, $p = 0.020$), likely reflecting its distinct conceptual framing compared to the rest of the attributes that were correctly obtained through the methodology developed in this research.

	Attribute	Spearman_rho	p_value	Significance
0	Diversity of authentic local restaurants	0.441	0.0000	****
1	Density of touristic attractions within walkin...	0.470	0.0000	****
2	City located outside major tourist circuits	0.208	0.0201	*
3	Availability of tourist information, both digi...	0.582	0.0000	****
4	Availability of centrally located accommodation	0.414	0.0000	****
5	Availability of museums and art venues	0.596	0.0000	****
6	Accommodation services, such as flexible cance...	0.451	0.0000	****
7	Heritage ambience of the historic center	0.526	0.0000	****
8	Proximity to natural areas or outdoor routes	0.529	0.0000	****
9	Flexible scheduling of guided tours	0.465	0.0000	****

Figure 3.18: *Spearman Correlation Between Rankings and Likert-scale Responses. Source: Author's own elaboration.*

Although these results validate the reliability of the ranking data, the moderate level of correlation suggests that the Likert scale is less informative for segmentation purposes. As illustrated in Figure 4.12 (see Appendix O), most respondents tended to rate nearly all attributes as important or very important (values 3–5 on the scale). This pattern of limited variability indicates that the Likert responses alone would not provide sufficient discrimination power for clustering, as the majority of participants expressed uniformly positive evaluations across attributes.

Determining the Optimal Number of Clusters (K)

Based on the results obtained from the clustering validation methods, the optimal number of clusters was determined to be $k = 4$. As shown in Figure 3.19, the Elbow method did not display a sharply defined inflection point; however, a noticeable change in the slope was observed starting at $k = 4$, with a similar variation at $k = 5$. This suggested that four and five clusters might represent a reasonable balance between model simplicity and internal cohesion. Using multiple validation methods allowed for a more robust determination of the optimal value of k.

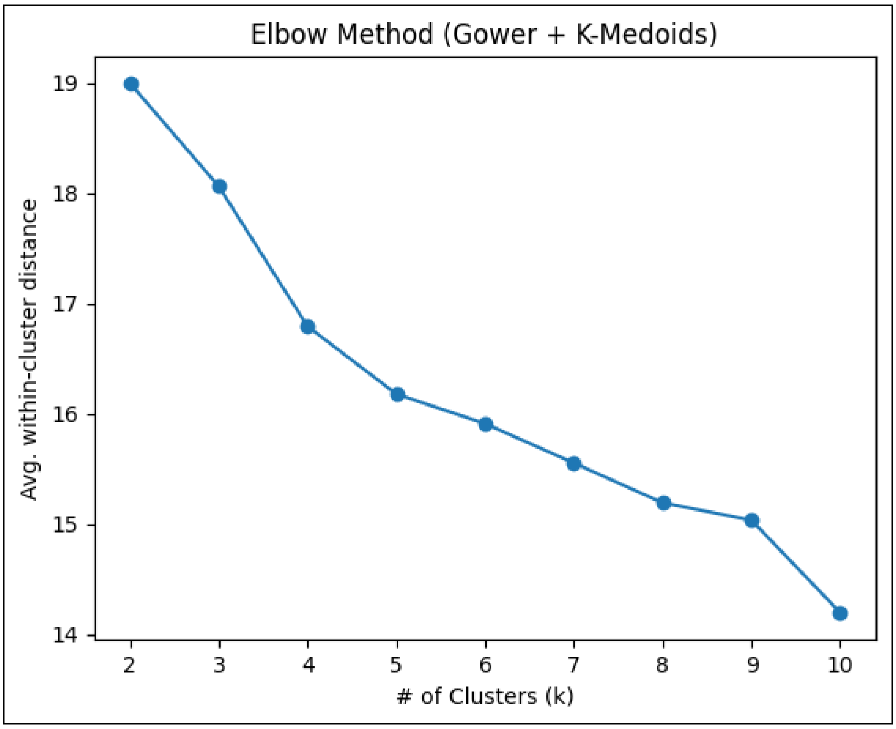


Figure 3.19: *Elbow plot. Source: Author's own elaboration.*

The Silhouette score analysis (see Figure 3.20) provided clearer evidence, indicating $k = 4$ as the most appropriate choice, with the highest score of 0.087. This result shows that, although the clusters are not perfectly separated (an expected outcome in behavioural data), the overall structure still reflects a meaningful segmentation of tourist preferences.

The Davies–Bouldin Index (DBI) supported this conclusion, with the lowest index value also occurring at $k = 4$ (DBI = 3.088), compared with higher values obtained for other configurations. Taken together, the three methods consistently

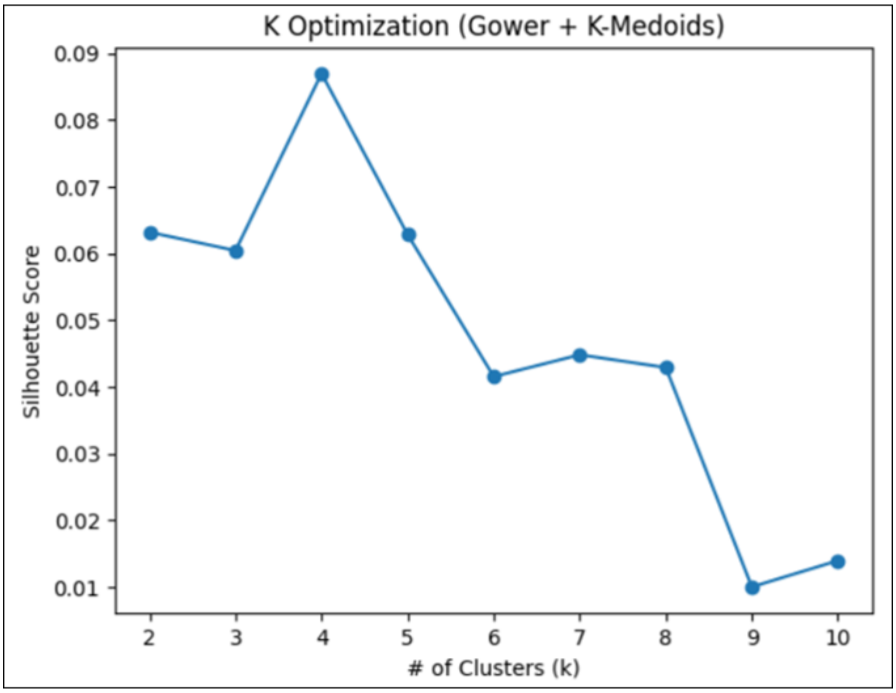


Figure 3.20: *Silhouette Scores and Plot. Source: Author's own elaboration.*

indicated that the most coherent and stable partition of the data corresponds to four clusters, which was therefore adopted for the subsequent analyses.

Overall, after applying multiple validation methods to determine the optimal number of clusters, the analysis concluded that the most appropriate solution was $k = 4$. The final segmentation included the following cluster sizes: Cluster 1 contains 35 respondents (28%), Cluster 2 contains 36 respondents (28%), Cluster 3 contains 23 respondents (18%), and Cluster 4 contains 31 respondents (24%).

Figure 3.21 presents the mean values of the segmentation variables by cluster, illustrating how the importance assigned to each attribute varies among groups. As shown, Cluster 1 exhibits relatively low mean values across most attributes, suggesting a segment that places less emphasis on cultural and informational aspects of the destination. Cluster 2, on the other hand, scores higher on attributes related to culture and heritage, particularly the availability of museums and the historic ambience, indicating a preference for culturally rich destinations. Cluster 3 stands out for prioritising convenience and flexibility, showing higher scores for accommodation services and the diversity of local restaurants, and moderately high values for proximity to outdoor areas. Finally, Cluster 4 is characterised by high scores across almost all variables, especially for attributes such as the density of tourist attractions, flexible guided tours, and accommodation amenities, suggesting a highly engaged segment that values accessibility, variety, and comfort during travel.

cluster	Population	1	2	3	4
Diversity of authentic local restaurants	3.92	3.29	4.17	5.61	3.10
Density of touristic attractions within walking distance	5.89	3.77	4.36	7.39	8.94
City located outside major tourist circuits	7.12	7.66	8.28	6.83	5.39
Availability of tourist information, both digital and on-site	6.41	5.71	7.42	5.39	6.77
Availability of centrally located accommodation	4.28	4.23	4.36	3.26	5.00
Availability of museums and art venues	5.04	6.66	4.14	5.65	3.81
Accommodation services, such as flexible cancellation, digital check-in, amenities, etc.	5.67	6.77	4.89	2.78	7.48
Heritage ambience of the historic center	3.23	3.77	2.33	5.26	2.16
Proximity to natural areas or outdoor routes	5.29	4.80	7.06	5.43	3.68
Flexible scheduling of guided tours	8.15	8.34	8.00	7.39	8.68

Figure 3.21: *Means of Segmentation Variables by Segment. Source: Author's own elaboration.*

Principal Component Analysis (PCA)

Although PCA was initially performed to explore potential data reduction and visualisation, its explanatory power proved insufficient for clear interpretation. The first six principal components, as shown in Figure 4.13 (see Appendix T), together explained most of the variance in the dataset, however, no two-dimensional or three-dimensional projection provided a meaningful representation of the underlying structure. For this reason, PCA was not useful for graphical visualisation of clusters (see Figure 4.14 in Appendix T), and the interpretation relied instead on clustering metrics and attribute-level analysis.

Random Forest for Attribute-Importance Estimation

To assess the relative influence of each attribute on the differentiation of tourist clusters, the Random Forest classifier was trained using the ten attributes as predictor variables and the cluster membership as the target variable. Figure 3.22 presents the importance scores obtained for each variable, ranked in descending order according to their contribution to the model (see Figure 4.15 in Appendix U).

The results show that the "Density of touristic attractions within walking distance" emerged as the most influential attribute (0.21), followed by "Accommodation services, such as flexible cancellation, digital check-in, and amenities" (0.16) and "Proximity to natural areas or outdoor routes" (0.13). These three factors appear to be the most discriminant variables for cluster assignment, suggesting that accessibility to attractions, accommodation comfort, and contact with nature play a key role in distinguishing traveller profiles.

Secondary importance levels were observed for "Availability of museums and art venues" (0.11), "City located outside major tourist circuits" (0.08), and "Heritage ambience of the historic centre" (0.08), indicating that cultural and spatial context still contribute to defining specific preferences. The remaining variables exhibited lower relevance in predicting cluster membership, and do not appear to be decisive factors in differentiating between tourist segments.

Overall, the distribution of importance values shows a moderate but clear hierarchy among the attributes, confirming that certain factors, especially accessibility, accommodation services, and natural environment proximity, play a leading role in shaping tourists' decision-making and their perception of destinations.

	Importance
Density of touristic attractions within walking...	0.21
Accommodation services, such as flexible cancel...	0.16
Proximity to natural areas or outdoor routes	0.13
Availability of museums and art venues	0.11
City located outside major tourist circuits	0.08
Heritage ambience of the historic center	0.08
Availability of tourist information, both digit...	0.07
Diversity of authentic local restaurants	0.07
Availability of centrally located accommodation	0.06
Flexible scheduling of guided tours	0.03

Figure 3.22: *Variable Importance in Random Forest Model. Source: Author's own elaboration.*

Predictive Modelling

Before running the Multinomial Logit model, potential multicollinearity among predictors was assessed. As shown in Figure 4.16 (see Appendix V), the correlation matrix indicates moderate correlations among some predictors, particularly between `historical_architecture` and `contemporary_cultural_interest`, both related to architectural interest, with a correlation of 0.735. This relationship was further confirmed by the VIF analysis (see Figure 4.17 in Appendix W), where both variables presented relatively high values compared to others (VIF > 3). To avoid collinearity effects and enhance model stability, `historical_architecture (architectural_interest)` was removed from the final specification.

The initial model was estimated using the `MNLogit` function from the `statsmodels` package. The formula included 19 independent variables, combining sociodemographic and behavioural factors. The model specification was as follows:

$$
\begin{aligned}
\texttt{cluster} \sim\ & \texttt{age\_group} + \texttt{gender} + \texttt{residence} + \texttt{occupation} \\
& + \texttt{define\_accommodation} + \texttt{less\_crowded\_willingness} \\
& + \texttt{extra\_travel\_time} \\
& + \texttt{city\_tour\_type} + \texttt{eating\_preference} \\
& + \texttt{digital\_platforms (accommodation\_search)} \\
& + \texttt{traditional\_channels (accommodation\_search)} \\
& + \texttt{historical\_architecture (architectural\_interest)}
\end{aligned}
$$

$$
\begin{aligned}
&+ \texttt{contemporary\_cultural\_interest (architectural\_interest)}\\
&+ \texttt{personal\_choice (restaurant\_choice)}\\
&+ \texttt{social\_influence (restaurant\_choice)}\\
&+ \texttt{Beach (natural\_environment\_type)}\\
&+ \texttt{Natural\_parks (natural\_environment\_type)}\\
&+ \texttt{Mountains (natural\_environment\_type)}\\
&+ \texttt{Lake/Rivers (natural\_environment\_type)}
\end{aligned}
$$

After the first estimation (see Appendix X), several variables exhibited unstable coefficients, large standard errors, and non-significant contributions to the prediction of clusters. Based on theoretical relevance and statistical performance, the optimized model retained the following predictors:

- `age_group`
- `gender`
- `residence`
- `occupation`
- `city_tour_type`
- `digital_platforms (accommodation_search)`
- `traditional_channels (accommodation_search)`
- `contemporary_cultural_interest (architectural_interest)`

The refined specification of the final model is presented below:

$$
\begin{aligned}
\texttt{cluster} \sim\ & \texttt{age\_group} + \texttt{gender} + \texttt{residence} + \texttt{occupation} + \texttt{city\_tour\_type}\\
&+ \texttt{digital\_platforms (accommodation\_search)}\\
&+ \texttt{traditional\_channels (accommodation\_search)}\\
&+ \texttt{contemporary\_cultural\_interest (architectural\_interest)}
\end{aligned}
$$

The final model converged successfully after six iterations (see Figure 3.23), indicating a statistically significant improvement compared to the null model. Although the pseudo R-squared value remains modest (typical for behavioural

MNLogit Regression Results			
Dep. Variable:	cluster	No. Observations:	125
Model:	MNLogit	Df Residuals:	98
Method:	MLE	Df Model:	24
Date:	Sat, 25 Oct 2025	Pseudo R-squ.:	0.1104
Time:	18:36:21	Log-Likelihood:	-152.58
converged:	True	LL-Null:	-171.53
Covariance Type:	nonrobust	LLR p-value:	0.03563

Figure 3.23: *MNLogit Regression Results (Final Model). Source: Author's own elaboration.*

multinomial models) it suggests that the included predictors capture meaningful variation in cluster membership.

As presented in Figure 3.24, the confusion matrix reveals an acceptable level of predictive accuracy, with clusters 1 and 2 achieving the highest correct classification rates. In contrast, clusters 3 and 4 exhibit higher levels of overlap, indicating partial similarity in respondent profiles and less sharply defined behavioural boundaries. Furthermore, the model's estimated coefficients (see Appendix Y) indicate that:

- Gender is a consistent discriminator, being statistically significant ($p < 0.05$) for clusters 2 and 4, suggesting differences in destination preferences between male and female travellers.

- Age group shows a significant effect ($p < 0.05$) for cluster 3, implying that younger respondents are more likely to belong to this group, characterised by exploratory travel behaviour.

- Residence is significant across multiple clusters ($p < 0.05$), reinforcing the influence of geographical origin on travel decision-making.

- Cultural engagement, represented by `contemporary_cultural_interest` `(architectural_interest)`, displayed a near-significant effect ($p = 0.07$) for cluster 3, indicating that interest in modern cultural experiences differentiates a specific segment of travellers.

In summary, the final Multinomial Logit model achieved convergence with statistically robust predictors and interpretable results, highlighting the relevance of age, gender, residence, and cultural orientation in explaining tourist segmentation.

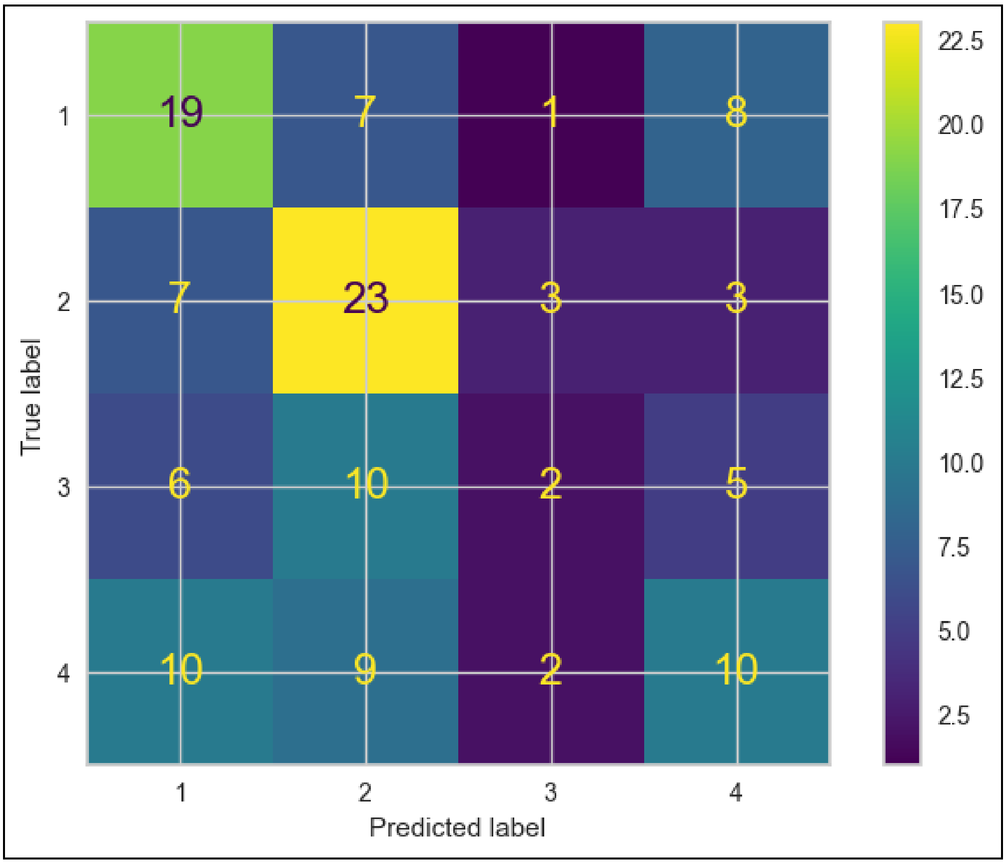


Figure 3.24: *Confusion Matrix of Final Model. Source: Author's own elaboration.*

Van Westendorp Price-Sensitivity Method

The Van Westendorp analysis was applied to estimate respondents' perceived price thresholds for the proposed tourism experience. Following the same methodological approach presented in Section 3.3, Sub-section 3.3.1 (see Appendix K), four key indicators were obtained: the Point of Marginal Cheapness (PMC), Point of Marginal Expensiveness (PME), Indifference Price Point (IPP), and Optimal Price Point (OPP). These values provide a reference framework for understanding perceived value and acceptable price ranges among different tourist segments.

Considering the full sample ($n = 125$), results indicate an Optimal Price Point (OPP) of €200.18 and an Indifference Price Point (IPP) of €250.20, with the acceptable price range extending approximately between €200 and €250. This suggests that most respondents perceive prices within this interval as fair and aligned with their value expectations for the proposed experience.

When examining results by cluster, meaningful variations emerge across the four groups:

- Cluster 1 shows the lowest willingness to pay, with an OPP of €157.19 and an IPP of €200.34. The perceived acceptable range (from €150 to €240) is notably narrower, indicating greater price sensitivity and a higher likelihood of perceiving offers above €250 as expensive or prohibitive.

- Cluster 2 exhibits similar patterns, with a PMC of €150.20 and a PME of €240.00. However, the Optimal Price Point could not be determined due to the crossing behaviour of cumulative distributions, suggesting inconsistent or highly heterogeneous perceptions of value within this group. This is consistent with the higher standard deviations observed in individual price responses, ranging from €49.2 for the "too cheap" threshold to €254.3 for "too expensive" responses.

- Cluster 3 presents the widest acceptable price range, with a PMC of €200.10, a PME of €300.02, and an OPP of €200.10. This segment demonstrates lower price sensitivity and greater tolerance towards higher price levels, aligning with profiles that emphasise experiential or cultural value over cost considerations.

- Cluster 4 reflects a pattern similar to the total sample, with an OPP of €200.05 and an IPP of €250.26, suggesting moderate price flexibility and a balanced perception of value relative to price.

Overall, the results reveal a general convergence around €200–€250 as the perceived fair price interval, although clusters differ in their sensitivity and upper price tolerance. Clusters 1 and 2 are the most price-constrained, whereas Cluster 3 exhibits the highest willingness to pay.

Cluster Profiles: Socio-Demographic Composition

After applying the clustering procedure and validating the optimal solution at $k = 4$, the resulting segments were analysed to describe their socio-demographic composition and distinctive attitudinal patterns. Appendix Z presents the distribution of age, gender, residence, and professional status across the four clusters, illustrating clear structural differences among groups.

- Cluster 1 ($n = 35$; 28% of total respondents) is mainly composed of individuals aged between 26–35 years, mostly females. This group is dominated by employed respondents (60%), mostly residing in South America.

- Cluster 2 ($n = 36$; 28%) is characterised by a younger demographic, concentrated in the 18–25 and 26–35 age ranges, with a slight predominance of male respondents. The majority are students or early-career professionals, primarily from Italy and other European countries.

- Cluster 3 ($n = 23$; 18%) includes a relatively younger population, again dominated by respondents aged between 18 and 35 years, with balanced gender representation. Geographically, it is mostly composed of respondents from Italy and South America, reflecting a mix of international perspectives.

- Cluster 4 ($n = 31$; 24%) presents a more balanced demographic composition, including respondents from all age groups but slightly older on average. Gender distribution is even, with respondents primarily from South America. Most participants are employed professionals or self-employed.

Cluster Profiles: Top and Bottom Attributes

To further interpret the clustering outcomes, each group's attribute rankings were analysed to identify their top three most valued and bottom three least valued attributes (see Appendix AA). This comparison highlights the distinct preferences and perceptions across the four identified clusters, providing additional insight into their motivational profiles.

Across all clusters, certain attributes consistently ranked high, particularly those related to organisation, accessibility, and convenience. Conversely, attributes associated with historical ambience or centrally located accommodation tended to receive lower scores, suggesting these are perceived as baseline expectations rather than differentiating factors.

- Cluster 1 prioritises "Flexible scheduling of guided tours", "Accommodation services (such as flexible cancellation, digital check-in, amenities)", and "City located outside major tourist circuits". These preferences reflect a pragmatic traveller profile that values operational flexibility and less crowded destinations. The least valued attributes are "Availability of centrally located accommodation", "Heritage ambience of the historic center", and "Diversity of authentic local restaurants", indicating less emphasis on location centrality and traditional heritage.

- Cluster 2 also ranks "Flexible scheduling of guided tours" and "Availability of tourist information (both digital and on-site)" among its top priorities, alongside "City located outside major tourist circuits". This combination suggests a preference for autonomy and information-rich experiences. On the other hand, "Availability of museums and art venues", "Heritage ambience of the historic center", and "Accommodation services" appear among the lowest-rated attributes, indicating that this segment seeks more experiential and less traditional cultural engagement.

- Cluster 3 places the highest value on "Density of tourist attractions within walking distance", "Flexible scheduling of guided tours", and "Availability of tourist information", showing a clear orientation towards cultural exploration and mobility. The lowest-ranked attributes ("Availability of centrally located accommodation", "Accommodation services", and "City located outside major tourist circuits") indicate that accessibility to attractions outweighs the importance of accommodation characteristics for this group.

- Cluster 4 exhibits similar patterns, with the top three being "Flexible scheduling of guided tours", "Density of tourist attractions within walking distance", and "Accommodation services". This cluster values efficiency and comfort when visiting destinations, reflecting a more structured and convenience-oriented profile Its lowest rated attributes include "Heritage ambience of the historic center", "City located outside major tourist circuits", and "Availability of museums and art venues", pointing to limited interest in heritage or traditional cultural offerings.

Cluster Profiles: Accommodation-Booking Behaviour

The survey results indicate that the majority of respondents tend to book their accommodation in advance, while only a smaller proportion prefer to make last-minute reservations. This pattern reflects a predominant orientation towards planned travel behaviour, which is consistent with the exploratory nature of leisure trips analysed in this study.

Figure 3.25 shows that, when booking in advance, the most frequently used channels across all clusters are online platforms such as Booking.com and Airbnb, followed by hotels' official websites and, to a much lesser extent, travel agencies or personal recommendations. Clusters 2 and 3 display the highest reliance on digital platforms (around 48–51%), suggesting a more digitally engaged and self-directed travel planning behaviour. Cluster 1, while also showing a strong preference for online channels (42%), presents slightly more diversity in platform use, including Airbnb (30%) and hotel websites (14%), which may indicate higher experimentation or comparison across options.

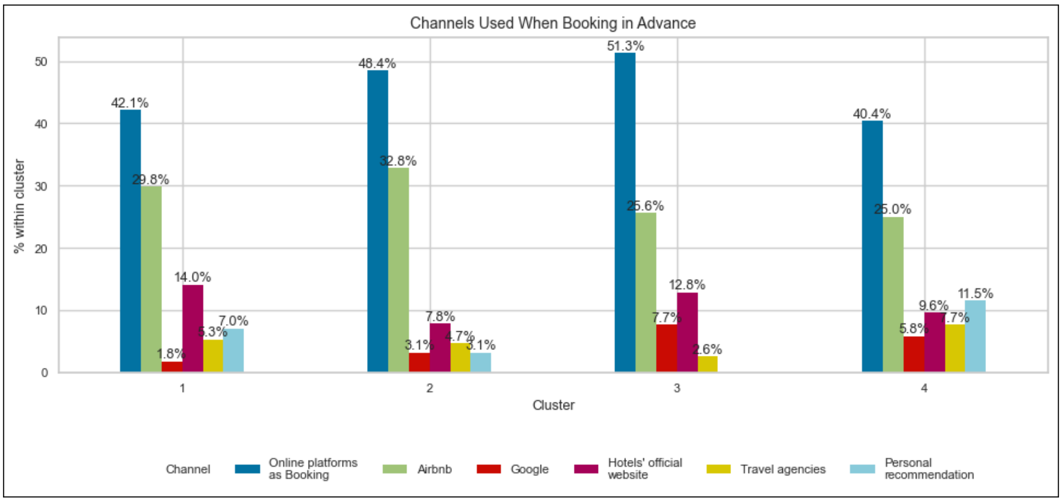


Figure 3.25: *Booking Channels Used in Advance. Source: Author's own elaboration.*

When focusing on last-minute reservations (Figure 3.26), exhibit the strongest relative appreciation, although with a narrower set of channels. In these cases, platforms as Booking and Airbnb again emerge as the leading options, while the relevance of personal recommendations increases slightly, particularly in Clusters 1 and 2, highlighting a tendency towards informal and opportunistic booking behaviours when travel plans are made spontaneously.

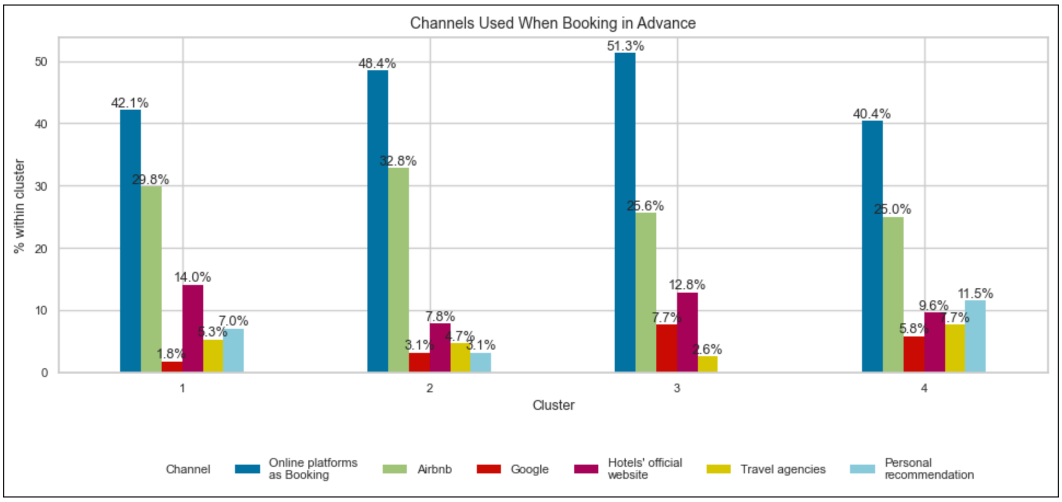


Figure 3.26: *Booking Channels Used Last Minute. Source: Author's own elaboration.*

Cluster Profiles: Willingness to Visit Less-Known Destinations and Extra Travel Time

One of the variables analysed to better understand behavioural differences among clusters was the willingness to visit destinations located outside major tourist circuits. As illustrated in Figure 4.27 (see Appendix AB), the results reveal that travellers' interest in lesser-known cities is closely associated with their tolerance for longer travel times. Nevertheless, this relationship varies considerably across clusters, highlighting distinct motivational patterns among traveller groups.

- Cluster 1 displays a relatively low and heterogeneous willingness, with most respondents scoring between 2 and 3. This group tends to avoid travelling longer distances for this reason.

- Cluster 2 exhibits a more balanced distribution, with moderate willingness levels (mostly between 3 and 4). These respondents appear open to exploring alternative locations but still within manageable travel distances, suggesting a balance between curiosity and practicality.

- Cluster 3 shows the highest willingness to travel longer distances to visit less crowded or lesser-known cities. Median values are above 4 for shorter travel times and remain relatively high even for longer durations, reflecting a strong exploratory orientation and motivation for authenticity and uniqueness in travel experiences.

- Cluster 4 reveals a mixed pattern, with moderate willingness for shorter additional travel times. This behaviour may indicate a preference for alternative destinations only when they are easily accessible or conveniently located.

Cluster Profiles: City-Tour Preferences and Architectural Interest

The analysis of respondents' preferences regarding city tours revealed notable differences across clusters, as shown in Figure 3.27. Overall, guided walking tours stand out as the most popular option, preferred by the majority of respondents in all clusters, particularly in Cluster 1 (75%) and Cluster 2 (72%). This highlights a general interest in immersive, on-foot experiences that enable direct interaction with local environments and communities. Conversely, private tours and tourist trains are less common choices, appealing mainly to Cluster 3, which displays a preference for structured and comfort-oriented activities. The relatively low proportion of respondents choosing bicycle tours indicates that this mode is still a niche option, likely influenced by accessibility and perceived physical effort.

To further explore the motivations of travellers who avoid guided tours, an additional comparison was conducted examining their architectural and cultural interests (see Figure 4.28 in Appendix AC). The results indicate that respondents who do not typically engage in city tours tend to express a stronger interest in historic public spaces and religious architecture. However, it is important to highlight that Cluster 4 exhibits the most diversified architectural appreciation, combining

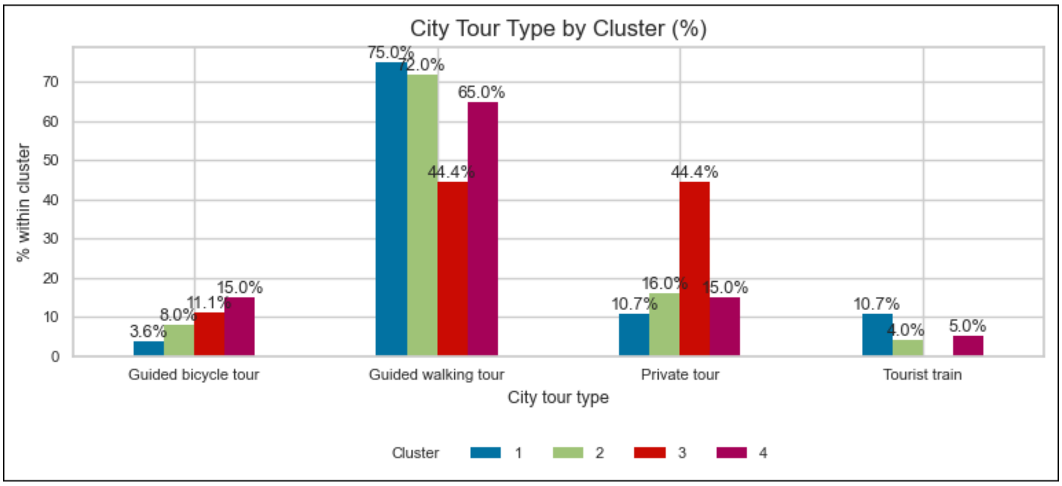


Figure 3.27: *Preferred City Tour Type by Cluster. Source: Author's own elaboration.*

interest in medieval and historical elements with a notable openness towards modern and contemporary structures. Also, Cluster 2 shows greater interest in religious and classical sites, consistent with a more traditional form of cultural tourism.

Cluster Profiles: Restaurant-Choice Behaviour

The analysis of eating preferences across clusters revealed a clear inclination towards less crowded, local neighborhoods as preferred dining areas. Further exploration of restaurant selection criteria (see Figure 3.28) provides additional insights into how respondents make dining decisions. The results show that the most common strategy, especially within Clusters 1 and 3, involves relying on online reviews like Google Maps, indicating the importance of digital validation. Meanwhile, Clusters 2 and 4 demonstrate a higher preference for personal recommendations and on-site decision-making, often deciding while walking by.

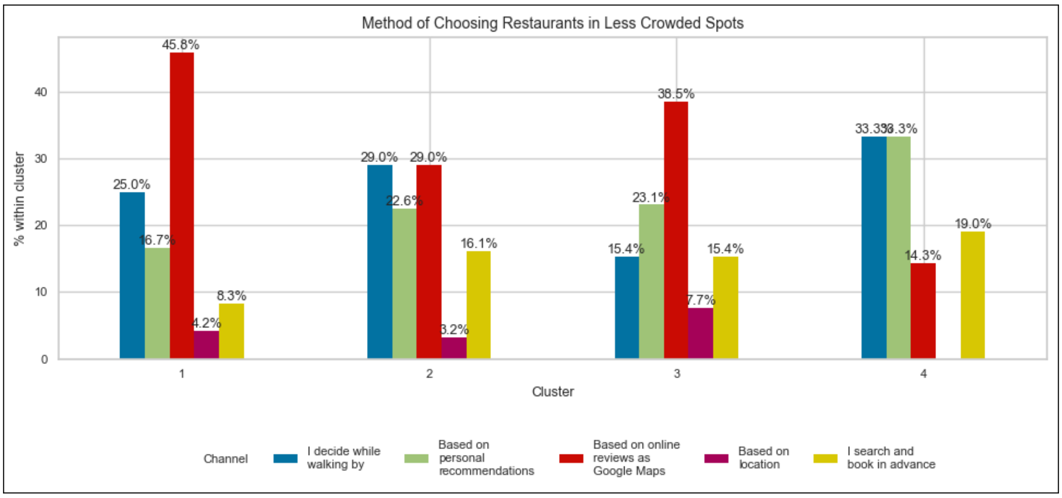


Figure 3.28: *Restaurant-selection Method by Cluster. Source: Author's own elaboration.*

Cluster Profiles: Natural Spots

The analysis of natural environment preferences revealed relatively homogeneous patterns across clusters (see Figure 3.29). Overall, beaches were the most frequently preferred natural setting, selected by approximately one-third of respondents in each group.

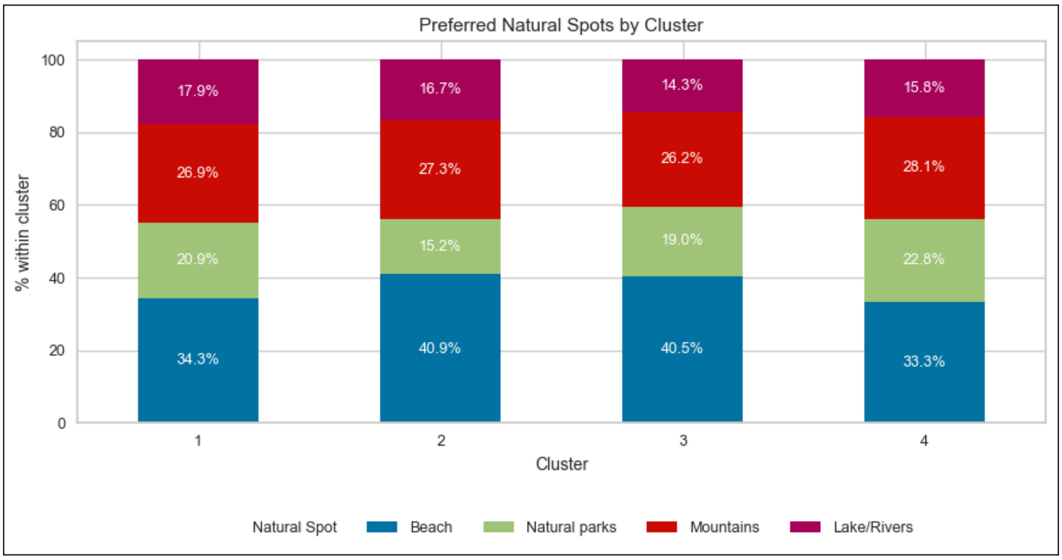


Figure 3.29: *Preferred Natural Spot by Cluster. Source: Author's own elaboration.*

Clusters 2 and 3 showed the highest proportion of beach-oriented travellers (around 41%), followed by moderate interest in mountain and natural parks, indicating a balanced combination of relaxation and outdoor exploration. In contrast, Cluster 1 displayed a slightly greater diversity of choices, with a more even distribution between beaches, mountains, and green areas, pointing to more generalist preferences. Cluster 4, meanwhile, exhibited the strongest relative appreciation for mountain and nature-based experiences.

Summary of Cluster Profiles

Based on the combined analysis of demographic, attitudinal, and behavioural variables, four distinctive visitor segments emerged. Each cluster reflects a different balance between cultural motivation, practical considerations, and experiential expectations. The following profiles summarize the main characteristics of each group:

- Cluster 1 (the culturally-oriented traditionalist):
  - This group consists mostly of female respondents aged 26–35, primarily from South America and with stable employment. They tend to book accommodation in advance, mainly through online platforms such as

Booking.com, reflecting organized planning habits. Their interest lies in heritage and cultural experiences, valuing historic ambience, museums, and accessible city layouts.

  - In terms of gastronomy, they favor less crowded, local neighborhoods and often rely on online reviews to choose restaurants, indicating digital trust combined with a search for authenticity. Their moderate willingness to travel longer distances suggests that comfort and familiarity play a role in their decisions.

- Cluster 2 (the young and explorative planner):
  - This group consists mostly of male respondents, mainly young professionals and students (18–35 years old). This segment is characterised by a strong presence of Italian and European residents. They are highly digitally engaged, often using online booking tools and showing the highest participation in guided tours, particularly walking tours.
  - Their top priorities include cities located outside major tourist circuits and authentic local gastronomy, reflecting curiosity and openness to new places. Although they are price-conscious, as suggested by their Van Westendorp thresholds, they balance budget constraints with a genuine interest in immersive cultural experiences. This group represents the experiential explorer, guided by curiosity and efficient digital habits.

- Cluster 3 (the independent culture-seeker):
  - This group is almost evenly male and female, mainly 26–35 years old and older people (> 55). This cluster values flexibility and independence in trip planning. They are less likely to pre-book accommodation or follow established tourist routes, preferring private or alternative tours. Their high appreciation of contemporary cultural elements (modern architecture, art spaces) differentiates them from the more traditional clusters.
  - In gastronomic behaviour, they rely on online reviews and a have a moderate preference for less crowded areas. Their higher willingness to travel extra time for unique destinations suggests a motivation driven by freshness and cultural meaning.

- Cluster 4 (the comfort-oriented urban visitor):
  - This segment includes a more heterogeneous age range with both young professionals and mature travellers, mainly from South America. They prefer centrally located accommodation and organized environments, often showing lower interest in visiting less-known destinations.

- They still appreciate cultural authenticity, particularly heritage ambience and local dining options, but their choices are marked by comfort and accessibility rather than exploration. They value guided walking tours, and show moderate engagement with natural environments such as mountains or lakes. This group reflects the pragmatic urban tourist, who values cultural identity but prioritises convenience and reliability.

Chapter 4

Discussion

This chapter discusses how the proposed research framework compares with previous work and with the main studies presented in the literature review. It also highlights the strengths and weaknesses of the approach, and on how it contributes to the study of tourist behaviour.

The main goal of this thesis was to create and test a step-by-step methodological process to find the most relevant attributes behind tourist choices, and to check whether these attributes help identify meaningful clusters of tourists. The approach combined qualitative interviews, text mining, an open-ended survey, and a final quantitative survey. The main idea was to improve reliability, reduce subjectivity, and obtain attributes that cities can work with in real situations.

4.1 Comparison to Previous Work

This thesis is a continuation of an earlier project. That project used short and very structured interviews, which limited the discovery of rich patterns in tourist behaviour. Many answers were short, and new or unexpected topics rarely appeared. The defined attributes were too general, such as price or food, making it difficult to understand how tourists actually make choices in a destination. Moreover, most interviews were not recorded, which made it harder to verify interpretations or extract more insights.

In this thesis, interviews were redesigned to encourage participants to describe real travel experiences. Questions avoided suggesting what the researcher expected to hear, and participants were not asked to predict hypothetical behaviour. This approach follows the logic of the Mom Test and helped reduce compliance, offset, and speculation biases. Asking about real trips produced clearer and more useful information. Recording, checking, and translating the interviews also helped limit misinterpretation. In addition, this work applied text mining to the interview transcriptions, using TF-IDF and BERTopic to identify relevant themes.

This strengthened the methodology and supported the discovery of new and more accurate attributes.

In addition, this thesis adds more steps after the interviews. Text mining of Booking and Airbnb reviews supported and refined the attributes using a larger body of spontaneous feedback from real travellers. Then, an open-ended survey, supported by sentence embeddings and TF–IDF, helped check whether other attributes were missing. These steps led to a more solid list of final attributes.

Another improvement concerns clustering. The first project used a small sample and very general attributes, which led to unstable clusters. In the present work, the sample was larger, and a more suitable clustering method (K-Medoids with Gower distance) was used. This method works better with ranking data and mixed variables. The clusters were evaluated using common quality measures, and their interpretation was supported with Random Forest and Multinomial Logit models. This helped understand how characteristics such as nationality or age influence preferences.

Overall, this thesis improves the previous work by using deeper qualitative methods, more structured refinement steps, a stronger clustering approach, and a better explanation of results.

4.2 Comparison to Literature

4.2.1 Qualitative and Quantitative Methods

The first stage of this thesis aligns with the literature, which highlights qualitative interviews as a useful way to identify new behaviours and generate attributes. The interviews allowed participants to describe real travel experiences, revealing priorities and behaviours that are not always visible through closed questions. These early insights helped avoid relying only on assumptions or pre-defined attribute lists found in previous work.

The latter stage, based on the quantitative survey, complemented the interviews by testing the relevance of these attributes with a larger sample. This made it possible to check whether the attributes identified in the whole process were consistent across a broader group of respondents, and whether they could be used to form meaningful clusters. In this sense, the mixed-method approach helped reduce the weaknesses of each individual method: qualitative work may lack generalisability, while quantitative surveys may miss unexpected behaviours if attribute selection is poor. Combining both methods therefore increases reliability and offers a more complete view of tourist preferences.

The literature also mentions several risks that can reduce the quality of interview data. One of them is compliance bias, and it was addressed by avoiding any disclosure of the project's purpose and by formulating questions without references to what other people might do. Interviewees were simply invited to participate in an interview, and once the session began, they were explained only that the topic was tourism before moving directly to the questions. The approach followed the principles of the "Mom Test" described by Rob Fitzpatrick, where participants are asked "questions that even your mom could answer honestly" [12]. Another potential risk concerned offset bias,which was mitigated by avoiding the introduction of unfamiliar terms and by ensuring that the evaluation of a problem remained entirely with the interviewee. Last but not least, to avoid speculation bias, participants were asked about past and present experiences, and to tell "stories" about what they normally do and don't do. In this way, deeper questions were asked progressively and with patience, as individuals are not able to correctly determine the underlying reasons for their own behaviour [12]. It is important to note that this exercise did not involve speculation bias, since participants were not asked to predict hypothetical future behaviours. Instead, they were asked to apply their usual, real-life trip planning process to the case of Ascoli Piceno, following the same steps, tools, and resources they normally use.

In addition, recording, checking, and translating the transcripts also reduced interpretation errors. This transcripts were then analysed using text mining techniques, including TF-IDF and embedding-based topic exploration. This helped identify recurring themes across interviews and served as a second check on the manually extracted attributes. As a result, the final list of attributes was informed by both human interpretation and language models, improving precision and reducing overlap among attributes.

4.2.2 Destination Image Formation

Baloglu and McCleary (1999) show that destination image has cognitive and emotional components. The attributes defined in this thesis mostly reflect the cognitive part, such as museums, heritage, accommodation, and proximity to nature. These attributes refer to tangible or factual elements of a destination and can often be influenced directly by destination managers. The emotional or affective side was not studied directly. Including emotional elements could offer new insights, especially because feelings such as excitement, comfort, or safety are important parts of destination image.

Measuring the affective dimension would require asking respondents to evaluate how a place makes them feel using emotional scales or sentiment analysis. This was outside the scope of the present survey. Nonetheless, emotional elements did appear indirectly in some qualitative responses, for example when participants described enjoyable or disappointing travel experiences. Future applications of this framework could include affective questions to capture this dimension more explicitly.

4.2.3 Segmentation

Pesonen and Honkanen (2014) warned that segmentation based on rating scales alone can be distorted by response styles, such as always choosing high or low values. To reduce this problem, the present work asked respondents to rank attributes instead of relying only on Likert scales. Ranking forces respondents to prioritise, making it harder to give the same answer to everything. This reduces the chance of forming artificial clusters driven by response patterns rather than true preferences. The use of Gower distance with K-Medoids also helped produce more realistic clusters. Gower distance is appropriate for ordinal and mixed data, while K-Medoids is more robust to noise than K-Means.

Wu, Zhang, and Fujiwara (2011) model travel decisions as a sequence. This thesis did not attempt to represent the order in which decisions are made, since doing so would require a more complex structure such as a nested model. The Multinomial Logit model used here focused instead on how socio-demographic and behavioural variables relate to preference groups. Although this model does not capture decision order, it still provides useful insights on the characteristics that distinguish clusters.

If this framework is expanded in the future, latent-class modelling could help capture unobserved heterogeneity by identifying hidden groups of tourists. Combined with a nested structure, such an approach could also represent different decision sequences across segments, extending the scope of the present analysis.

4.2.4 Discrete Choice and Conjoint Analysis

Discrete choice experiments and conjoint analysis are widely used to estimate preferences and willingness to pay. Both methods require defining a structured set of attributes and levels, building efficient choice sets, and analysing the resulting data with specialised models. Because of their design complexity and the time needed to test and refine attribute levels, they were not applied in this thesis.

Nonetheless, the methodology developed here can be interpreted as a preparatory stage for such experiments. The definition of attributes was not assumed a priori but was built through an iterative process combining qualitative interviews, text mining, and two survey stages. This approach helps reduce one of the recurring challenges in conjoint studies: choosing attributes that are too broad, overlapping, or poorly grounded in real behaviour. By narrowing the list to ten well-defined and actionable attributes, the present framework provides a stronger foundation for designing future conjoint tasks.

Another advantage is that the survey results indicate how respondents prioritise these attributes and how they cluster into different preference groups. These insights can guide the selection of attribute levels and help ensure that future conjoint experiments test realistic trade-offs. For example, results showed that accommodation services, proximity to nature, and attraction density were particularly relevant for differentiating clusters. These attributes could therefore be central components in a follow-up discrete choice or conjoint design.

While no willingness-to-pay estimates were derived here, the Van Westendorp analysis shows that price is meaningful in this context and could be included as an attribute level in future experiments. A conjoint design would allow researchers to quantify price trade-offs more precisely and understand how much travellers value specific improvements in attributes.

4.2.5 Hypothetical Bias

Hensher (2010) explains that willingness-to-pay estimates can be affected by hypothetical bias. In this thesis, the Van Westendorp method was used only to explore acceptable price ranges, not to produce monetary valuations. For this reason, no mitigation strategies (such as scripts or oaths) were applied. Future research could add such treatments to obtain more reliable estimates.

4.2.6 Text Mining

Several studies show that text mining is useful for discovering new attributes. This thesis used TF-IDF and sentence embeddings to support interview results. Unlike some recent studies that rely on deep learning models, this thesis prioritised interpretability and reproducibility. While more advanced models could reveal more complex relationships, the simpler techniques used here are easier for practitioners to understand and replicate. Access to additional data sources, such as paid APIs, could further improve attribute extraction.

4.3 Methodological Implications

The main contribution of this thesis is its step-by-step structure. Each stage complements the others: interviews help build the first list of attributes, text mining refines that list, and surveys test whether the attributes explain real behaviour. Using ranking questions, Gower distance, and K-Medoids made segmentation more coherent. Supporting clustering results with Random Forest and Multinomial Logit models increased interpretability. This combination of qualitative and quantitative approaches shows that it is possible to extract actionable attributes and validate them with data. The process is flexible and can be adapted to other destinations.

4.4 Limitations

This study presents several limitations that should be considered when interpreting the results. Although the mixed-method design supported a solid definition of attributes, there were practical and methodological constraints that affected data collection and analysis. The main limitations are:

- The final sample, while reasonably aligned with the intended target, did not reach a broader audience. A larger and more diverse sample would improve external validity.

- Access to some relevant text sources, sucha as TripAdvisor, was not possible due to API costs, reducing the amount of review data.

- Attribute refinement could have benefited from additional rounds of open-ended surveys.

- The final survey did not include some potentially useful socio-demographic variables, as household size, marital status, travel budget, among others.

- Conjoint or discrete-choice experiments were not performed, so willingness to pay or trade-offs between attributes were not estimated.

- Predictive modelling was exploratory and did not focus on destination specific outcomes such as likelihood of visiting or recommending. Nevertheless, some questions were included in the final survey as an example of how it could be used this methodology.

4.5 Future Directions

Based on the limitations described above, several opportunities for further development can be identified. These directions could help strengthen both the methodological framework and its practical applications. Potential improvements include:

- Conducting a conjoint or discrete-choice experiment to quantify attribute trade-offs and willingness to pay.
- Introducing additional rounds of qualitative or open-ended surveys to refine and validate attributes.
- Expanding text sources and collecting multi-language responses to improve representativeness.
- Applying latent-class or hierarchical models to capture deeper heterogeneity between tourist groups.
- Developing predictive models that focus directly on destination outcomes, as likelihood of visiting, recommending, or returning.
- Adapting behavioural questions to destination-specific contexts to support practical decision-making.

Conclusion

The results of this thesis show that the proposed methodological framework is effective and reliable for discovering, refining, and validating destination attributes. The different stages of the process (qualitative interviews, text mining, open-ended analysis, and quantitative testing) converged consistently, producing a stable set of attributes and a clear segmentation structure. This confirms that the method works as intended: it reduces subjectivity in attribute discovery, increases transparency in the refinement process, and provides attributes that remain valid when tested quantitatively. Moreover, the stability of the final attributes suggests that the framework is suitable for replication and that it can be further strengthened in future applications through conjoint or discrete-choice experiments, which would allow more precise estimation of trade-offs and willingness to pay.

The framework proved effective for three main reasons. First, the combination of qualitative interviews and automated text-based techniques reduced subjectivity and allowed the detection of meaningful patterns that manual coding alone would have missed. Second, the attributes derived through this process showed strong discriminant power in the quantitative stage, as evidenced by the clustering analysis, Random Forest results, and the Multinomial Logit model. Third, the final attribute set remained stable across different analytical tools, which indicates internal consistency and supports the reliability of the method.

From a theoretical perspective, this thesis contributes to the literature in three main ways. First, it strengthens the early stages of attribute discovery by integrating automated text-based techniques with qualitative interviews. Previous studies often rely solely on manual coding or predefined attribute lists, which can introduce subjectivity or overlook emerging patterns. By combining TF–IDF, topic exploration, and interview analysis, this work provides a more systematic and replicable way to identify meaningful destination attributes.

Second, the thesis shows that mixed methods can improve segmentation quality. Ranking-based clustering using Gower distance and K-Medoids produced four consistent tourist segments, overcoming the issues reported in earlier work where clusters were unstable due to broad attributes and limited sample size. Supporting the segmentation with Random Forest and Multinomial Logit models adds transparency, showing which attributes and socio-demographic variables matter the most in defining each group.

Third, the study highlights the value of linking attribute discovery with behavioural variables. By analysing information search, accommodation behaviour, and willingness to visit less-known cities, the thesis provides a more complete view of how tourists interpret attributes in real situations. This complements previous segmentation studies that often rely only on attitudinal or rating-scale data.

The framework also offers practical implications for destination managers and tourism organisations. The resulting attribute set is concise, actionable, and grounded in real behaviour rather than assumptions. Managers can use these attributes to understand what different segments value the most. In particular, the findings indicate that accommodation services, attraction density, and proximity to nature have a strong influence on tourist preferences.

The segmentation results can support strategic decisions such as communication targeting, product development, or the design of visitor experiences. For instance, the most exploratory cluster shows openness to less-known destinations and longer travel times, while other clusters value comfort, accessibility, or cultural depth. Understanding these profiles can help destinations diversify their offer and attract different types of visitors. The proposed process also serves as a template that other destinations can replicate, adapting it to their local context with minimal adjustments.

Beyond the methodological contribution, the empirical results show that the process effectively identified ten attributes that are both meaningful and discriminant for segmentation. The clusters differ in their cultural interests, practical behaviour, and willingness to explore alternative destinations. The Random Forest results confirm that accessibility, accommodation services, and natural surroundings are key drivers of differentiation. The Multinomial Logit model shows that age, gender, residence, and cultural orientation significantly influence cluster membership, meaning that socio-demographic factors still play an important role in shaping preferences. The Van Westendorp analysis adds an initial view of price perception, showing that most respondents consider €200–€250 a reasonable price for the proposed experience, although each cluster displays different levels of price sensitivity. These findings suggest that the attribute set can support more advanced valuation studies, such as conjoint or discrete-choice experiments. Taken together, the results indicate that the framework achieved its purpose: it produced a validated and interpretable segmentation based on attributes extracted through a transparent, multi-stage discovery process.

Compared with previous work, this thesis addressed several limitations. The interview process was redesigned to reduce bias and improve reliability, the attribute extraction combined qualitative and text mining techniques, and the quantitative stage incorporated more robust analytical tools. These improvements resulted in a more coherent and replicable framework. While the study was constrained by time, resources, and sample size, it provides a strong foundation for future applications and further methodological refinement.

The study also reveals several opportunities for future research. A structured agenda includes:

- Additional rounds of qualitative or open-ended surveys to refine and test attributes over time and across destinations.
- Expansion of text sources, including multi-language reviews and social media data, to improve representativeness and capture cultural differences.
- Conjoint or discrete-choice experiments to quantify trade-offs and willingness to pay using the validated attributes.
- Advanced segmentation techniques, such as latent-class models or hierarchical clustering, to capture hidden behavioural patterns.
- Destination-specific behavioural modelling, including likelihood of visiting or recommending, to support operational decision making.
- Inclusion of emotional or affective dimensions in future attribute discovery, complementing the cognitive focus of this work.

In conclusion, the thesis achieved its main objective: to design and test a structured process for attribute discovery and validation that enhances the understanding of tourist behaviour. The results show that integrating text mining techniques with qualitative interpretation and quantitative testing leads to more accurate and actionable findings. Future applications can build on this foundation by expanding data sources, refining attribute testing, and introducing conjoint or discrete choice experiments to measure trade-offs and economic value. Ultimately, this approach contributes to more data-driven and evidence-based decision making in tourism marketing and destination management.

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Appendix

A. Example of Qualitative Interview from Previous Project

Interviewee's information: 27 years, Female, MSc student.

Interviewer: What are the main characteristics you consider from a place, when deciding your holidays?

Interviewee: I consider mainly the season of the place I'm going. Also I prefer to go to new places, rather than already known ones. Depending on my budget, I'll look a place abroad or the country in which I'm living.

Interviewer: What makes a city unique/different from another, for you?

Interviewee: For me, the experience/tours that the city offers. Historical monuments that I can visit are also important. Taking into account this, also the transportation available to move around and get to different points of it. I also like the food experiences that I may have, so I'll look for specific dishes of the city I'm going to.

Interviewer: Do you prefer being autonomous when looking for things to do/see or rely on agencies and organize activities by a third party?

Interviewee: Depends, if I'm doing a short trip, I would rely on an agency because I have very short time to go to places, so if it is already organized it would be better. If it is a long trip, I prefer to be into the culture of the place, going around, loosing myself in the city, so I prefer to be autonomous. Also, if my trip includes more than one city, I usually search some combo of more than one city tour or experience. For example, in Naples I saw some ship experience that takes you to different Islands, or I know that Firenze and Pisa have some convention that sells some pack that includes both cities.

Interviewer: Do you choose tourists packs (accommodation, meals, transport, activities included) or plan out everything yourself?

Interviewee: I plan where to sleep and nothing else. When I'm in the city, I start searching for packs that it can provide me. I look for packs and guides that offer me some city tour, get to know more from inside the city, recommendations from a local person.

Interviewer: When choosing a spot, do you research its popularity and then decide whether to go or not?

Interviewee: No, if it is a place I want to go, it doesn't matter if other people go or not.

Interviewer: Give me three characteristics that your vacation spot must have.

Interviewee: History/Traditions, Food, Accessibility/Transport.

B. Example of Qualitative Interview from Actual Thesis

Interviewee's information: Italian, male, 24 years old.

[Interviewer] Can you describe your last vacation? Not so much what you did while you were on vacation, but how you organized it? Starting from home, deciding to go on vacation and the whole planning phase.

[Interviewee] What I usually do is open... I don't know if you know Notion. Yes, it's a note-taking app, let's say, in an organized way. And I create a file for the trip I want to take—in this case, it was Rome. And I write down day by day what I want to do. To do that, I do research online, maybe look for ideas from other itineraries I can find. And then I adjust it based on how much I want to walk, what I want to see, what I'm interested in. But more or less that's the approach, and it also depends on price, how to get there, if I can go by public transport or not, or if I can drive there...

[Interviewer] Describe in detail this Rome trip. So how many days you stayed, what you did the first day, where you found the information... Actually, if you can share your screen, maybe we can see exactly... practically how you searched, also for sleeping arrangements. To see the other options you mentioned, it's really useful. Thank you.

[Interviewee] So, this is the schedule. This is the trip to Rome. It says seven days here, but I think it was actually eight, right? Seven, okay. So for each day there's also the arrival date. In the end I drove, so I didn't have to worry about transportation. As for attractions in Rome, it wasn't very difficult, I mean... it's a very famous city... there are lots of places one can go, so I just tried to make the itineraries a bit organized, so the Pantheon is not too far from Campo...

[Interviewer] So you went by memory for the Rome attractions? You didn't have to search what to see in Rome?

[Interviewee] No, I searched for that too. I looked up "one-week itinerary in Rome". And I compared various ones, but I don't remember exactly which itinerary I used. But I searched, opened Google, looked up ”Rome itinerary”, or another thing I did was ask ChatGPT to help me compare them.

[Interviewer] So to give you the actual itinerary and maybe see what was closer, what was farther?

[Interviewee] Yes, even though I must say that when it comes to geographical distance it’s not very good?

[Interviewer] Oh no?

[Interviewee] I didn’t really find it helpful. Sorry, do you hear the noise? Can you hear it?

[Interviewer] No, we don’t hear anything.

[Interviewee] Perfect. Yes, like if you tell it “make me an itinerary considering distance,” sometimes it gets it wrong.

[Interviewer] It makes a few blunders.

[Interviewee] Exactly. But if you give it itineraries that already consider distance, the ones written by people who actually took the trip and maybe give you tips, then I find them useful—you can give those a try.

[Interviewer] Okay, so you searched on Google and what did you find?

[Interviewee] Well, usually blogs. I usually go to independent blogs, so travelers—maybe families who like to travel and write—or posts on Reddit.

[Interviewer] Families, but were you traveling with family, with friends, alone?

[Interviewee] A friend who hosted me.

[Interviewer] Okay, so maybe it didn’t feel strange to take advice from families? Or maybe you thought: ”Well, in the end, the things to see...” Or maybe you were more interested in advice from people your age or travel style?

[Interviewee] Let’s say what I wanted to do in Rome was a city visit, the cultural side. See monuments, maybe go to a few museums, so I think that kind of trip is very popular, also with tourists from abroad. Maybe, I don’t know, nightlife, stuff like that—obviously I would’ve looked for other kinds of blogs, I would’ve looked specifically...

[Interviewer] You said independent blogs—why independent? What do you mean? Not affiliated with... tell me, tell me.

[Interviewee] Usually agencies charge money. It’s a bit tricky—maybe you find fairly generic itineraries on agency websites. I had to find it myself. Then, I don’t know, I get a bit... how to say... maybe unfairly... when I realize it’s an agency, or maybe even if it’s a supposedly independent blog, when I feel like maybe they’re recommending a place not because they genuinely liked it, but maybe because they’re getting paid, or they get commission for sending people there. It’s not necessarily a bad thing.

[Interviewer] But it puts you in a different mindset, you’re saying?

[Interviewee] I sense a bias from them.

[Interviewer] Sure. Listen, do you ever get inspired by images? Do you search on social media through hashtags or things like that?

[Interviewee] I don't think I've ever used hashtags. I don't even know if people use them. Like I don't go looking for them—maybe it's happened... not so much for Rome, but maybe for mountain hikes or things like that where you get a view... it shows you the scenery. So yeah, it's happened, but I didn't look for it on purpose—it's more something that just popped up, so I got curious and maybe decided to visit it.

[Interviewer] Okay. So you're saying if it pops up in front of me, sure, but it's not a method I use to search—right? Now try to make a little memory effort if you can. Try to imagine when you were younger and maybe went with your parents. Did you ever go somewhere with them? Can you remember how the process worked—maybe your mom or dad took care of it, maybe they went to a travel agency... I don't know, can you remember anything, even vaguely? Obviously not the whole process but anything at all?

[Interviewee] I think so—like yes... but I was really little. They probably relied on agencies. Then there was a period where we did another kind of trip, and it was more based on word-of-mouth. My father also... I mean he had internet access quite early on compared to... like in the early 2000s I think it wasn't that common yet. And even then, back then there were blogs too, though fewer than now. I think he did more or less what I do now, but with fewer tools. But he probably also relied a lot on personal word-of-mouth. Advice from a friend who had the same passions. But yes, surely for a couple of trips they used agencies, though I think only a few.

[Interviewer] Okay, now I'll have you play a role. If you can share your screen again. I'll give you the name of a city. Imagine we're friends and I suggest this city to you... and you show me what process you would follow to decide if it's a place to go or not, if you would go or not. What would you search, what would you look at—photos, text, you tell me. I give you the city name. We're friends and I say "Hey, I think a good place to visit is Ascoli Piceno."

[Interviewee] How many days?

[Interviewer] I don't know. You decide.

[Interviewee] What to see in five days. So... I'd go by car because in winter with transportation... driving is best. So what I would do is check out all these websites I saw. No, this one for example I wouldn't look at... not right now at least.

[Interviewer] Say everything out loud—even what comes to mind. Like "No, ugly," "Nice," "This is boring"—tell me everything that pops into your head. I'm the voice in your head—go, speak!

[Interviewee] Like TripAdvisor! It's really useful in my opinion, maybe for finding... I don't know... I usually use it more for checking if a restaurant is good, if it's rated well. I'd close it for now in this phase because I wouldn't use it. This site though, which reviews places, tells me opening times, if you need tickets... this is the kind of site I'd look at. Same here, I imagine they're doing about the same thing. And here's ChatGPT giving me more information. And what I would do then... you can't see because I didn't share my screen properly—I only shared the page—but I'd reopen the Notion file from before and start writing notes.

[Interviewer] Okay, but after this quick glance... would you say "Alright, let's go, let's do this trip," or "No, boring, I'll look for something else"?

[Interviewee] Ah, for Ascoli Piceno in particular?

[Interviewer] Yes, see if I convinced you. I threw it out there—we're friends and now you have to decide.

[Interviewee] Then I also need to check one more important thing.

[Interviewer] Go ahead, tell me.

[Interviewee] How long it takes to get there. Everything counts, so... yeah, it takes a while!

[Interviewer] Have you ever been there? Have you ever looked around that area? Have you ever been in those parts?

[Interviewee] I was in the car a lot as a kid, but I was in a seaside area. So, let's say... the first thing I think is that I feel kind of in the middle of nowhere. A positive thing in terms of, maybe, if you're interested in hiking—though it's not even easy to find someone willing to do it—if I were interested in that type of excursion, I'd say let's go. One element that makes me think 'maybe I shouldn't go' is that the town of Ascoli Piceno seems like a fairly small city, and the monuments don't seem unforgettable. So, usually... when do I go to that kind of little village? When there are a lot of them around. So maybe I can go to Ascoli Piceno for a couple of days: visit it the first day, do some hikes nearby on the second day, and then go somewhere else on the third. This type of trip would require a lot of time to organize, also because for sure there are far fewer websites that recommend these kinds of trips.

[Interviewer] What do you mean by "there are fewer websites that recommend these kinds of trips"?

[Interviewee] Not necessarily that there are ones that discourage these kinds of trips—not in the sense of being positive or negative. Rome, Ravenna... these are more popular places, so it's easier to find more websites of that kind.

[Interviewer] And Ascoli instead... a small village, a little town near other little towns—do you imagine it more or less like that?

[Interviewee] I would only go if I managed to organize—another trip comes to mind that I planned, maybe not on Notion... maybe it's somewhere on Drive—in Tuscany, where we did a road trip.

[Interviewer] Kind of on the road, going where it happens.

[Interviewee] Exactly, and of course we visited the main cities, the ones you usually include as tourist destinations—Pisa, Florence... but we also went to Livorno, for example, which I would have never visited intentionally if I hadn't been on a road trip.

[Interviewer] So you passed by, saw Livorno and said, "okay, let's stop," right? Not that...

[Interviewee] We organized it specifically to see as many cities as possible in Tuscany. But if someone had told me, "Let's go to Livorno"... I would have said absolutely not. We went to Livorno because it was part of the bigger Tuscany trip... yes.

[Interviewer] And then you went to Livorno—did you like it?

[Interviewee] No, not really. I don't think it's... I mean, we went, we stayed for a few hours, because we didn't feel like there was much else to see. We went to the seafront.

[Interviewer] Continuing that Tuscany tour, where else did you go besides Florence, Lucca, Pisa? Besides Livorno?

[Interviewee] We went to Rosignano Solvay, for example. The white beach... we wouldn't have gone there on purpose, but since we were nearby, it was interesting.

[Interviewer] For example, Rosignano—the beach of Rosignano. Did you discover it while already there by doing a search, or did you look it up before leaving?

[Interviewee] Before leaving, we had the entire itinerary.

[Interviewer] The whole itinerary planned beforehand. And how did you find Rosignano? Did you search something like "beaches," "unforgettable places," or was it suggested by someone?

[Interviewee] Now I couldn't tell you the first time I heard about it. But it's very interesting because the beach turned white due to industrial waste dumped into the water. So I remembered it, and since our goal was a tour of Tuscany, I said, "Let's pass through there, and have a beach day."

[Interviewer] While organizing this Tuscany tour, did you discover anything on blogs—something you didn't know before, but after seeing the pictures, felt inspired to include?

[Interviewee] Yes, yes. There are various little Tuscan villages, I forget their names now... I'm terrible with names.

[Interviewer] Don't worry about it.

[Interviewee] There are these tiny villages, unreachable by public transport. I probably wouldn't have gone from Bergamo to that place. But they were beautiful places. Even within the overall Tuscany trip, they were a great added value—also inland.

[Interviewer] And you chose them based on images, or reviews you read?

[Interviewee] Both... beautiful images—the Tuscan inland is very scenic, anywhere you go you find beautiful things—and also based on reviews from people who recommended them. So even there, we...

[Interviewer] Listen, I have to ask—since you were in Tuscany, you definitely ate well. I have no doubt! How did you find where to eat? Or did you just stop wherever? Like, "I want to eat Florentine steak."

[Interviewee] Yes, for example, the Florentine steak... since I have friends who've traveled and some who live in Tuscany, I got some advice from them. Then sometimes we just stopped wherever. In those villages, there are very few places anyway—we just sat down and that was it. Other times we searched...

[Interviewer] Have you ever gone to a place specifically just to eat something?

[Interviewee] Yes, for example, there was this restaurant someone recommended to eat Florentine steak at a decent price. And we went there on purpose.

[Interviewer] Since you're talking about price—going back to Ascoli, from the two images you saw earlier... would you expect it to be more expensive, cheaper, or the same as, say, Ravenna?

[Interviewee] That's a good question... well... since I'd be going by car...

[Interviewer] Show me an example of how you'd estimate costs—also for accommodation. I don't know, would you look on Airbnb, Booking... Show me how you'd roughly estimate the costs.

[Interviewee] Usually the first thing I do is go on Airbnb and Booking. I prefer Airbnb for these types of trips, because usually I travel with a group and we like to have one meal out and one at home. The nice thing about going by car is that you don't need to book in the city center...

[Interviewer] Open Airbnb and show me—you'd look at the Airbnb map, right? Because that gives you a better idea of prices in the area since, as you said, you can drive around.

[Interviewee] Exactly. A bit outside, for example here I see 42 euros per night. And yeah, what I'd do is say, "okay, that's the price." It's true a lot is already booked.

[Interviewer] The fact that it's mostly booked—does that give you a positive or negative impression of the place?

[Interviewee] About the quality of the place—it tells me nothing. But in itself, it gives me a negative feeling because I think, if I see that many places are already booked, the risk is that what's left might be quite expensive. So it might be hard to find good prices.

[Interviewer] So, looking at these prices—you put the end of August, right? With these prices, would you think, "No way, it's too much"? You put three people... no—five people for four nights.

[Interviewee] No, I'd have to check what that place looks like. 35 euros per night, this one, which is the cheapest per person.

[Interviewer] Stefano, I was just saying thanks because you're dedicating so much time to this—really, thank you. Very kind. So you were saying this is more or less the price. And for food, you said you'll decide once there, right? I think you said you do Airbnb also to optimize the fact that sometimes you eat in, sometimes out?

[Interviewee] Yes, though... I mean... it's not even guaranteed. It depends a lot. For example, I might consider staying near a place and only need a few hours to visit it all. So maybe I just pass through...

[Interviewer] You wouldn't even stay the night, you mean?

[Interviewee] No, I'd go somewhere else—toward the next city.

[Interviewer] Sure, absolutely, okay. So, imagine a nearby city—would you go on Maps... what would you do to see what else to pair with it?

[Interviewee] I'd look it up on Google. For example, for these types of trips... just for no particular reason... maybe "what to see in the Marche," or in Abruzzo too—maybe we could combine those two regions and decide based on that...

[Interviewer] ...based on a mix of things. So not a single city. Maybe a city fits in because you associate two or three nearby things and say, "Hey, since we're here and it takes just a few hours to see it, let's go." Right? More or less?

[Interviewee] Exactly. I wouldn't plan a trip just for a city like that.

[Interviewer] Let me ask you a couple of direct questions about the city. Have you ever heard of olive all'ascolana? You know what they are? Have you eaten them?

[Interviewee] I've eaten them, even from the place itself.

[Interviewer] So you know they're from Ascoli? That they're called Ascolane because of the city? Okay. Oh, you ate them from the place—maybe when you were a kid on holiday?

[Interviewee] No, someone brought them to me.

[Interviewer] Ah, someone brought them—like a friend from there? Someone from the city?

[Interviewee] A friend from the Marche region.

[Interviewer] Have you ever heard of the Quintana?

[Interviewer] Okay. Can you quickly look it up on Google and tell me what comes up and what you think it is? Like if it sparks any thoughts.

[Interviewee] I don't know… it makes me think of something like the Palio di Siena, in the sense of: a tradition handed down from the past, something that's stayed around. Then the horses—it seems like a game...

[Interviewer] Is it something that might interest you to see? Or do you absolutely not care?

[Interviewee] Yes, I'd be interested. I think… in that famous Tuscany trip, we also included the Palio di Siena… We even planned it that way—we timed our Siena visit for when the Palio was happening.

[Interviewer] And how did you find out it was during the Palio? I doubt you knew the dates by heart—I don't even know them.

[Interviewee] No, let's say that when…

[Interviewer] So when planning the Siena part, you remembered the Palio and thought "Let's see when it is." You already knew about the Palio, it wasn't something you had to look up? You just knew it.

[Interviewee] Yes, I knew it.

[Interviewer] Alright. Stefano, thank you. You've been really, really kind. Thank you so much for also sharing your screen and showing everything. I really appreciate the chat. Thank you so much.

[Interviewee] No problem. Thank you guys.

[Interviewer] Thanks a lot. Bye everyone. Franco, I'll text you on WhatsApp.

C. Qualitative Interview Notes

Interview 1 (Male, Italian, 24 years)

- Some important points, and insights:
  - "...look for ideas from other itineraries...". **ONLINE RECOMMENDATION**
  - "...it also depends on price, how to get there, if I can go by public transport or not, or if I can drive there..." . **COST TO REACH, ACCESIBILITY BY PUBLIC TRANSPORT**
  - "As for attractions in Rome, it wasn't very difficult, I mean... it's a very famous city... there are lots of places one can go...". **HISTORICAL ATTRACTIONS**
  - "...opened Google, looked up "Rome itinerary", or another thing I did was ask ChatGPT to help me compare them". **AI RECOMMENDATION**
  - "I usually go to independent blogs...". **ONLINE RECOMMENDATION**
  - "...city visit, the cultural side. See monuments, maybe go to a few museums...". **CULTURE , MUSEUMS, HISTORIC MONUMENTS.**
  - "...I feel like maybe they're recommending a place not because they genuinely liked it, but maybe because they're getting paid...". **GENUINE ONLINE RECOMMENDATIONS.**
  - "...it was more based on word-of-mouth.". **PERSONAL RECOMMENDATION**
  - "Like TripAdvisor! It's really useful in my opinion..." . **ONLINE RECOMMENDATION**
  - "This site though, which reviews places, tells me opening times, if you need tickets... this is the kind of site I'd look at.". **INFORMATIVE PAGES**
  - "How long it takes to get there.". **DISTANCE TO REACH, TIME TO REACH**
  - "A positive thing in terms of, maybe, if you're interested in hiking ..." **HIKING, NATURAL LANDSCAPE**
  - "...Ascoli Piceno seems like a fairly small city, and the monuments don't seem unforgettable.". **CITY SIZE, HISTORICAL MONUMENTS**
  - "When there are a lot of them around.". **NEAR TO OTHER TOURISTIC DESTINATIONS**
  - "Rome, Ravenna... these are more popular places, so it's easier to find more websites of that kind.". **INFORMATION AVAILABILITY**
  - "...but we also went to Livorno, for example, which I would have never visited intentionally if I hadn't been on a road trip.". **ROAD TRIP**
  - "We went to the seafront.". **NATURAL LANDSCAPE**
  - "...and also based on reviews from people who recommended them.". **ONLINE RECOMMENDATION**
  - "...someone recommended to eat Florentine steak at a decent price.". **FOOD'S PRICE, LOCAL CUISINE**
  - "...I prefer Airbnb for these types of trips, because usually I travel with a group, and we like to have one meal out and one at home.". **ACOMMODATION TYPE, AVAILABLE RESTAURANTS**
  - "The nice thing about going by car is that you don't need to book in the city center.". **CITY'S ACCESIBILITY**

    - "…in that famous Tuscany trip, we also included the Palio di Siena…". **LOCAL EVENTS**

Interview 2 (Male, Argentinian, 25 years)

- Some important points, and insights:
    - "…right now going to the Argentine coast, with how expensive it is…". **PRICE**
    - "…how hard it is to get around, especially during New Year's—it's much harder to find good lodging at a fair price.". **ACCESSIBILITY, PRICE**
    - "On one hand, TikTok—honestly, there's a ton of tourist information there, not so much about prices or packages, but more from people sharing their experiences. And on the other hand, I also use recommendations from friends…". **SOCIAL MEDIA RECOMMENDATION, PERSONAL RECOMMENDATION**
    - "…we looked a lot into what the transportation is like there, food prices, which beaches are recommended.". **PUBLIC TRANSPORT, FOOD PRICE, BEACHES**
    - "What interests me most about Italy is the history, and of course everything that comes with it—architecture, culture…". **HISTORIC PLACES, ARCHITECTURE, CULTURE**
    - "…I always look for places that have either beaches or nature." **NATURAL LANDSCAPE**
    - "If I don't know anyone, I usually go to TripAdvisor.". **PERSONAL RECOMMENDATION, ONLINE RECOMMENDATION**
    - "I feel like there are lots of content creators who, in one minute, give you a good summary of the destination's essence…". **ONLINE RECOMMENDATION, CONCISE RECOMMENDATION**
    - "And sometimes Airbnb…I feel like it gives you more of a sense of "hey, I'm actually living here,"…". **LOCAL EXPERIENCE, ACOMMODATION'S SERVICES**
    - "…when it comes to accommodation, what I really look at is three things: location, price, and then what the place offers.". **ACOMMODATION LOCATION, PRICE, LOCAL EXPERIENCE**
    - "…when planning a trip to Italy, one wants to visit as many cities as possible.". **NEAR TO OTHER TOURISTIC DESTINATIONS**
    - "…I like that it has the Tronto River running right through it.". **NATURAL LANDSCAPE**
    - "…food tourism spots with more traditional dishes…". **LOCAL CUISINE**
    - "…opening Google, searching for restaurants, reading reviews, ratings, opinions, checking menu photos.". **GOOGLE REVIEWS**
    - "The fact that it's close to Florence, which is a place I really want to visit.". **NEAR TO POPULAR CITIES**

Interview 3 (Male, Argentinian, 31 years)

- Some important points, and insights:

- “The way a trip starts is by thinking about the cost to get to that place, or how much it costs.”. **COST TO REACH**
- “...I look more at the distance...”. **DISTANCE TO REACH**
- “Hotels, I usually go to Booking.com, because I’m more into hotels. I prefer hotels over Airbnb because of the built-in conveniences.”. **ACOMMODATION TYPE, ACOMMODATION’S SERVICES**
- “I’m all in on ChatGPT for itineraries.”. **AI RECOMMENDATION**
- “So I’d look up... literal ways to get to Ascoli.”. **POSSIBILITIES TO REACH**
- “I’d also ask people I know or places like that.”. **PERSONAL RECOMMENDATION**
- “Something I don’t do, but my wife does, is travel agencies.”. **TRAVEL AGENCIES**
- “Also just to get a price range based on the service quality too...”. **ACOMMODATION PRICE, ACCOMMODATION QUALITY**
- " Based on a mix of days where I’d prioritize... I don’t know... Museums and culture, and others where I’d prioritize nature and gastronomy.”. **MUSEUMS, CULTURE, NATURAL LANDSCAPE, CUISINE**
- “But it seems like a typical place where you’d want to set up a base and then do some kind of excursions to nearby spots.”. **NEAR TO OTHER TOURISTIC DESTINATIONS**
- “Look, for example, 27 minutes from Ascoli Piceno to the coast. I want to believe the coast must be beautiful just because it’s Italy. So here, it’s very likely I’d want to do this kind of excursion.” **BEACHES**
- “It was such a tiny place that it wasn’t prepared for people to go by car...”. **CAR FIRENDLY**
- “First and foremost is Google Maps. Restaurants. I look by distance from where I’m standing, and the stars and reviews.”. **ONLINE RECOMMENDATION, RESTAURANTS’ PROXIMITY**
- “...they usually have a central theme, like: “This is... people come here for this.” If this falls into that category, I’d probably go...”. **LOCAL EVENTS**
- “Since its traditional food, I’d try it 100%...”. **LOCAL CUISINE**
- “The first thing I’d do is check photos, because I’m way more excited about a mix of city and coast—or water or scenic natural stuff—than just city.”. **NATURAL LANDSCAPE**

Interview 4 (Male, Spain, 25 years)

- Some important points, and insights:
  - “...visit all the beaches in Menorca, which are very beautiful.”. **BEACHES**
  - “Because it was cheaper that way.”. **COST TO REACH**
  - “...the distance we had from the house we rented to the place to visit...”. **CITY’S ACCESSIBILITY**
  - “...when it comes to looking on Airbnb, then I’d use 2 or 3 different websites...”. **ACOMMODATION TYPE**
  - “So we can go to the supermarket without any problem, which is important.”. **SUPERMARKETS AVAILABILITY**

    - “I’d look for a place near a city. That’s reasonably priced.”. **ACOMMODATION PRICE**
    - “…with good reviews—of course, reviews are essential. You always have to check them.”. **ONLINE RECOMMENDATION**
    - “…I’d ask if there are museums to visit. Also about restaurants there.”. **MUSEUMS, RESTAURANTS**
    - “…I’d look for places around it that are also interesting—like natural parks, hiking routes…”. **NEAR TO OTHER TOURISTIC DESTINATIONS, NATURAL LANDSCAPE, HIKING**
    - “…visiting nearby mountain areas or coastal areas.”. **NATURAL LANDSCAPE, MOUNTAINS, BEACHES**
    - “…but it’s also very important to know the price. Of everything it says you can do.”. **LOCAL PRICES**

Interview 5 (Male, Argentina, 26 years)

- Some important points, and insights:
    - “…the decision to travel to Brazil has to do first with financial reasons.”. **LOCAL PRICE**
    - “…we believe it’s a trip that includes several destinations…”. **NEAR TO OTHER TOURISTIC DESTINATIONS**
    - “So everything in terms of entertainment, food, and drink is covered—and accommodation, of course—is covered with everything the cruise includes.”. **PRICE, ACOMMODATION SERVICE**
    - “…they had prices—or rather packages—that were quite affordable for what they offered…”. **TOURISTIC PACKAGE**
    - “…and also to attend San Fermín—the San Fermín festival.”. **LOCAL EVENTS**
    - “…to organize the trip in such a way that we had a practical triangle in terms of logistics—that is, in terms of travel.”. **ACCESIBILITY**
    - “That route is much cheaper.”. **COST TO REACH**
    - “there’s a lot of traffic and so on. I mean people-traffic.”. **OVERCROWDED**
    - “Well, Google has this attractions option…”. **ATTRACTION AVAILABILITY**
    - “This bridge here seems very nice. I think there's a fort or something similar around here. And it seems like a kind of medieval city, but at the same time very well preserved…”. **HISTORIC BUILDINGS**
    - “It’s closer to Rome, which is a major city.”. **NEAR TO POPULAR CITIES**
    - “…I’m obviously going to prioritize pasta or some traditional local food.”. **LOCAL CUISINE**
    - “…I generally look for something near the center. Especially when the city’s attractions or most important sites are close to the center.”. **CITY’S ACCESIBILITY**

Interview 6 (Male, Iran, 25 years)

- Some important points, and insights:

- “...the ticket for going back is... Straightly, it was too expensive. And I decided to maybe... find a way to be cheaper. Like, going somewhere else, and then have a flight back to Milan.”. **COST TO REACH, PROXIMITY TO OTHER DESTINATIONS**
- “...it was more a historic place, a more historic city.”. **HISTORIC BUILDING, HISTORICAL DESTINATION**
- “...the cheap trip we could have there. Like, even, uh, the food. Or, uh, the price of the Airbnb, the place of living.”. **FOOD PRICE, ACCOMMODATION PRICE**
- “And for historic places, I mean... maybe ChatGPT.”. **ONLINE RECOMMENDATION**
- “...best place for us, uh, the castle. They had on the top of the hill... Also, the whole, like, it was, like, an old town there, right? All over the hill.”. **HISTORICAL BUILDING**
- “And then...It'll be the location... to see if... I mean, which one is nearer to me.”. **DISTANCE TO REACH**
- “Like, an Airbnb or Booking.com, or any other websites regarding this.”. **ACCOMMODATION TYPE**
- “...and of course, it's so close to Florence. Because, for example. I can be, like, one day in Pistoia, and then I go to Pisa or Florence to have more fun there.”. **NEAR TO POPULAR CITIES, NEAR TO OTHER DESTINATIONS**
- “...it has a better... uh, like, route by bus, or... uh, train, I don't know, uh... any vehicle that, uh, we... I can use to get there.”. **AVAILABILITY OF TRANSPORT TO DESTINATION**

Interview 7 (Female, Spain, 21 years)

- Some important points, and insights:
    - “There we looked for hostels...”. **ACCOMMODATION TYPE**
    - “Like, finding the cheapest way to make it all work.”. **COST TO REACH**
    - “Then for hotels, we looked on both Booking and Trivago...”. **ACCOMMODATION TYPE**
    - “We also checked they weren’t super far from Vienna, that we could get there by bus or walking in 15 or 20 minutes.”. **CITY’S ACCESSBILITY**
    - “And they’re much cheaper than hotels or any hostel. Cleaner and quieter. Breakfast included—also important.”. **ACCOMMODATION PRICE, ACCOMMODATION SERVICES**
    - “...we used ChatGPT, we asked it to make us a list of everything.”. **ONLINE RECOMMENDATION**
    - “So we do something near the convent.”. **ACCESSIBILITY TO TOURISTIC POINTS**
    - “You’d get these huge lines...And I’m like, not worth saving 3 euros.”. **OVERCROWDED**
    - “Okay, for that we’d use a mix of Google Maps and TripAdvisor. We’d look at price ranges...”. **ONLINE RECOMMENDATION, FOOD PRICES**
    - “Then we look at typical local places...”. **LOCAL RESTAURANTS**

    - "…we'd decide based on what was best depending on where we were, depending on the monuments and everything we had to see.". **RESTAURANTS' PROXIMITY**
    - "So yeah, for me, I always go for the traditional. I really value culinary culture.". **LOCAL CUISINE**
    - "So for a weekend, it's not worth it to go that far.". **DISTANCE TO REACH**
    - "Which has more historical stuff…". **HISTORICAL DESTINATION**
    - "But you also have to get to the airport 2h early, and then there's the trip afterward to get wherever.". **AVAILABILITY OF TRANSPORT TO DESTINATION, DISTANCE TO REACH**
    - "Like, if someone told me, "Hey, there's this event in this place," and the tickets are affordable—like, I don't know, 30 or 40€, then yeah, I'd go.". **LOCAL EVENT, EVENT PRICE**
    - "It's a city that—I mean, architecturally, it's beautiful. The churches inside were gorgeous.". **HISTORIC BUILDING**
    - "…people are friendly. But in Italy… that's not really a barrier.". **LOCAL FRIENDLINESS**

Interview 8 (Male, Colombia, 41 years)

- Some important points, and insights:
    - "I always travel for an electronic music festival.". **MUSIC EVENTS**
    - "…hopefully, that city will also have something in terms of modern art.". **CULTURE**
    - "…and I try to explore nearby areas as well.". **NEAR TO OTHER DESTINATIONS**
    - "…I'd look at how much I'd spend on the festival.". **MUSIC EVENT COST**
    - "I go to Booking or Airbnb.". **ACCOMMODATION TYPE**
    - "…I look at prices. Based on what I like, the quality.". **ACCOMMODATION PRICE, ACCOMMODATION QUALITY**
    - "Then online, I also looked up rankings…". **ONLINE RECOMMENDATION**
    - "…everything is always full all the time.". **OVERCROWDED**
    - "…was super expensive—it didn't matter, I was working, so no problem. Now it's a bit different.". **FOOD'S PRICE**
    - "…some popular local food places. I also love to visit those.". **LOCAL CUISINE**
    - "…I'll open Google and check the top-rated places nearby—what Google Maps recommends.". **ONLINE RECOMMENDATION**
    - "The museums… this right here. That's enough to win me over.". **MUSEUMS**
    - "Then I'd look up the train.". **ACCESIBILITY BY PUBLIC TRANSPORT**
    - "That's awesome. Cool. I'd go… No, I really like it because it has to do with culture.". **LOCAL EVENTS, CULTURE**

Interview 9 (Female, Argentina, 30 years)

- Some important points, and insights:

- “We wanted to visit a lot of places in a short time, so we went for capitals or more touristy, well-known places.”. **NEAR TO OTHER DESTINATIONS, PROXIMITY TO POPULAR CITIES, POPULAR CITY**
- “So, we picked locations based on proximity and timing.”. **NEAR TO OTHER DESTINATIONS, DISTANCE TO RECH**
- “We searched a lot of websites from people who had already traveled and had tips or recommendations.”. **ONLINE RECOMMENDATION**
- “So I click on them as a first guide, but then I try to find people I think have more genuine content—not people who were paid to promote.”. **GENUINE ONLINE RECOMMENDATION**
- “I’d go directly to Booking or sites that are focused specifically on accommodation, and another site just for transport.”. **ACCOMMODATION TYPE, AVAILABILITY OF TRANSPORT TO DESTINATION**
- “The first thing I would do is check how long it would take to get there and back from where I’m staying.”. **DISTANCE TO REACH**
- “…I’d try to make the trip out as early as possible to have the whole day to explore. And the return trip as late as possible to also get the most out of that day.”. **VARIETY OF HOURS FOR TRANSPORTATION**
- “…first, look at the location… So first, I check that the area is recommended because it’s close to tourist spots.”. **ACCOMMODATION LOCATION, ACCESSIBILITY TO TOURISTIC POINTS**
- “And also because if I’m going by bus or public transport and won’t have a car, it should be easy to get around…”. **CITY’S ACCESSBILITY**
- “Then I’d set a price range for lodging.”. **ACCOMMODATION PRICE**
- “If there’s a well-known museum, I’d try to dedicate part of the trip to visiting it, but mainly I’d focus on the historical spots.”. **MUSEUMS, HISTORICAL PLACES**
- “If they have any kind of city tour… I’d check if they have local tour guides that give you a general overview of the city.”. **AVAILABILITY OF CITY TOURS**
- “But I try to research the city a bit before going so when I arrive, I know what I’m seeing, what I’m visiting.”. **INFORMATION AVAILABILITY**
- “Maybe it’s known more for food, so I’d look into where to try the local cuisine.”. **LOCAL CUISINE**

Interview 10 (Male, Argentina, 30 years)

- Some important points, and insights:
  - “…before organizing everything was to see what was cheapest in terms of flying in and out.”. **COST TO REACH**
  - “We booked entry and exit through different countries so we could plan a route going from one place to another.”. **NEAR TO OTHER DESTINATIONS**
  - “I had found some different blogs.”. **ONLINE RECOMMENDATIONS**
  - “We mainly stayed in hostels…”. **ACCOMMODATION TYPE**

- “We used Omio to compare if it was better to travel by bus, plane, or train between cities. If we could travel overnight to save on lodging, we did that too.”. **AVAILABILITY OF TRANSPORT TO DESTINATION, VARIETY OF HOURS FOR TRANSPORTATION**
- “What we did a lot of was checking which historical places, castles, museums, and such we could book in advance…”. **HISTORICAL PLACES, HISTORICAL BUILDING, MUSEUMS**
- “Then we’d plan the day around that activity we’d already scheduled.”. **ACCESSIBILITY TO TOURISTIC POINTS**
- “…they’re very cultural cities, very rich in culture, especially.”. **CULTURE**
- “Not so many fast-food chains, more traditional food.”. **LOCAL CUISINE**
- “Also, culturally, being out on the street—the people are more laid-back, if you will.”. **LOCAL FRIENDLINESS**
- “So we’d try to eat something simple and cheap at lunchtime.”. **FOOD’S PRICES**
- “A lot from TikTok, social media, blogs.”. **ONLINE RECOMMENDATIONS**
- “…how much it costs to get to each one—on Omio or something like that.”. **COST TO REACH**
- “I’d start looking at it all on Google Maps to see if there are actually things to do and how far away everything is, because if it’s 20 kilometers from the city…”. **ACCESSIBILITY TO TOURISTIC POINTS**
- “I’d search by area—somewhere central, or where I’ll be arriving (train or bus).”. **ACCOMMODATION LOCATION**
- “If it looks good or has the services I need, like towels, a bed, that’s enough.”. **ACCOMMODATION SERVICES**
- “They’re smaller cities, you know, it’s not like Rome or that I’m missing out on Venice.”. **POPULAR CITY**

D. Pseudocodes for Text Mining

Algorithm 3 text_pipeline

Input: DataFrame *df*, input column name *input_col*, and stopword removal flag

```
for each row in df do
    t ← df[input_col]
    if t is missing then
        append ∅ to processed list
        continue
    end if
    lowercase t
    remove non-alphanumeric characters
    tokenize and lemmatize
    if stopwords removed then
        filter out stopwords
    end if
    append tokens to processed list
end for
add processed list as new column final_tokens
```

Output: *df* with cleaned token list

Algorithm 4 compute_tfidf_ngrams

Input: cleaned interviews *df* with token column `final_tokens`; custom stopword list

```
concatenate tokens of each row into list docs
for k ∈ {1, 2, 3} do                                  ▷ ngram sizes
    build TF–IDF vectorizer for k-grams
    transform docs into TF–IDF matrix X_k
    compute mean TF–IDF scores per k-gram
    extract feature names
    rank terms by average TF–IDF
    keep top n_k terms
    store results in table T_k
end for
```

Output: tables T_1, T_2, T_3 with top unigrams, bigrams, and trigrams

Algorithm 5 chunk_text

Input: text string *text*; chunk size $n = 6$

```
sentences ← sent_tokenize(text)
chunks ← [ ]
for i ← 0 to len(sentences) step n do
    chunk ← concat(sentences[i : i + n])
    if len(chunk) > 30 then
        append chunk to chunks
    end if
end for
```

Output: list *chunks*

Algorithm 6 BERTopic implementation

Input: data frame *df* with column `text`

```
documents ← [ ]
origin ← [ ]
for each row (i, t) in df do
    c ← chunk_text(t)
    append c to documents
    record i in origin for each chunk
end for
embed documents
extract n-gram representations
initialize BERTopic with embeddings and text representations
fit BERTopic on documents
obtain topic assignments and probabilities
```

Output: BERTopic model and topic distribution

E. Prompts for ChatGPT

Prompt used:

I'm going to send you several text files. I want you to read them carefully and then give me a detailed analysis that includes:

- *A list of specific tourism attributes (*in English*) found in each interview (not general ones like price or accommodation).*
- *The frequency of each attribute (across a total of 10 interviews).*
- *A table showing the attributes in English, their frequency, and which interviews they appear in.*

When I send you the files, confirm that you've read them, and then start the analysis. Don't write anything until I confirm that I've sent all the files.

Refinement prompt:

Continuing with this, it's important to understand that sometimes an attribute can be ambiguous for different respondents. For example, if I include "Historical architecture & monuments," it's not clear whether it refers to the availability of monuments, the presence of a historic center, etc.

That's why I need you to conclude with more detailed, clearly defined attributes that avoid ambiguity and don't overlap with other attributes.

F. Manual Attribute Identification

#	Attribute	Frequency	#	Attribute	Frequency
1	accommodation type	9	23	overcrowded	3
2	online recommendations	9	24	restaurants availability	3
3	proximity to other destinations	9	25	accessible by public transport	2
4	local cuisine	8	26	accommodation quality	2
5	accessibility to touristic points	7	27	city's accessibility	2
6	cost to reach	7	28	genuine online recommendations	2
7	accommodation price	6	29	hiking trails	2
8	distance to reach	6	30	local friendliness	2
9	historical buildings	6	31	popular city	2
10	local events	6	32	variety of hours for transportation	2
11	accommodation services	5	33	access to mountains	1
12	availability of transport to destination	5	34	attraction availability	1
13	cultural immersion	5	35	availability of city tours	1
14	food's price	5	36	availability of travel agencies	1
15	proximity to popular cities	5	37	car friendly	1
16	accommodation location	4	38	destination size	1
17	historical places	4	39	events price	1
18	local prices	4	40	local experience	1
19	natural landscape	4	41	local restaurants	1
20	access to beaches	3	42	music events	1
21	information availability online	3	43	music events cost	1
22	museums	3	44	supermarket availability	1

Figure 4.1: *Preliminary Attributes and Their Frequency. Source: Author's own elaboration.*

G. ChatGPT Attributes

Attribute (English)	Frequency	Interviews where mentioned
Local food / gastronomy	10	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Architecture / historical buildings	9	1, 2, 3, 4, 5, 6, 7, 9, 10
Museums / art exhibitions	7	1, 2, 4, 6, 8, 9, 10
Cultural festivals / events	7	1, 4, 5, 6, 7, 8, 10
City atmosphere / vibe / livability	6	1, 2, 3, 6, 7, 8
Traditional restaurants / family-run eateries	6	3, 4, 6, 7, 9, 10
Historic centers / old towns	5	2, 4, 5, 9, 10
Local people friendliness / hospitality	5	3, 4, 6, 7, 10
Cleanliness of the city	4	3, 4, 6, 7
Artistic heritage / Renaissance art	4	2, 5, 8, 10
Squares / piazzas / public spaces	4	4, 7, 8, 10
Natural landscapes / vegetation / parks	3	4, 6, 7
Religious sites (churches, convents)	3	2, 6, 10
Guided tours / city tours	3	1, 5, 9
Nightlife / music festivals	3	1, 5, 8
Traditional markets / local markets	2	5, 8
Local crafts / authenticity / non-commercial atmosphere	2	4, 7
Street food / casual local food	2	9, 10
Festivals or parades (La Quintana)	2	8, 10
Scenic views / viewpoints / aesthetic beauty	2	6, 7

Figure 4.2: *Attributes Suggested by ChatGPT. Source: ChatGPT.*

Attribute
Quality of local cuisine
Presence of heritage architecture
Ambience of the historic center
Availability of museums and art venues
Frequency of cultural events
Quality of natural scenery
Friendliness of local residents
Cleanliness and orderliness of the city
Authenticity of local experience
Availability of guided tours and information on-site
Attractiveness of main squares and public spaces
Ease of access from other destinations
Cost and comfort of transport connections
Quality of accommodation options
Usefulness of online information
Scale and diversity of attractions
Affordability of the destination
Availability of visitor services

Figure 4.3: *Refined Attributes from ChatGPT. Source: ChatGPT.*

H. Airbnb Top Words

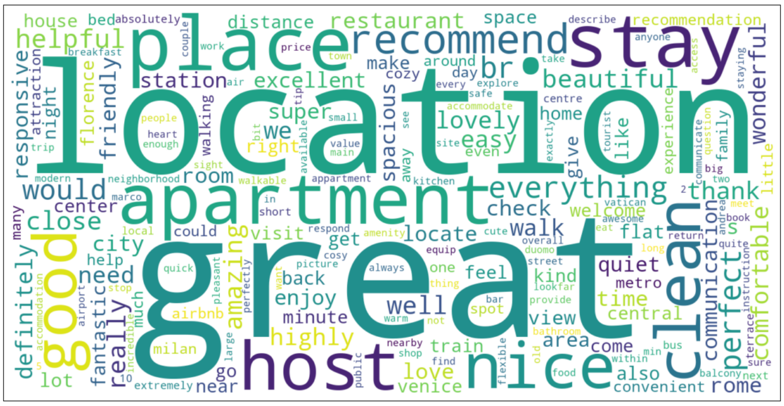


Figure 4.4: *Top Words from Airbnb Reviews. Source: Author's own elaboration based on Inside Airbnb dataset.*

I. Booking.com Top Words

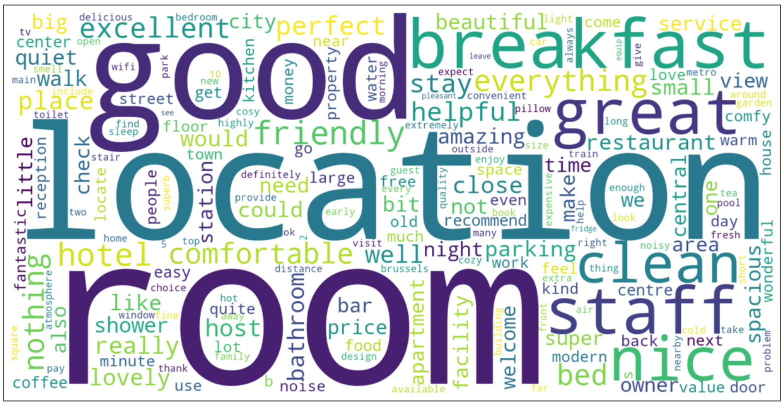


Figure 4.5: *Top Words from Booking.com Positive Reviews. Source: Author's own elaboration based on Kaggle dataset.*

Figure 4.6: *Top Words from Booking.com Negative Reviews. Source: Author's own elaboration based on Kaggle dataset.*

J. Open-ended Survey

WELCOME!

This survey is meant to study the wishes of tourists and make them as happy as possible during their trip. Every piece of information you share will be valuable to us. Please answer as thoroughly as you can to help us. There are no right or wrong answers. We are not interested in the "best destinations," but in your actual choices, even if they are impulsive or not well planned. The answers are anonymous and will be used only for research purposes.There are 10 simple questions (hopefully also fun ones), but it is important that you have at least ten minutes to answer them—so if you don't have time now, please come back when you can dedicate a moment to us!

PAST TRIPS

1. Tell us about a trip of yours that you were not satisfied with. What would you not do again, especially in the planning phase? **(OPEN-ENDED QUESTION)**
2. Which trip or vacation best represents your way of choosing how to travel? Tell us about the destination: how you chose it, why it reflects you, etc. **(OPEN-ENDED QUESTION)**

ORGANIZING A VACATION

3. Imagine organizing a vacation. Sketch a short story of your choice: what would be the 3–4 main scenes?
    a. Scene 1 **(OPEN-ENDED QUESTION)**
    b. Scene 2 **(OPEN-ENDED QUESTION)**
    c. Scene 3 **(OPEN-ENDED QUESTION)**
    d. Scene 4 (Optional) **(OPEN-ENDED QUESTION)**

MULTIPLE CHOICE

4. Which of these stories represents you the most?
    a. I see an offer, check very little, and book the same day.
    b. I compare 3–4 destinations, read reviews, and book after a few days.
    c. I check flexibility/free cancellation and decide when I find the right one.
    d. Others.

ALTERNATIVE CHOICES

Look at the two alternatives and answer the following questions.

Screen A	Screen B
Price: €420 (total) Reviews: 4.2/5 (based on 600 reviews) Free cancellation: NO	Price: €470 (total) Reviews: 4.6/5 (based on 320 reviews) Free cancellation: YES
Pantalla A Precio: 420 € (total) Reseñas: 4,2/5 (basado en 600 reseñas) Cancelación gratuita: NO	**Pantalla B** Precio: 470 € (total) Reseñas: 4,6/5 (basado en 320 reseñas) Cancelación gratuita: SÍ

5. If you had to choose right now, which of the two screens would you pick?
    a. A.
    b. B.
6. Which was the decisive piece of information you looked at last before choosing? **(OPEN-ENDED QUESTION)**

SUGGESTIONS

7. If you had to teach someone your method of choosing, which 3 fundamental steps would you tell them to follow?
    a. Step 1 **(OPEN-ENDED QUESTION)**
    b. Step 2 **(OPEN-ENDED QUESTION)**
    c. Step 3 **(OPEN-ENDED QUESTION)**
8. If instead you had to give 3 pieces of advice to your past self about organizing a vacation, what would you say?
    a. Step 1 **(OPEN-ENDED QUESTION)**
    b. Step 2 **(OPEN-ENDED QUESTION)**
    c. Step 3 **(OPEN-ENDED QUESTION)**

OPEN QUESTIONS

9. When you organized your last vacation, did you have a list of destinations in mind? How did you make your final choice? And among them, was there a destination you said, “No, this is not for me”? Tell us why. **(OPEN-ENDED QUESTION)**
10. When did you feel you had “enough” information to decide? What made you stop searching? And if you had had more time, what would you have added to your research? **(OPEN-ENDED QUESTION)**
11. Is there something important we haven’t asked you but that influenced your decision? Tell us about it. **(OPEN-ENDED QUESTION)**

ASCOLI PICENO (ADD PICTURE)

Ascoli Piceno is a historic city in central Italy, located about 2 hours from Rome and close to the Adriatic coast. It is famous for its medieval towers, Renaissance squares (such as Piazza del Popolo), local gastronomy (olive all'ascolana), and its well-preserved historic center surrounded by mountains and natural parks.

Reference map: Look at this map and continue to the next section. (ADD MAP)

PRICING

12. Imagine a weekend package in Ascoli Piceno: it includes 2 nights in centrally located accommodation, 1 dinner at a restaurant, and 1 activity of your choice (for example: a museum, guided tour, etc.). Exclude transportation.
    a. At what price would you consider the product to be priced so low that you would feel the quality couldn't be very good? In euros (€). **(Space to put only numbers)**
    b. At what price would you consider the product to be a bargain - a great buy for the money? In euros (€). **(Space to put only numbers)**
    c. At what price would you consider the product starting to get expensive, so that it is not out of the question, but you would have to give some thought to buying it? In euros (€). **(Space to put only numbers)**
    d. At what price would you consider the product to be so expensive that you would not consider buying it? In euros (€). **(Space to put only numbers)**

TYPE OF TRAVELER

13. When you travel, who takes care of organizing the trip?
    a. Myself.
    b. I organize it with others.
    c. Someone else.
14. What is your primary purpose for traveling? You can select up to 2 options.
    a. See new cities or explore new destinations.
    b. Relaxation.
    c. Outdoor activities.
    d. Cultural attractions.
    e. Events: festivals, sports, etc.
15. What type of trip do you usually choose: a short trip, a medium-length trip, or a long one? Please explain the reason for your choice. **(OPEN-ENDED QUESTION)**
16. What type of traveler are you? You can select more than one. You can select up to 2 options.
    a. Solo traveler.
    b. Family traveler.
    c. Group traveler.
    d. Couple traveler.
17. What is the first place you go to for travel information?
    a. Google or similar.
    b. ChatGPT or similar Ais.
    c. Online Platforms as Booking.com.
    d. Airbnb, TripAdvisor or similar.
    e. Social media.
    f. Official destination websites.
    g. Travel Blogs.

h. Travel agencies.
i. Personal recommendations.

18. What is your age group?
    a. <25.
    b. 25-35.
    c. 35-59.
    d. >60.
19. Which is your gender?
    a. Male.
    b. Female.
    c. Other/I prefer not to say it.
20. Where do you live?
    a. Italy.
    b. Europe, excluding Italy.
    c. Asia.
    d. Africa.
    e. Oceania.
    f. North America.
    g. Central America.
    h. South America.
21. What is your current occupation?
    a. Employed.
    b. Self-employed.
    c. Student.
    d. Retired.
    e. Unemployed.
    f. Other.

K. Pseudocodes for Van Westendorp Model

Algorithm 7 Van Westendorp intersection

Input: two price–response curves (x_1, y_1) and (x_2, y_2)
Output: list of intersection points

construct common grid x
interpolate both curves on x to obtain $\hat{y}_1, \hat{y}_2$
compute difference $d = \hat{y}_1 - \hat{y}_2$
find indices where d changes sign
initialize list I
for each sign change **do**
  estimate intersection point (x^*, y^*)
  append (x^*, y^*) to I
end forreturn I

Algorithm 8 Van Westendorp Analysis

Input: response vectors for price perceptions: `too_cheap`, `cheap`, `expensive`, `too_expensive`
Output: Van Westendorp intersection points

function CUM_PERCENT(x, invert)
  sort x; compute cumulative distribution
  if invert **then**
    return $1 - \text{CDF}(x)$
  else
    return $\text{CDF}(x)$
  end if
end function
compute cumulative curves: TC = CUM_PERCENT(*too_cheap*, true), CH = CUM_PERCENT(*cheap*, false), EX = CUM_PERCENT(*expensive*, false), TE = CUM_PERCENT(*too_expensive*, true)
compute intersection points:
    PMC = FIND_INTERSECTION(TC, EX)
    PME = FIND_INTERSECTION(TE, CH)
    IPP = FIND_INTERSECTION(CH, EX)
    OPP = FIND_INTERSECTION(TC, TE)
Output: PMC, PME, IPP, OPP

L. Open-ended Survey Results

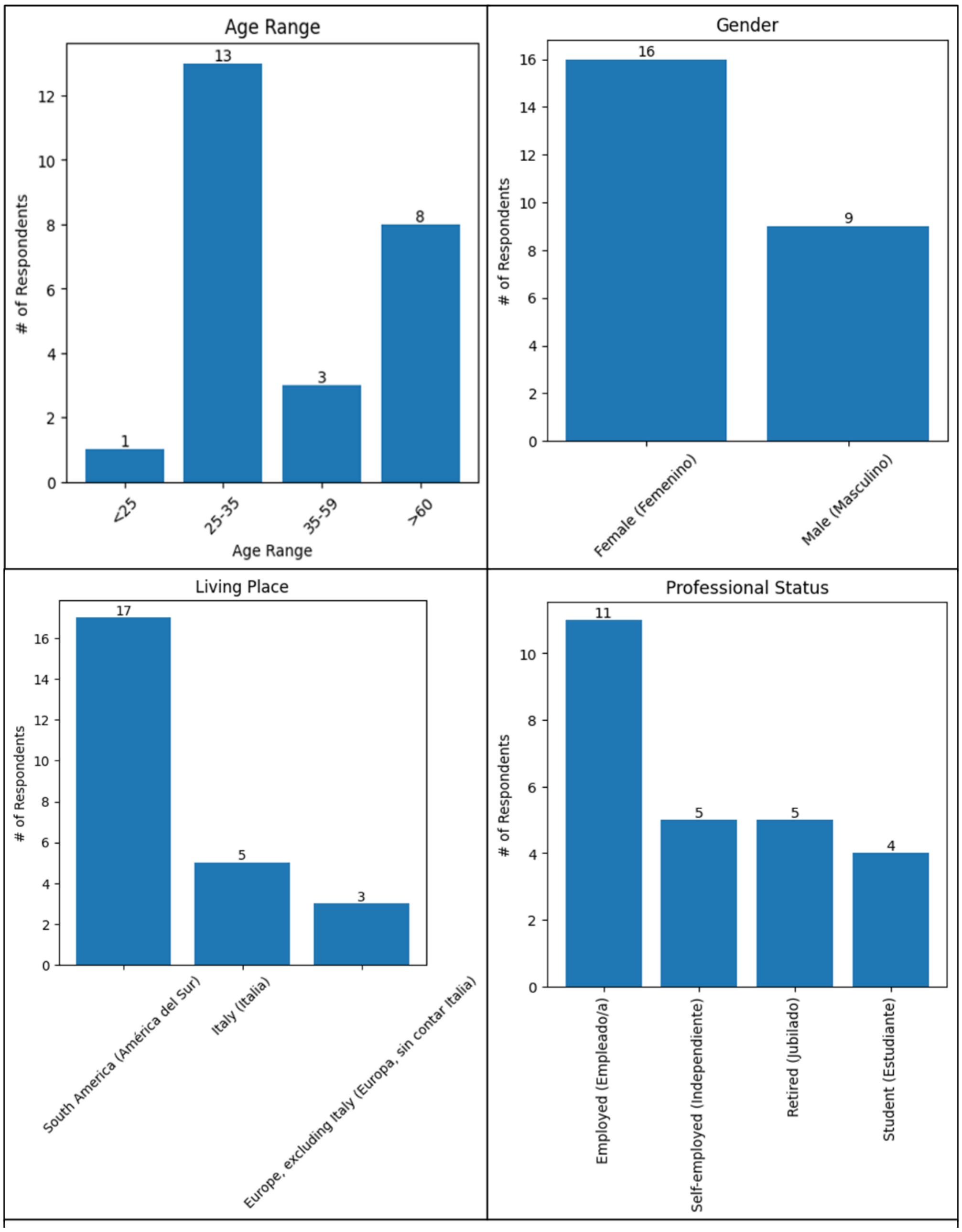


Figure 4.7: *Description of Open-ended Survey Respondents (I). Source: Author's own elaboration.*

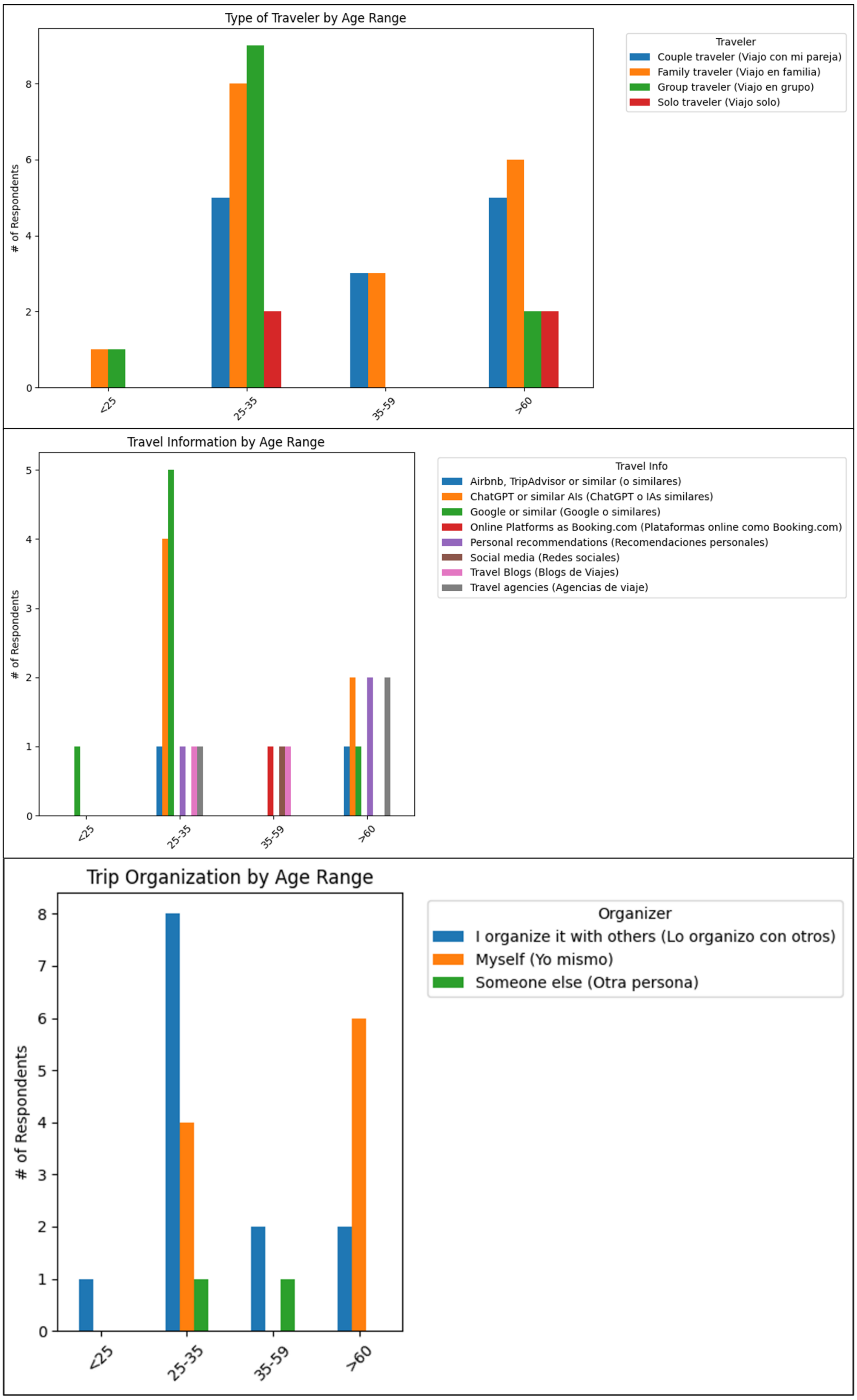


Figure 4.8: *Description of Open-ended Survey Respondents (II). Source: Author's own elaboration.*

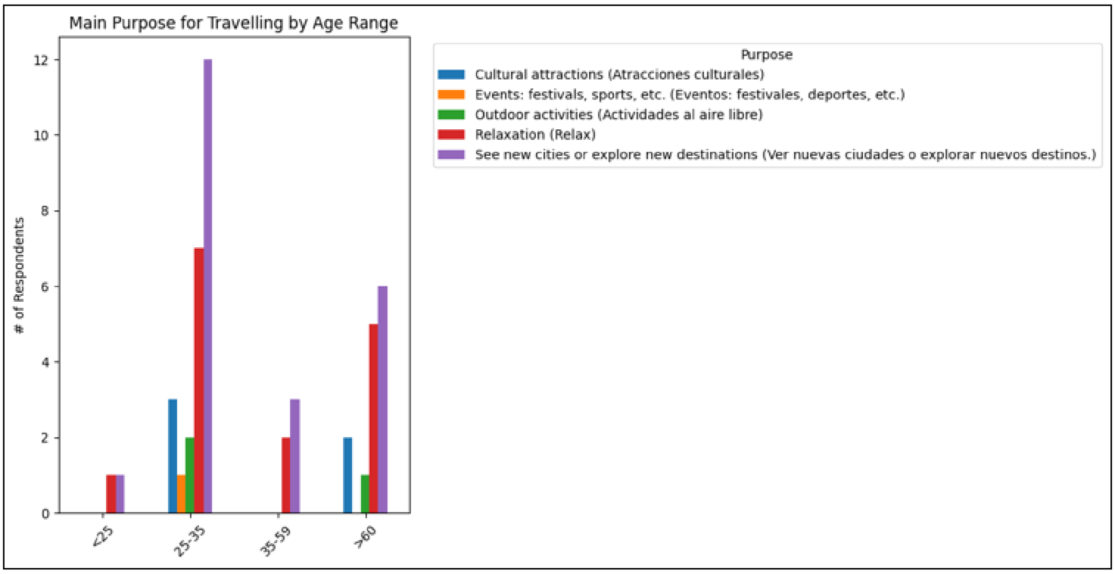


Figure 4.9: *Main Travel Purpose. Source: Author's own elaboration.*

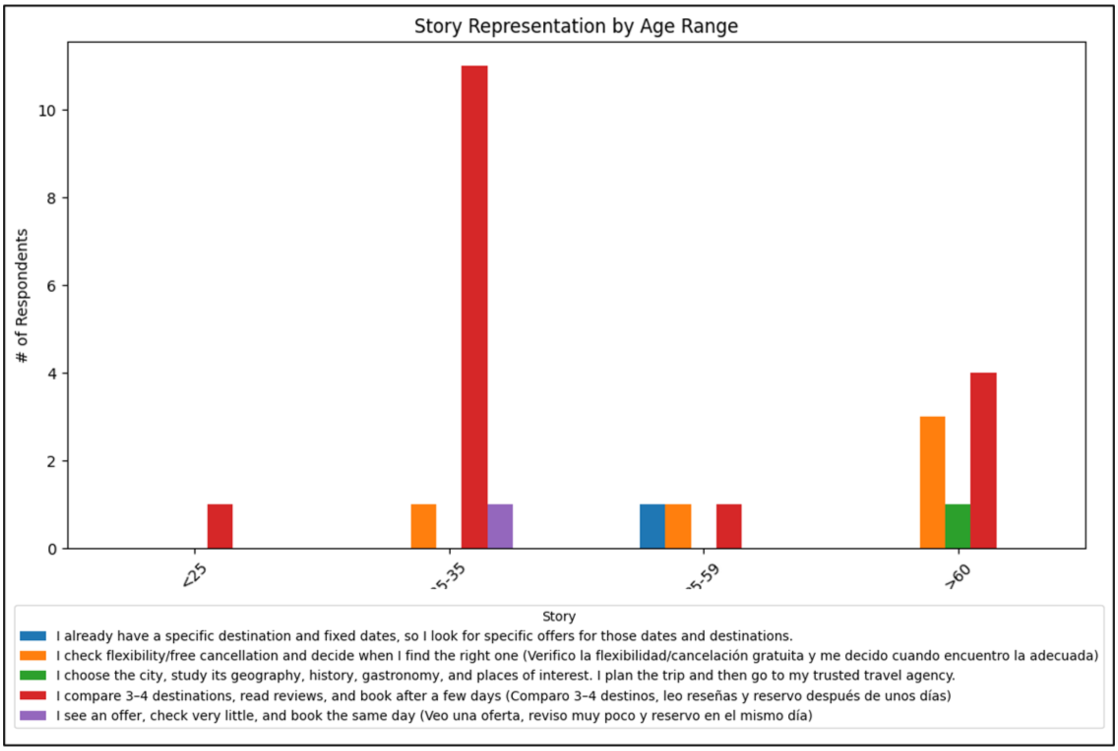
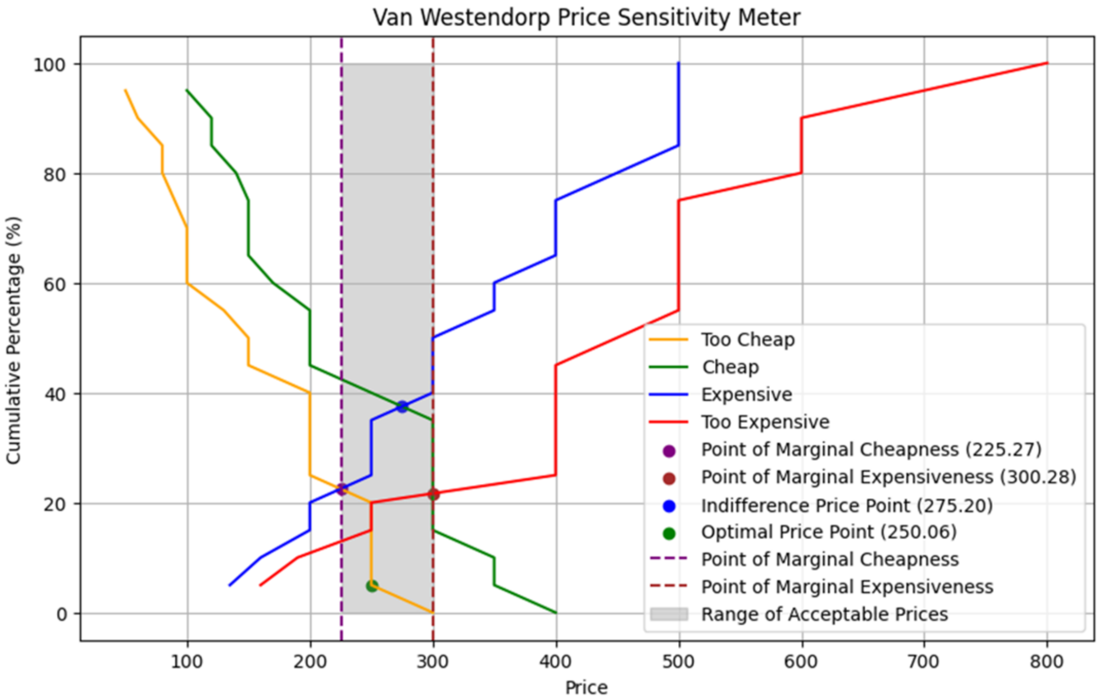


Figure 4.10: *Description of Open-ended Survey's Respondents. Source: Author's own elaboration.*

M. Van Westendorp Results by Age Group




```
Van Westendorp Analysis:
Point of Marginal Cheapness: 225.27
Point of Marginal Expensiveness: 300.28
Indifference Price Point: 275.20
Optimal Price Point: 250.06
```

Figure 4.11: *Van Westendorp Pricing Method – Full Sample. Source: Author's own elaboration.*

N. Quantitative Survey

WELCOME

APROX. DURATION: 6-8 minutes

Add some short descriptive message.

ATTRIBUTE PREFERENCES

1. **Order the following travelling attributes from MOST IMPORTANT (1) to LEAST IMPORTANT (10).** ***Drag and drop each one to the desired position.***
   Ordená los siguientes atributos de viaje, del MÁS IMPORTANTE (1) al MENOS IMPORTANTE (10). *Arrastrá/Mové cada uno a la posición deseada.*
   - **Diversity of authentic local restaurants** (Diversidad de restaurantes locales auténticos)
   - **Heritage ambience of the historic center** (Ambiente histórico del centro patrimonial)
   - **Availability of museums and art venues** (Disponibilidad de museos y espacios artísticos)
   - **Availability of centrally located accommodation** (Disponibilidad de alojamientos ubicados en el centro)
   - **Proximity to natural areas or outdoor routes** (Proximidad a zonas naturales o rutas al aire libre)
   - **City located outside major tourist circuits** (Ciudad ubicada fuera de los principales circuitos turísticos)
   - **Availability of tourist information, both digital and on-site** (Disponibilidad de información turística, tanto digital como presencial)
   - **Accommodation services, such as flexible cancellation, digital check-in, amenities, etc.** (Servicios de alojamiento, como cancelación flexible, registro digital, comodidades, etc.)
   - **Flexible scheduling of guided tours** (Programación flexible de visitas guiadas)
   - **Density of touristic attractions within walking distance** (Densidad de atracciones turísticas a poca distancia a pie)

ACCOMMODATION

2. **To what extent do you agree with the following statements?**
   ¿Qué tan de acuerdo estás con estas afirmaciones?
   - **“The availability of centrally located accommodation is important for my travel experience.”** (“La disponibilidad de alojamiento en una zona céntrica es importante para mi experiencia de viaje.”)
     - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).
   - **“Accommodation services such as flexible cancellation, digital check-in, and amenities are important for my travel experience.”** (“Los servicios de alojamiento, como la cancelación flexible, el registro digital y las comodidades, son importantes para mi experiencia de viaje.”)
     - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).

3. **When defining accommodation, I usually...**
   A la hora de definir el alojamiento, suelo...
   - **...book in advance during the planning phase** (reservar con tiempo durante la planificación)
   - **...book at the last minute** (reservar a último momento)
   - **...book directly at the destination** (reservar directamente en destino)

4. **Where do you usually search for accommodation?** ***You can select up to 2 options*****.**
   En general, ¿dónde buscas alojamiento? *Podés elegir hasta 2 opciones*.
   - **Online platforms as Booking** (En plataformas online como Booking)
   - **Airbnb**
   - **Travel agencies** (Agencias de viajes)
   - **Google**
   - **Social media** (Redes sociales)
   - **Hotels' official website** (Página oficial de los hoteles)
   - **Personal recommendation** (Recomendación personal)
   - **Others**

TOURISM DENSITY

5. **To what extent do you agree with the following statements?**
   ¿Qué tan de acuerdo estás con estas afirmaciones?
   - **“Visiting less crowded destinations improves my travel experience.”** “(Visitar destinos menos concurridos mejora mi experiencia de viaje.”)
     - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).

6. **How willing would you be to visit a destination that is less accessible by public transport if it means fewer tourists?**
   ¿Qué tan dispuesto/a estarías a visitar un destino con menor accesibilidad en transporte público si eso implica menos turistas?
   - Likert from **1 = Not at all willing** (Nada dispuesto/a) to **5 = Very willing** (Muy dispuesto/a)

7. **How much extra travel time would you accept to visit a destination with fewer tourists?**
   ¿Cuánto tiempo extra de viaje aceptarías para visitar un destino con menos turistas?
   - **Up to 30 minutes** (Hasta 30 minutos)
   - **1–2 hours** (1-2 horas)
   - **3–4 hours** (3-4 horas)
   - **More than 4 hours** (Más de 4 horas)
   - **I wouldn't travel longer for that reason** (No viajaría más por esa razón)

DISCOVERING THE CITY

---

8. **To what extent do you agree with the following statements?**
   ¿Qué tan de acuerdo estás con estas afirmaciones?
   - **"The heritage ambience of the historic center is important for my travel experience."** ("El ambiente patrimonial del centro histórico es importante para mi experiencia de viaje.")
     - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).
   - **"The density of touristic attractions within walking distance is important for my travel experience."** ("La densidad de atracciones turísticas a poca distancia a pie es importante para mi experiencia de viaje.")
     - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).
   - **"The flexible scheduling of guided tours is important for my travel experience."** ("La programación flexible de las visitas guiadas es importante para mi experiencia de viaje.")
     - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree)

9. **Would you take a city tour in the city you are visiting?**
   ¿Harías un city tour en la ciudad que visitás?
   - **Yes** (Sí) **→ Continues with section CITY TOUR**
   - **No → Jumps to section NO CITY TOUR**

CITY TOUR

---

10. **What type of city tour would you prefer?**
    ¿Qué tipo de city tour preferirías?
    - **Guided walking tour** (Tour guiado a pie)

- **Private tour** (Tour privado)
- **Guided bicycle tour** (Tour guiado en bicicleta)
- **Tourist train** (Tren turístico)
- **Others**

NO CITY TOUR

---

11. **What type of architectural sites are you most interested in visiting? *You can select up to 3 options*.**
    ¿Qué tipo de sitios arquitectónicos te interesa más visitar? *Podés elegir hasta 3 opciones*.
    - **Historical squares and public spaces** (Plazas históricas y espacios públicos)
    - **Religious buildings: churches, cathedrals, monasteries**. (Edificios religiosos: iglesias, catedrales, monasterios)
    - **Medieval architecture: towers, fortifications, bridges**. (Arquitectura medieval: torres, fortificaciones, puentes)
    - **Renaissance and Baroque palaces** (Palacios renacentistas y barrocos)
    - **Museums and cultural centers** (Museos y centros culturales)
    - **Modern or contemporary architecture** (Arquitectura moderna o contemporánea)

DISTINCTIVE LOCAL CUISINE

---

12. **To what extent do you agree with the following statements?**
    ¿Qué tan de acuerdo estás con estas afirmaciones?
    - **"The diversity of authentic local restaurants is important for my travel experience."** ("La diversidad de restaurantes locales auténticos es importante para mi experiencia de viaje.")
        - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).

13. **Where do you usually prefer to eat when traveling?**
    ¿Dónde preferís comer cuando viajás?
    - **In the historic center** (En el centro histórico)
    - **Near my accommodation** (Cerca de mi alojamiento)
    - **In tourist areas** (En zonas turísticas)
    - **In less crowded, local neighborhoods** (En barrios locales menos concurridos)
    - **It doesn't matter to me** (Me da lo mismo)

14. **How do you usually choose a restaurant when traveling? *You can select up to 2 options*.**
    ¿Cómo solés elegir un restaurante cuando viajás? *Podés elegir hasta 2 opciones*.
    - **I search and book in advance** (Lo busco y reservo con anticipación)

- **I decide while walking by** (Decido al pasar por el lugar)
- **Based on online reviews as Google Maps** (Según reseñas online como Google Maps)
- **Based on personal recommendations** (Según recomendaciones personales)
- **Based on location** (Según la ubicación)
- **Others**

NATURAL AREAS

---

15. **To what extent do you agree with the following statements?**
    ¿Qué tan de acuerdo estás con estas afirmaciones?
    - **"The proximity to natural areas or outdoor routes is important for my travel experience."** ("La proximidad a zonas naturales o rutas al aire libre es importante para mi experiencia de viaje.")
        - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).

16. **What type of natural environment do you prefer to visit when traveling? *You can select up to 2 options.***
    ¿Qué tipo de entorno natural preferís visitar cuando viajás? *Podés elegir hasta 2 opciones.*
    - Beach (Playas)
    - Mountains (Montañas)
    - Natural parks (Parques naturales)
    - Lake/Rivers (Lagos/Ríos)
    - Others

REMAINING ATTRIBUTES

---

17. **To what extent do you agree with the following statements?**
    ¿Qué tan de acuerdo estás con estas afirmaciones?
    - **"The availability of tourist information, both digital and on-site, is important for my travel experience."** ("La disponibilidad de información turística, tanto digital como presencial, es importante para mi experiencia de viaje.")
        - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).
    - **"The availability of museums and art venues is important for my travel experience."** ("La disponibilidad de museos y espacios de arte es importante para mi experiencia de viaje.")
        - Likert Scale from 1 (strongly diasgree) to 5 (strongly agree).

Van Westendorp Question

18. **Imagine a weekend package in an Italian city: it includes 2 nights in centrally located accommodation, 1 dinner at an authentic local restaurant, and 1 activity of your choice (for example: a guided tour). *Exclude transportation*.**
Imaginá un paquete de fin de semana en una ciudad Italiana: 2 noches en alojamiento céntrico, 1 cena en un restaurante local auténtico y 1 actividad a elección (como un tour guiado). *Excluí el transporte*.

- **At what price would you consider the product to be priced so low that you would feel the quality couldn't be very good? In euros (€).** ¿A partir de qué precio considerarías que el producto es tan barato que pensarías que su calidad no es muy buena? En euros (€).
- **At what price would you consider the product to be a bargain - a great buy for the money? In euros (€).** ¿A partir de qué precio considerarías que el producto es una ganga - una excelente compra por el dinero? En euros (€).
- **At what price would you consider the product starting to get expensive, so that it is not out of the question, but you would have to give some thought to buying it? In euros (€).** ¿A partir de qué precio considerarías que el producto empieza a resultar caro, de modo que no lo descartarías, pero tendrías que pensarlo antes de comprarlo? En euros (€).
- **At what price would you consider the product to be so expensive that you would not consider buying it? In euros (€).** ¿A partir de qué precio considerarías que el producto es tan caro que no lo comprarías? En euros (€).

ASCOLI PICENO

19. **Have you ever visited the Italian city Ascoli Piceno?**
¿Alguna vez visitaste la ciudad italiana de Ascoli Piceno?
    - **Yes** (Sí) **→ Continues with section VISITED ASCOLI PICENO**
    - No **→ Continues in section DID NOT VISITED ASCOLI PICENO**

VISITED ASCOLI PICENO →Jumps to section ASCOLI PICENO

20. **Thinking about your experience, how likely are you to recommend Ascoli Piceno to someone who hasn't traveled there?**
Pensando en tu experiencia, ¿qué tan probable es que recomiendes Ascoli Piceno a alguien que no ha viajado?
    - Likert scale of 5 (0=Not probable (Probablemente no) ; 5=Very Probable (Muy probable))

DID NOT VISITED ASCOLI PICENO

Ascoli Piceno is a historic city in central Italy, located about two hours from Rome and near the Adriatic coast. It is characterized by its medieval towers, Renaissance squares such as Piazza del Popolo, its local cuisine (especially olive all'ascolana), and its well-preserved historic center surrounded by mountains and natural parks. ADD A MAP FOR REFERENCE
Ascoli Piceno es una ciudad histórica del centro de Italia, ubicada a unas dos horas de Roma y cerca de la costa del Adriático. Se caracteriza por sus torres medievales, plazas renacentistas como la Piazza del Popolo, su gastronomía local (especialmente las olive all'ascolana), y su centro histórico bien conservado, rodeado de montañas y parques naturales. AGREGAR UN MAPA COMO REFERENCIA.

21. **How likely are you to consider visiting Ascoli Piceno if you were traveling through Italy on your next trip?**
    ¿Qué tan probable es que consideres visitar Ascoli Piceno si estuvieras viajando por Italia en tu próximo viaje?
    - Likert scale of 10 (0=Not probable (Probablemente no) ; 10=Very Probable (Muy probable))
22. **Before this survey, had you heard of Ascoli Piceno?**
    Antes de esta encuesta, ¿habías oído hablar de Ascoli Piceno?
    - **Yes** (Sí)
    - No

ASCOLI PICENO

23. Based on what you know or have read, how attractive does Ascoli Piceno seem to you as a tourist destination?
En base a lo que sabés o leiste, ¿qué tan atractiva te parece Ascoli Piceno como destino turístico?
- Likert scale of 10 (0= Not attractive (Nada atractiva); 10=Very attractive (Muy atractiva))

PERSONAL INFORMATION

24. **What is your age group?**
    ¿Cuál es tu rango etario?
    - < 18
    - 18-25
    - 26-35
    - 36-45
    - 46-55
    - >55

25. **Which is your gender?**
¿Cuál es tu género?
   - **Male** (Masculino)
   - **Female** (Femenino)
   - **Other/I prefer not to say it** (Otro/Prefiero no decirlo)

26. **Where do you live?**
¿Dónde vivís?
   - **Italy** (Italia)
   - **Europe without Italy** (Europa sin Italia)
   - **Asia**
   - **Africa**
   - **Oceania**
   - **North America** (América del Norte)
   - **Central America** (América Central)
   - S**outh America** (América del Sur)

27. **What is your current occupation?**
¿Cuál es tu ocupación actual?
   - **Employed** (Empleado/a)
   - **Self-employed** (Independiente)
   - **Student** (Estudiante)
   - **Retired** (Jubilado)
   - **Unemployed** (Desempleado)
   - **Others**

O. Distribution of Likert-Scale Responses

	1	2	3	4	5
att_local_restaurants_diversity	0	9	26	47	43
att_attractions_density	3	7	29	53	33
att_less_crowded_destinations	4	6	37	53	25
att_tourist_info_availability	3	11	29	53	29
att_central_accommodation	4	5	18	67	31
att_museums_art_availability	7	10	41	50	17
att_accommodation_services	4	5	27	58	31
att_heritage_ambience	4	6	20	55	40
att_natural_areas_proximity	1	6	40	55	23
att_guided_tours_flexibility	17	16	36	37	19

Figure 4.12: *Likert-scale Response Counts. Source: Author's own elaboration.*

P. Pseudocode for Spearman Correlation

Algorithm 9 Spearman Correlation Analysis

Input: paired datasets A and B with matching attributes

Output: table of Spearman correlations, p-values, and significance labels

initialize list R
for each attribute i **do**
    extract paired vectors $x \leftarrow A_i,\ y \leftarrow B_i$
    compute Spearman statistics (ρ, p)
    append (i, ρ, p) to R
end for
for each (i, ρ, p) in R **do**
    if $p < 0.0001$ **then**
        label ← "****"
    else if $p < 0.001$ **then**
        label ← "***"
    else if $p < 0.01$ **then**
        label ← "**"
    else if $p < 0.05$ **then**
        label ← "*"
    else
        label ← ""
    end if
    store label with record
end for

Output: results table containing attribute, ρ, p, significance

Q. Pseudocode for Elbow Method

Algorithm 10 Elbow Method with Gower + KMedoids

Input: dataset X, cluster range $k \in [2, 10]$
Output: estimated optimal number of clusters

```
compute Gower distance matrix D
function WITHIN_CLUSTER_DISTANCE(D, labels)
    compute sum of pairwise distances within each cluster
    return total within-cluster distance
end function
for each k in [2, 10] do
    apply K-Medoids to D to obtain labels_k
    score_k ← WITHIN_CLUSTER_DISTANCE(D, labels_k)
    record (k, score_k)
end for
determine elbow from (k, score_k) curve
```

Output: optimal k

R. Pseudocode for Silhouette Score

Algorithm 11 Silhouette Score with Gower + K-Medoids

Input: distance matrix D, cluster range $k \in [2, 10]$
Output: estimated optimal number of clusters

```
for each k in [2, 10] do
    apply K-Medoids to D to obtain labels_k
    compute Silhouette score s_k using (D, labels_k)
end for
select k maximizing s_k
```

Output: optimal k

S. Pseudocode for Davies Bouldin Index

Algorithm 12 Davies–Bouldin Index with Gower + K-Medoids

Input: distance matrix D, cluster range $k \in [2, 10]$
Output: estimated optimal number of clusters

function DBI($D, labels$)
    identify clusters and their medoids
    for each cluster i **do**
        compute internal dispersion S_i
    end for
    for each pair (i, j) **do**
        compute $R_{ij} = \frac{S_i + S_j}{d(m_i, m_j)}$
    end for
    return mean of $\max_j R_{ij}$ over all clusters
end function
for $k = 2$ to 10 **do**
    run K-Medoids on D to obtain $labels_k$
    $DBI_k \leftarrow \mathrm{DBI}(D, labels_k)$
end for
select k minimizing DBI_k
Output: optimal k

T. Principal Components Analysis (PCA)

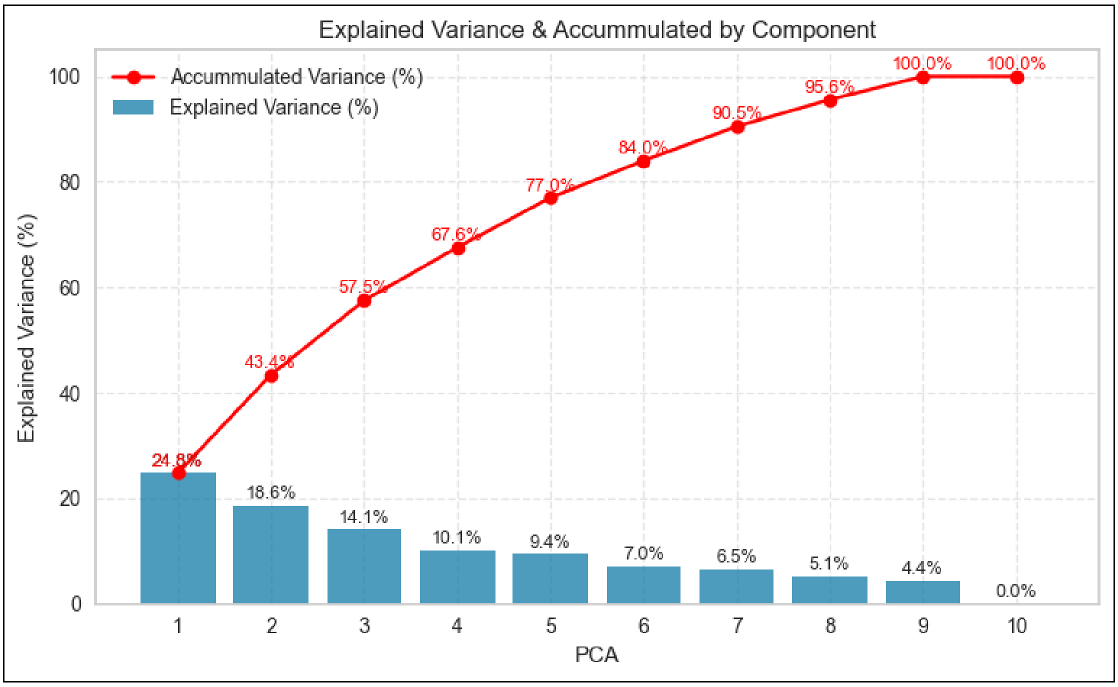


Figure 4.13: *Cumulative Explained Variance (PCA). Source: Author's own elaboration.*

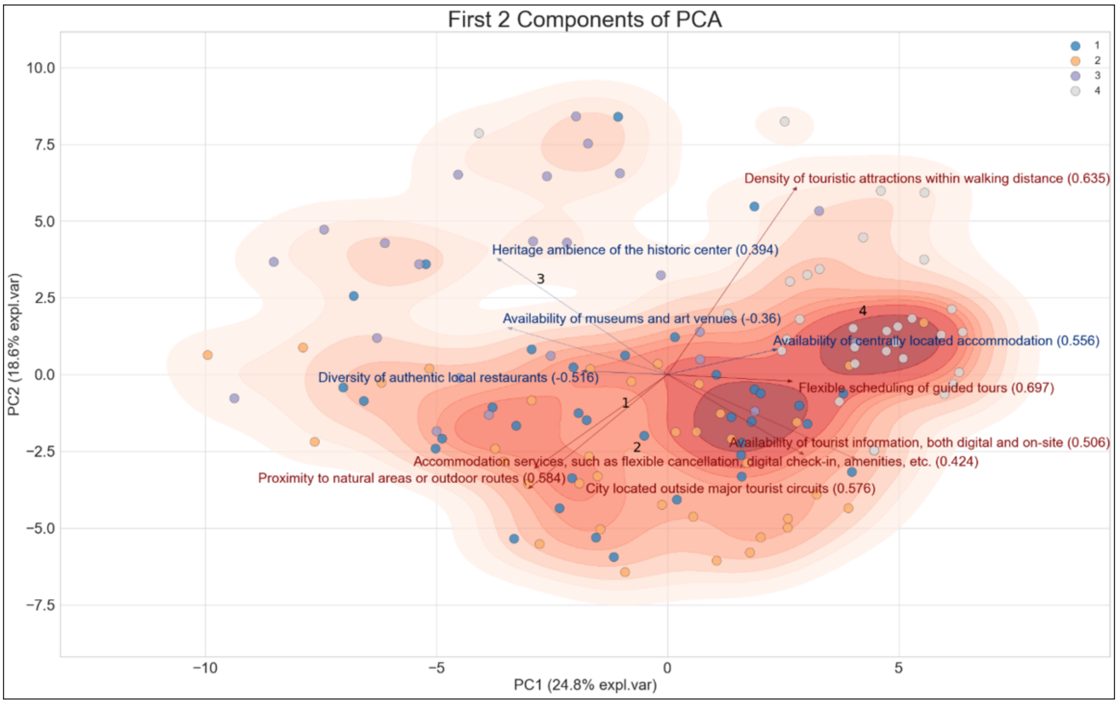


Figure 4.14: *Biplot of First Two Components. Source: Author's own elaboration.*

U. Random Forest Plot

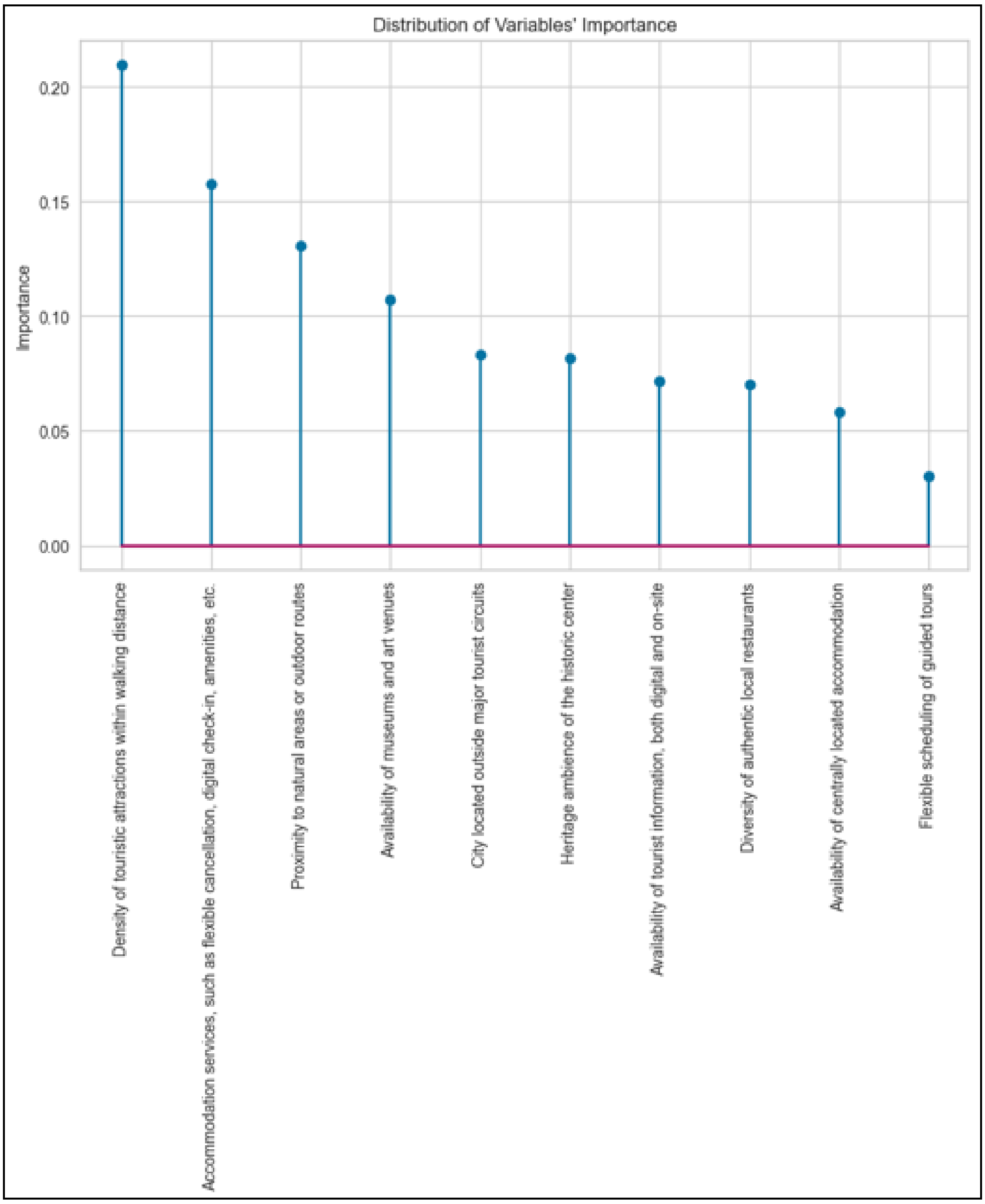


Figure 4.15: *Variable Importance in Random Forest Model. Source: Author's own elaboration.*

V. Correlation Heatmap

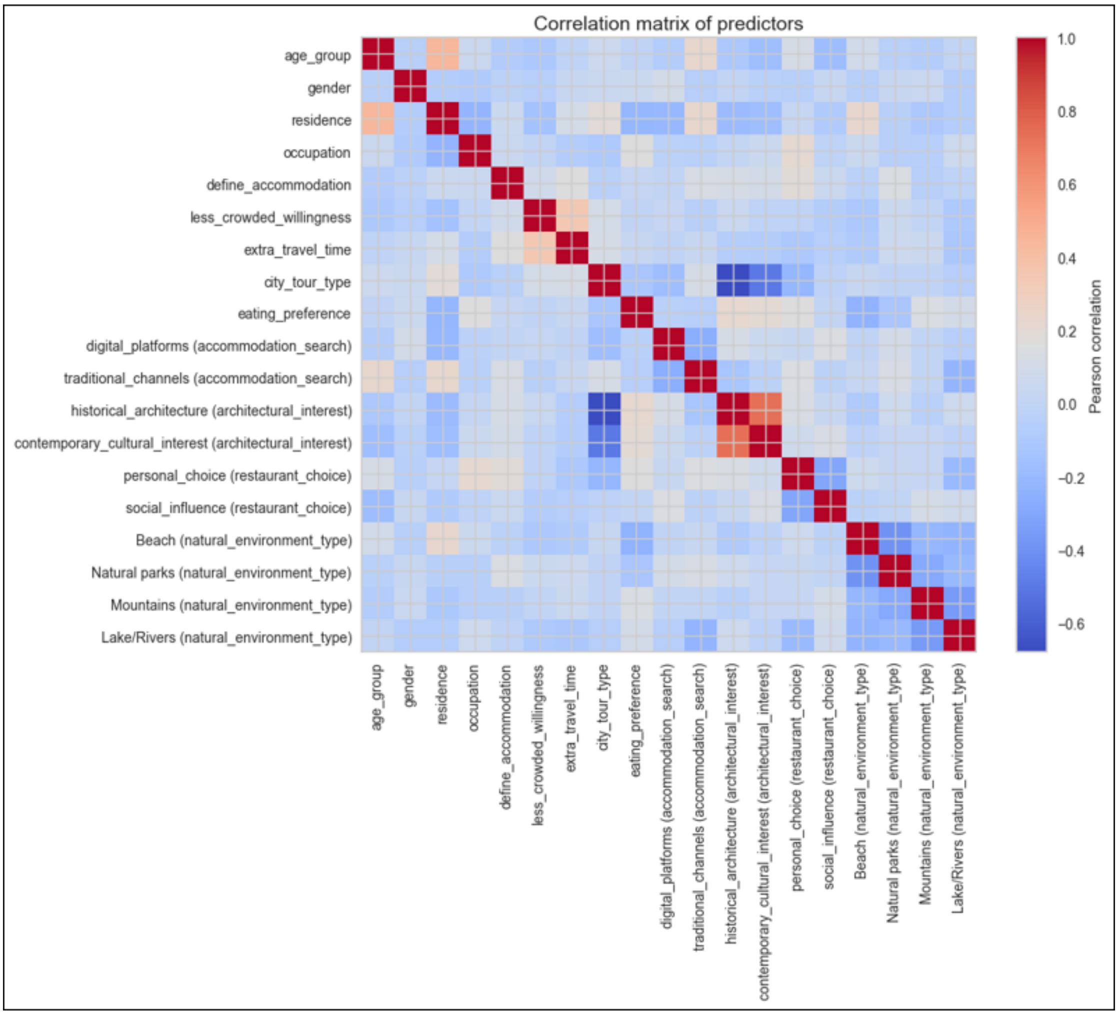


Figure 4.16: *Correlation Heatmap. Source: Author's own elaboration.*

W. Variance Inflation Factor (VIF)

	Variable	VIF
19	const	96.218293
11	historical_architecture (architectural_interest)	3.385057
15	Beach (natural_environment_type)	2.622538
16	Natural parks (natural_environment_type)	2.590555
18	Lake/Rivers (natural_environment_type)	2.477505
17	Mountains (natural_environment_type)	2.428294
12	contemporary_cultural_interest (architectural_...	2.376164
7	city_tour_type	2.124081
2	residence	1.677343
13	personal_choice (restaurant_choice)	1.448941
0	age_group	1.434796
10	traditional_channels (accommodation_search)	1.321366
8	eating_preference	1.320544
5	less_crowded_willingness	1.296000
6	extra_travel_time	1.266397
14	social_influence (restaurant_choice)	1.255486
9	digital_platforms (accommodation_search)	1.231135
3	occupation	1.211654
4	define_accommodation	1.165311
1	gender	1.059941

Figure 4.17: *Variance Inflation Factor (VIF) Scores. Source: Author's own elaboration.*

X. Initial Predictive Model

MNLogit Regression Results			
Dep. Variable:	cluster	No. Observations:	125
Model:	MNLogit	Df Residuals:	65
Method:	MLE	Df Model:	57
Date:	Sat, 25 Oct 2025	Pseudo R-squ.:	0.2252
Time:	18:36:21	Log-Likelihood:	-132.89
converged:	False	LL-Null:	-171.53
Covariance Type:	nonrobust	LLR p-value:	0.03827

Figure 4.18: *MNLogit Regression Results – Model 1. Source: Author's own elaboration.*

Y. Final Model Table

cluster=2	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	1.4212	1.617	0.879	0.380	-1.748	4.591
age_group	-0.0640	0.242	-0.264	0.792	-0.539	0.411
gender	1.2254	0.542	2.262	0.024	0.163	2.287
residence	-0.2761	0.097	-2.838	0.005	-0.467	-0.085
occupation	0.0579	0.272	0.213	0.831	-0.475	0.590
city_tour_type	-0.0946	0.314	-0.301	0.763	-0.710	0.521
Q("digital_platforms (accommodation_search)")	-0.5241	1.100	-0.477	0.634	-2.679	1.631
Q("traditional_channels (accommodation_search)")	-0.1734	0.569	-0.305	0.761	-1.288	0.942
Q("contemporary_cultural_interest (architectural_interest)")	0.1513	0.868	0.174	0.862	-1.549	1.852

Figure 4.19: *MNLogit Regression Results – Cluster 2. Source: Author's own elaboration.*

cluster=3	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-3.0774	1.913	-1.609	0.108	-6.826	0.671
age_group	0.5017	0.253	1.984	0.047	0.006	0.997
gender	0.7213	0.597	1.207	0.227	-0.450	1.892
residence	-0.2159	0.107	-2.027	0.043	-0.425	-0.007
occupation	0.1860	0.287	0.647	0.518	-0.377	0.749
city_tour_type	0.5088	0.318	1.599	0.110	-0.115	1.132
Q("digital_platforms (accommodation_search)")	0.5086	1.345	0.378	0.705	-2.127	3.144
Q("traditional_channels (accommodation_search)")	-0.3754	0.616	-0.609	0.542	-1.583	0.832
Q("contemporary_cultural_interest (architectural_interest)")	1.6924	0.932	1.815	0.069	-0.135	3.520

Figure 4.20: *MNLogit Regression Results – Cluster 3. Source: Author's own elaboration.*

cluster=4	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-0.8706	1.478	-0.589	0.556	-3.766	2.025
age_group	0.2759	0.221	1.250	0.211	-0.157	0.709
gender	1.1129	0.552	2.015	0.044	0.030	2.195
residence	-0.1746	0.098	-1.776	0.076	-0.367	0.018
occupation	0.2716	0.262	1.036	0.300	-0.242	0.785
city_tour_type	-0.0879	0.322	-0.273	0.785	-0.719	0.543
Q("digital_platforms (accommodation_search)")	-0.7909	0.928	-0.852	0.394	-2.610	1.028
Q("traditional_channels (accommodation_search)")	0.6393	0.570	1.122	0.262	-0.478	1.757
Q("contemporary_cultural_interest (architectural_interest)")	0.6726	0.869	0.774	0.439	-1.031	2.376

Figure 4.21: *MNLogit Regression Results – Cluster 4. Source: Author's own elaboration.*

Z. Socio-Demographic Composition by Cluster

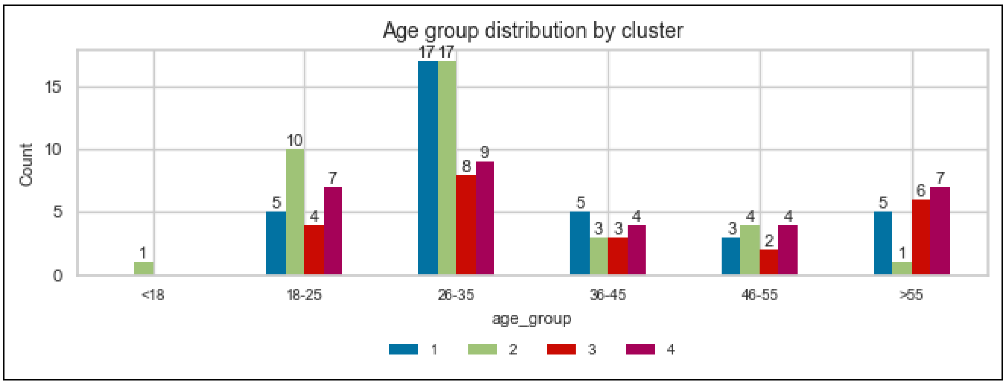


Figure 4.22: *Age Range by Cluster. Source: Author's own elaboration.*

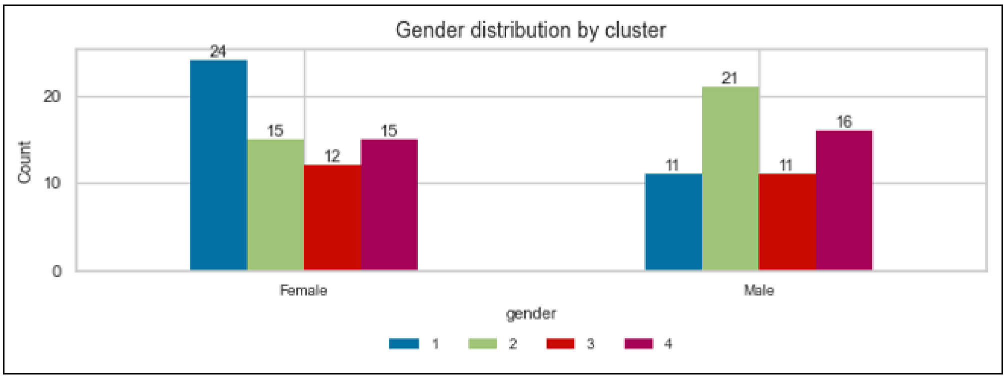


Figure 4.23: *Gender by Cluster. Source: Author's own elaboration.*

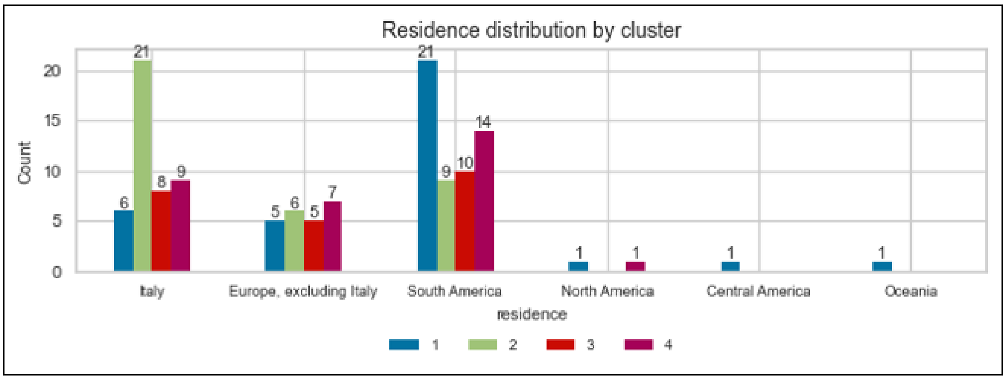


Figure 4.24: *Residence by Cluster. Source: Author's own elaboration.*

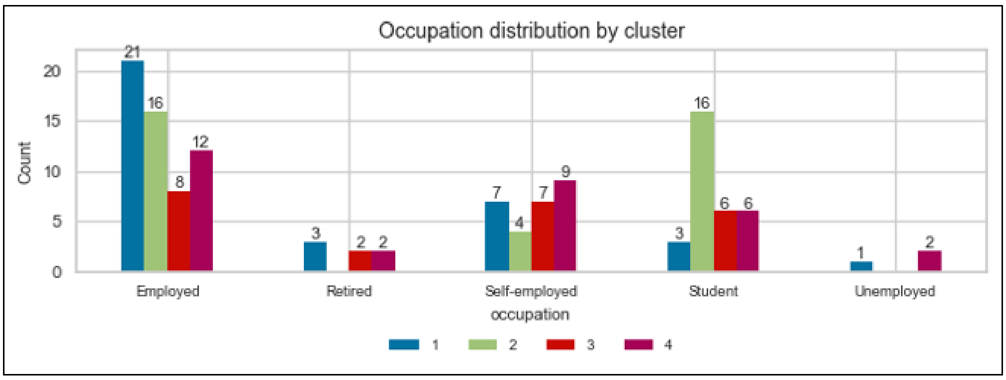


Figure 4.25: *Professional Status by Cluster. Source: Author's own elaboration.*

AA. Top and Bottom Attributes by Cluster

cluster	Diversity of authentic local restaurants	Density of touristic attractions within walking distance	City located outside major tourist circuits	Availability of tourist information, both digital and on-site	Availability of centrally located accommodation	Availability of museums and art venues	Accommodation services, such as flexible cancellation, digital check-in, amenities, etc.	Heritage ambience of the historic center	Proximity to natural areas or outdoor routes	Flexible scheduling of guided tours
1	3	3	8	6	3	6	7	3	4	9
2	4	4	9	8	4	4	5	1	7	9
3	6	8	7	5	2	5	2	5	6	8
4	2	10	6	7	5	3	8	1	4	9

Figure 4.26: *Top and Bottom Attributes by Cluster. Source: Author's own elaboration.*

AB. Willingness to Visit Less-Known Destinations and Extra Travel Time

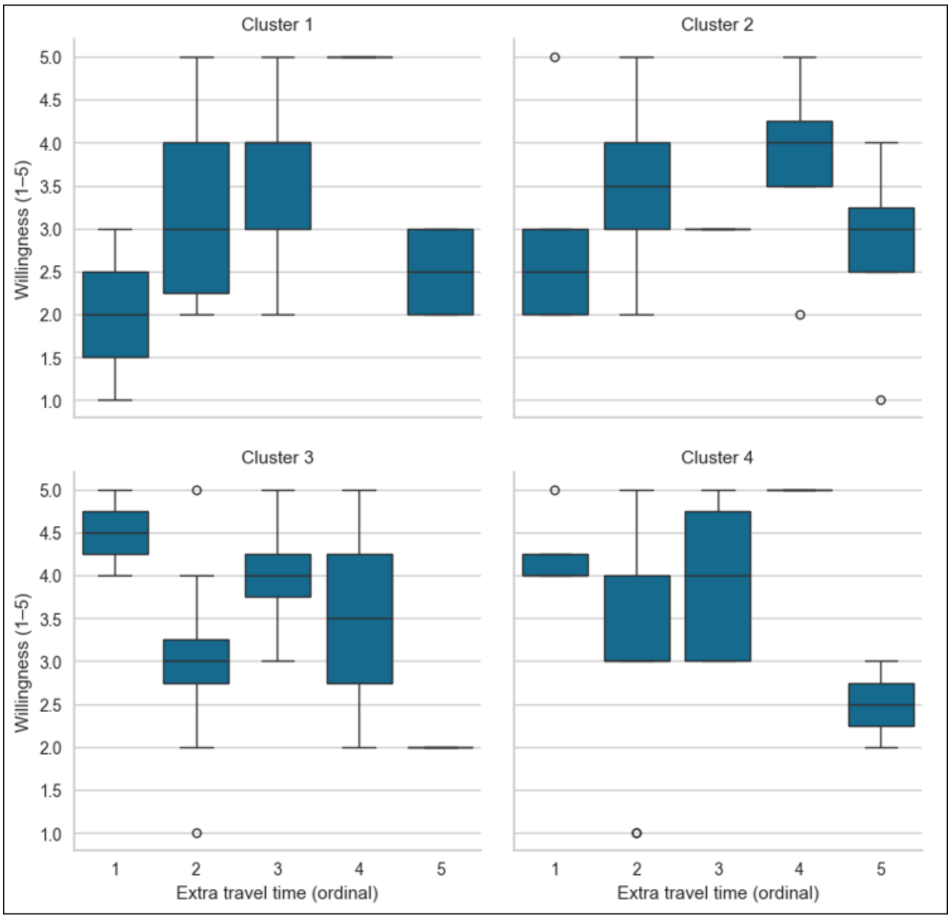


Figure 4.27: *Willingness to Visit Less-known Destinations vs. Extra Travel Time, by Cluster (1–5 Likert Scale). Source: Author's own elaboration.*

AC. Architectural Interest by Cluster

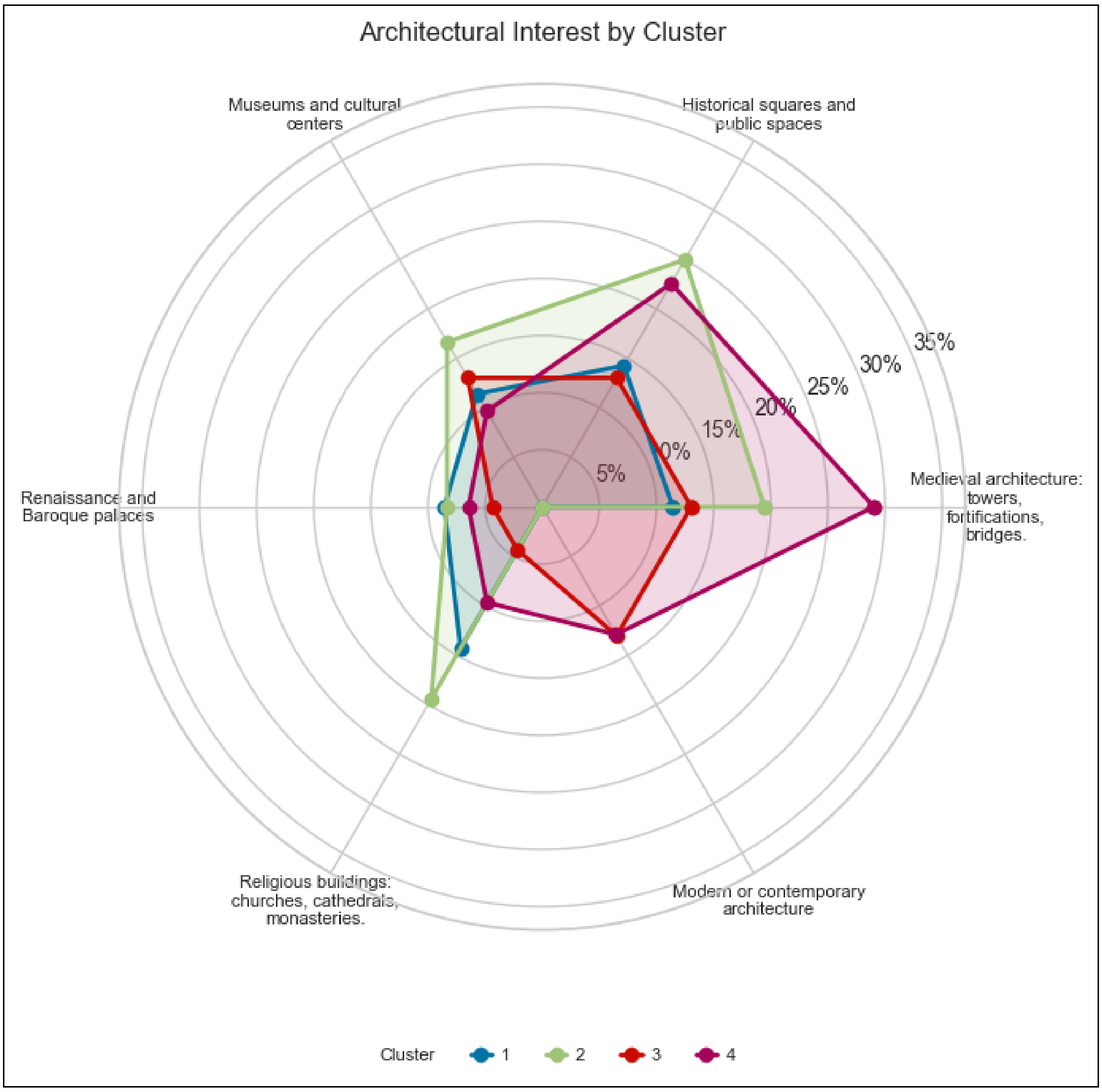


Figure 4.28: *Architectural Interest by Cluster. Source: Author's own elaboration.*